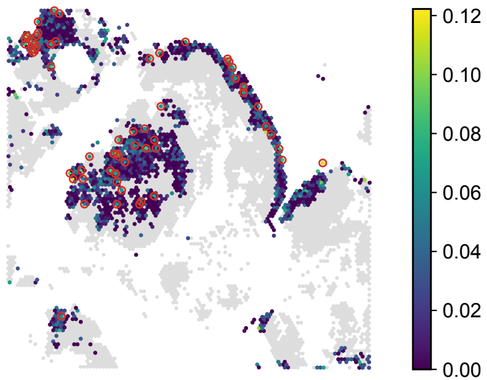
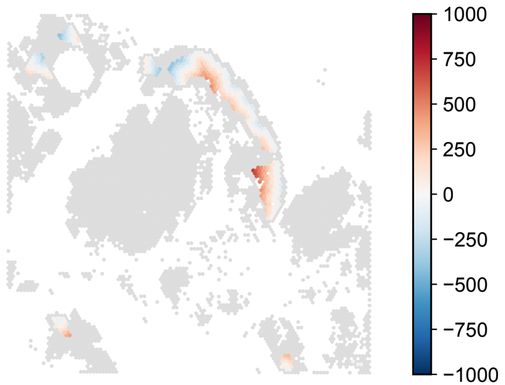
GSE319536_BC01 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

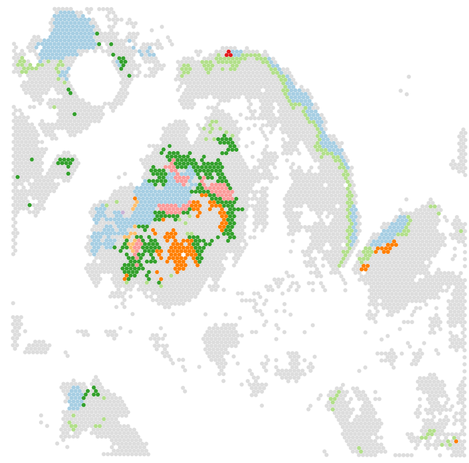
MP10 AUCell and frozen P95



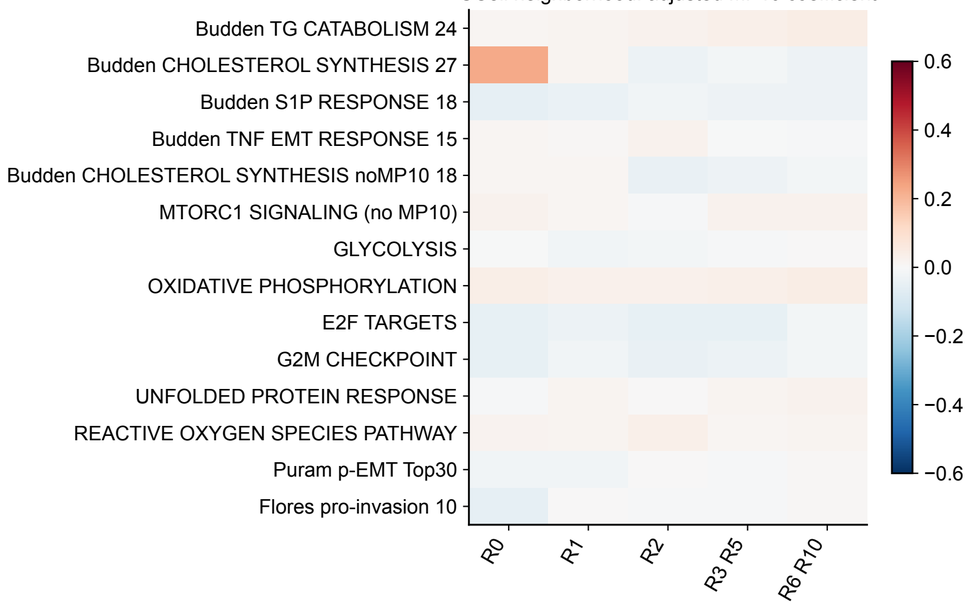
Strict v3.4 signed distance (µm)



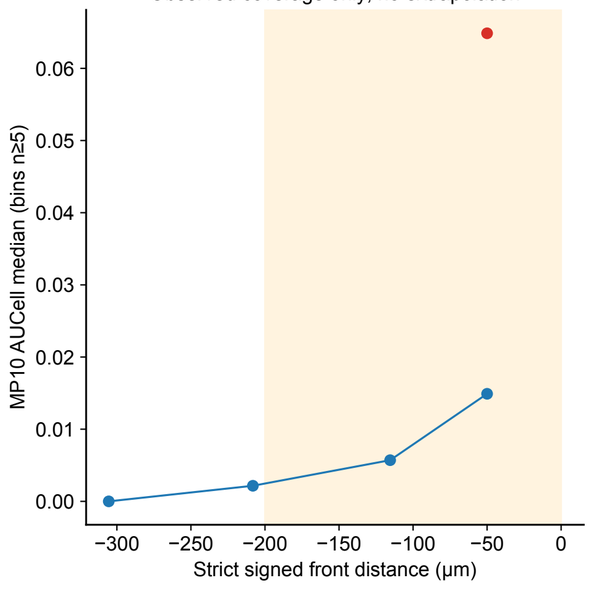
Formal R12 0–8; grey = unlabelled



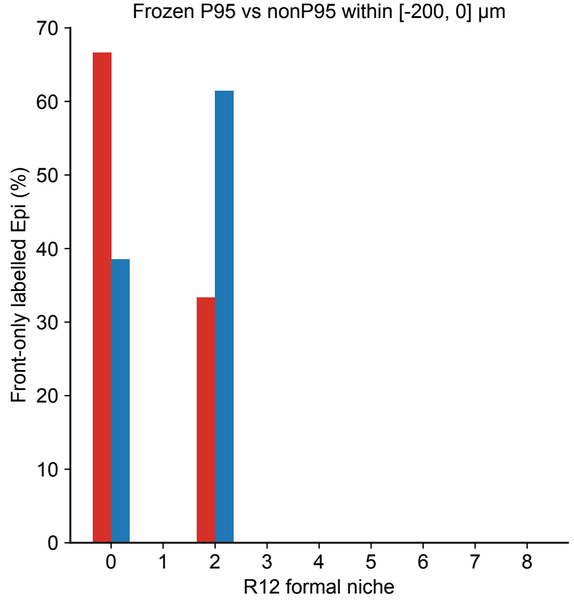
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

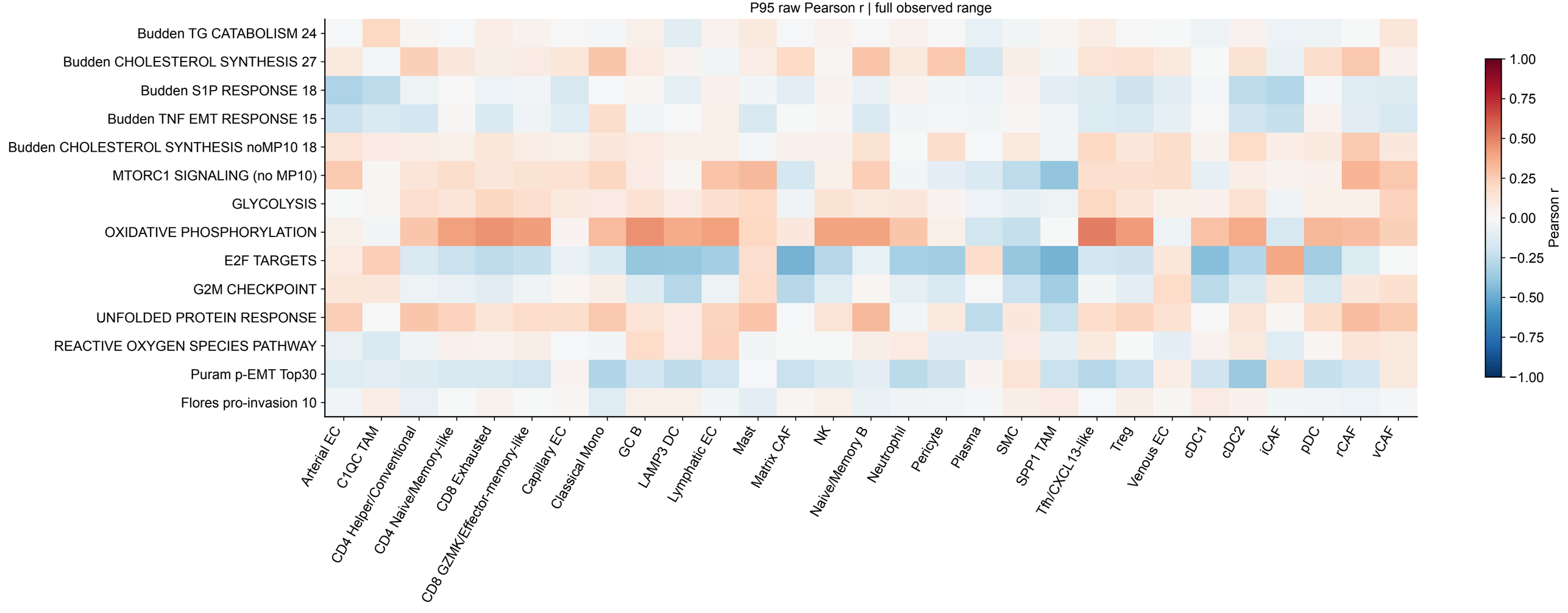


Tissue 5278 | Epi 1567 | internal 1173 | P95 59
Exact front in Epi 187 | R12 labelled 1191

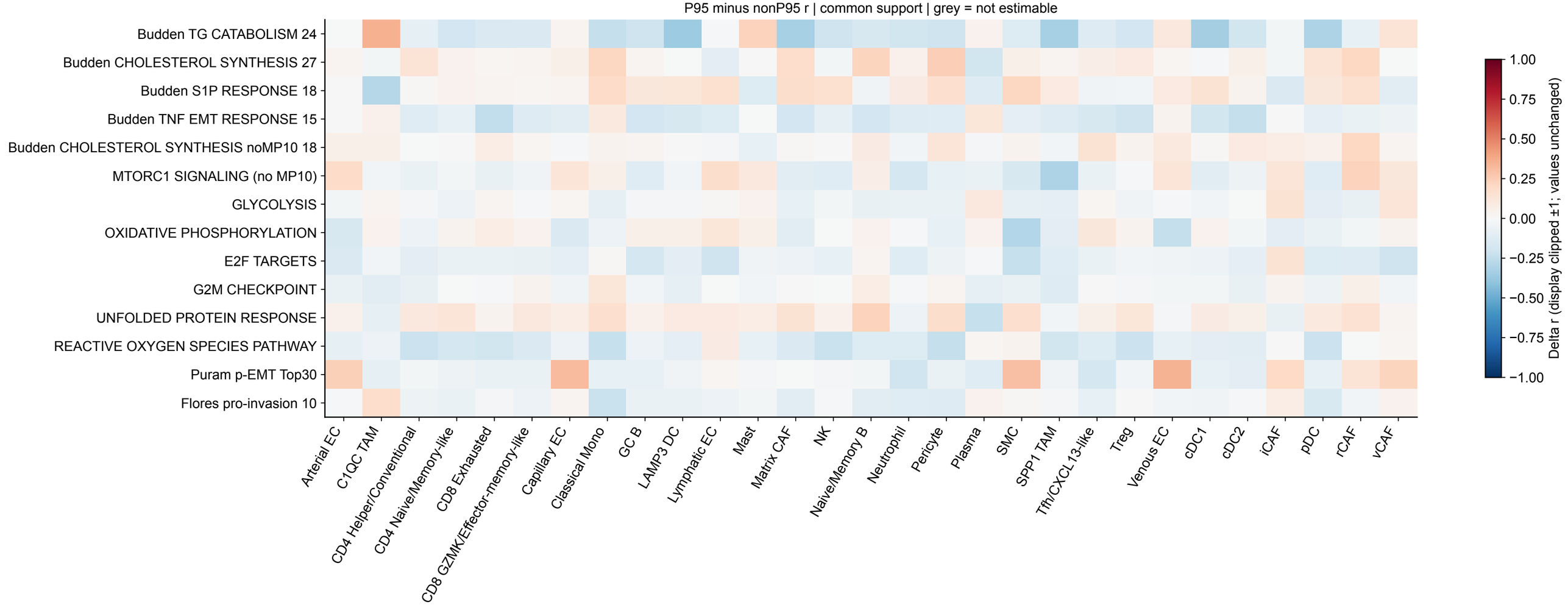
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC01 | all 29 candidates; fixed core programs

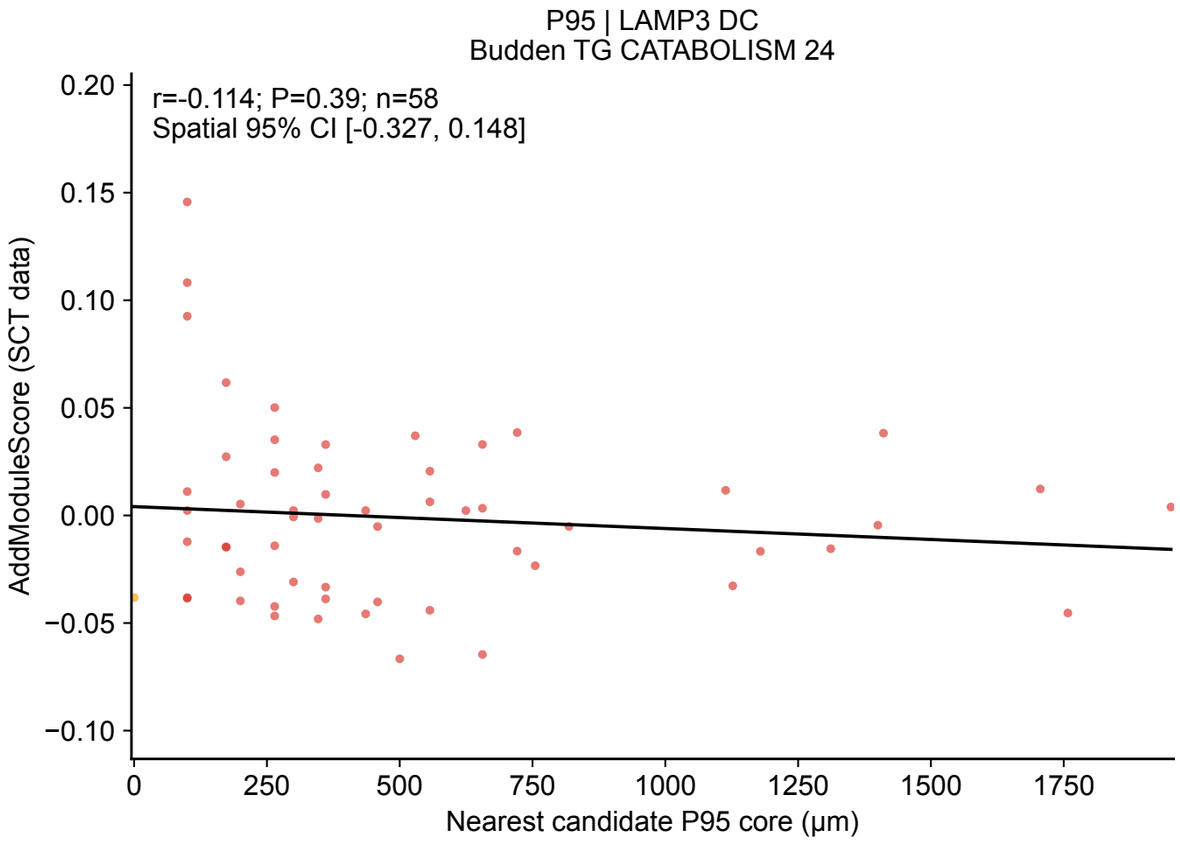
P95 raw Pearson r | full observed range

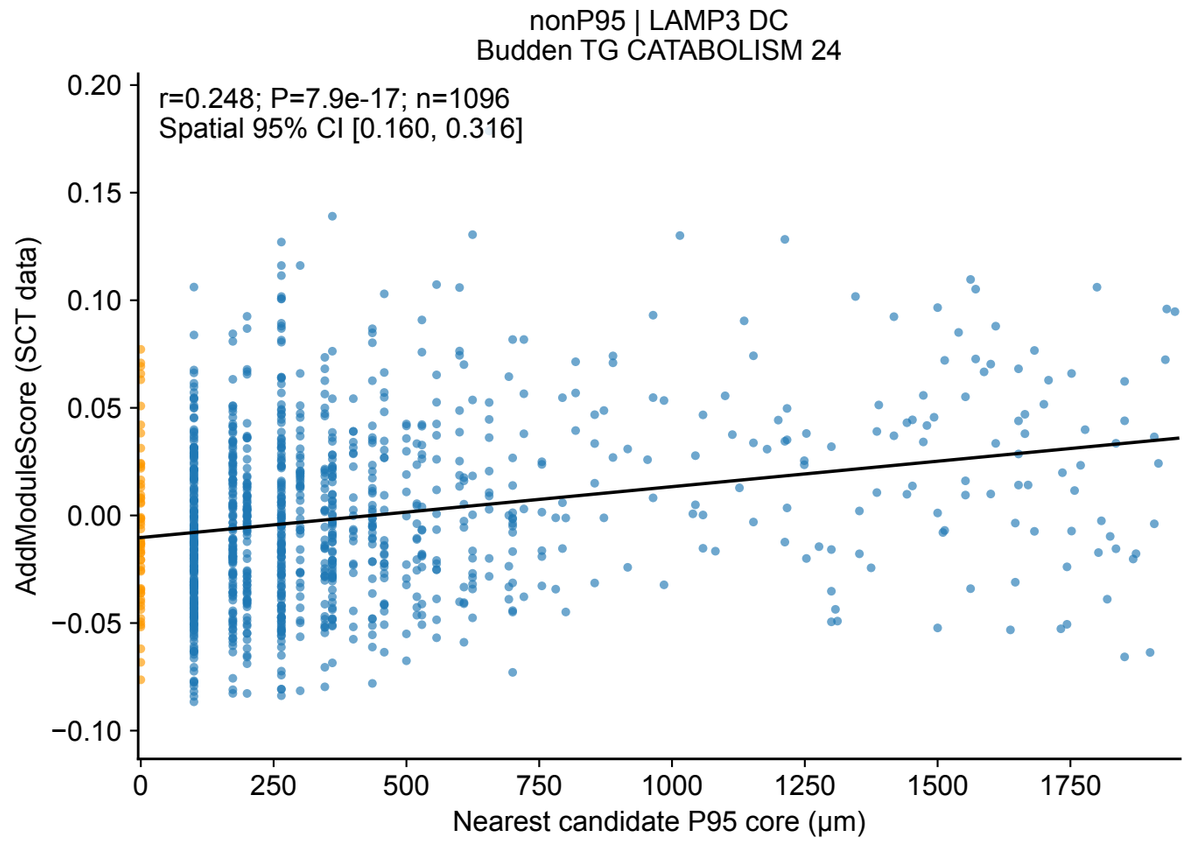


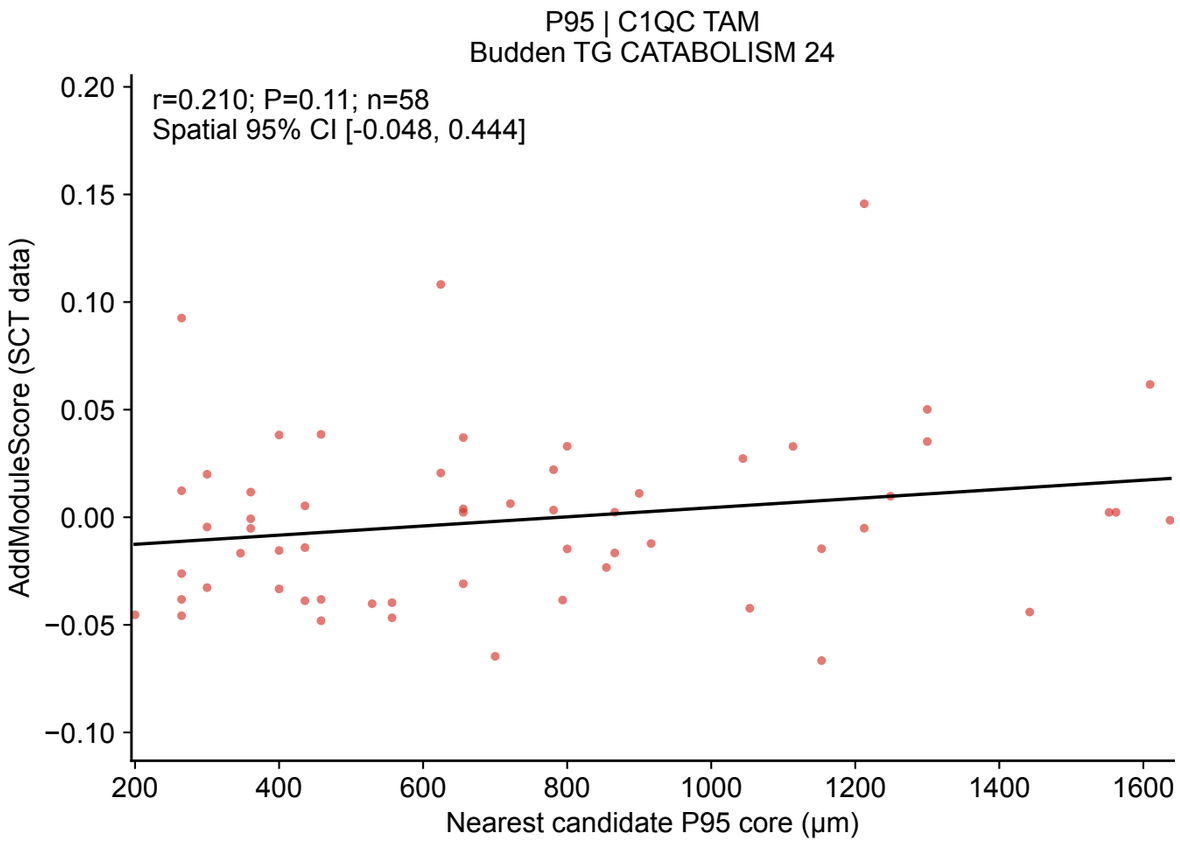
P95 minus nonP95 r | common support | grey = not estimable

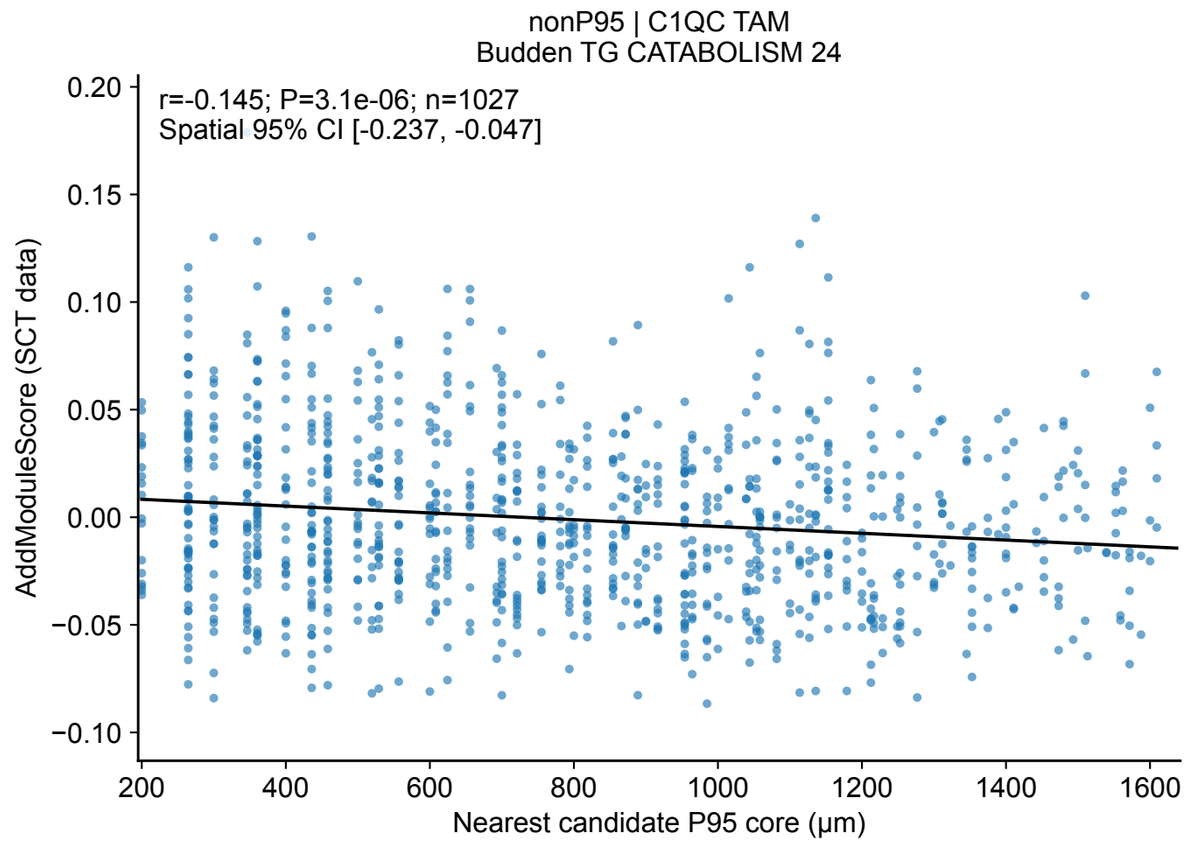


GSE319536_BC01 | exploratory scatter display selected by coverage and |delta r|, not P





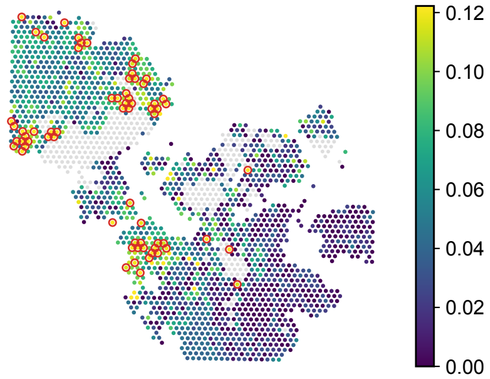


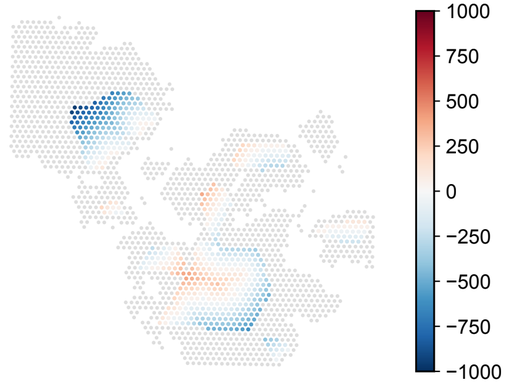


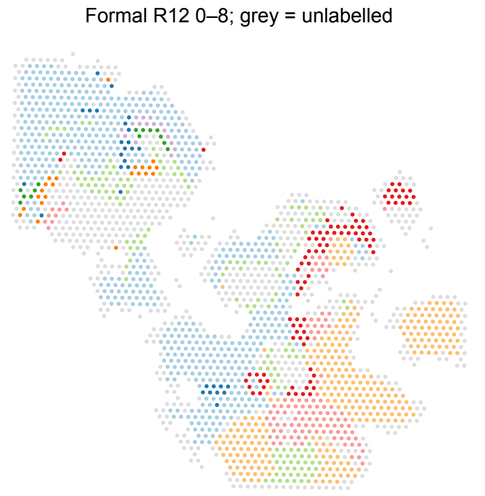
LAMP3 DC / Budden TG CATABOLISM 24: delta r=-0.362 [-0.593, -0.089]; common 0–1952 µm; spatial P=0.0039
C1QC TAM / Budden TG CATABOLISM 24: delta r=0.355 [0.085, 0.605]; common 200–1637 µm; spatial P=0.0084
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

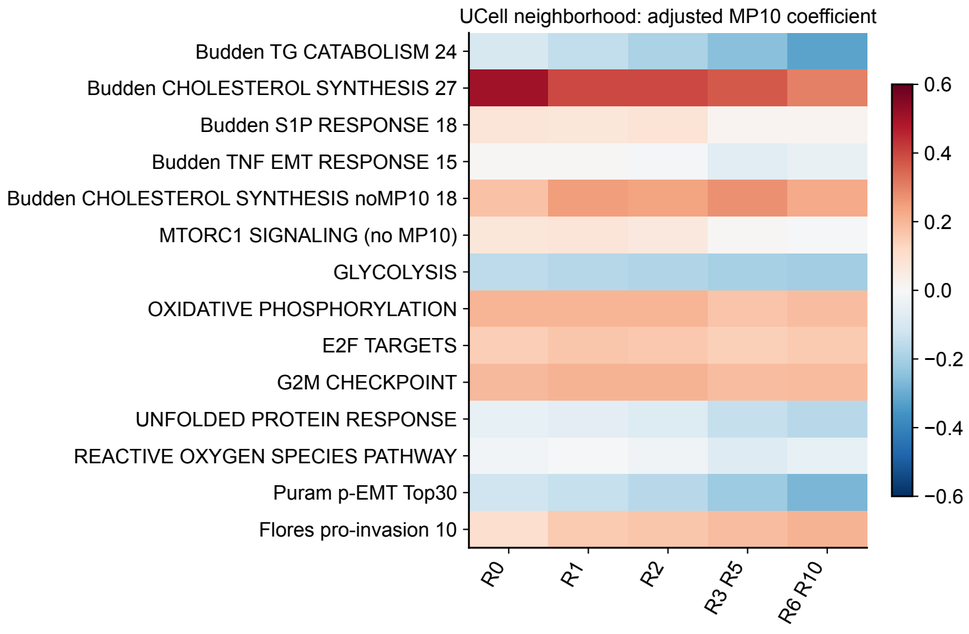
P95 nonP95 distance 0

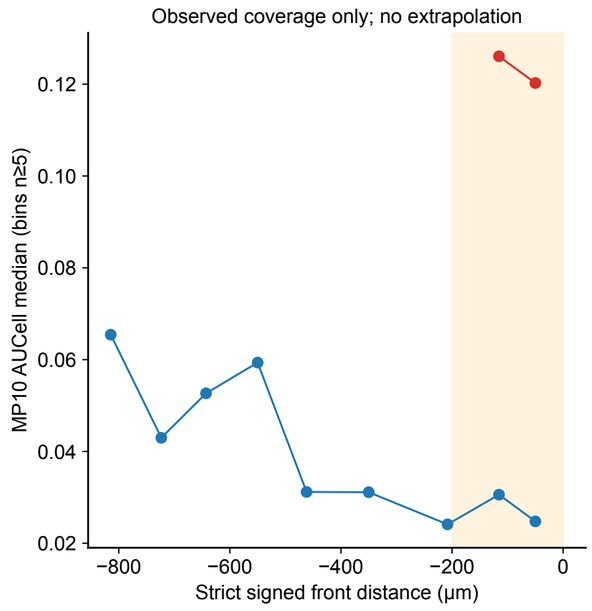
GSE319536_BC02 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

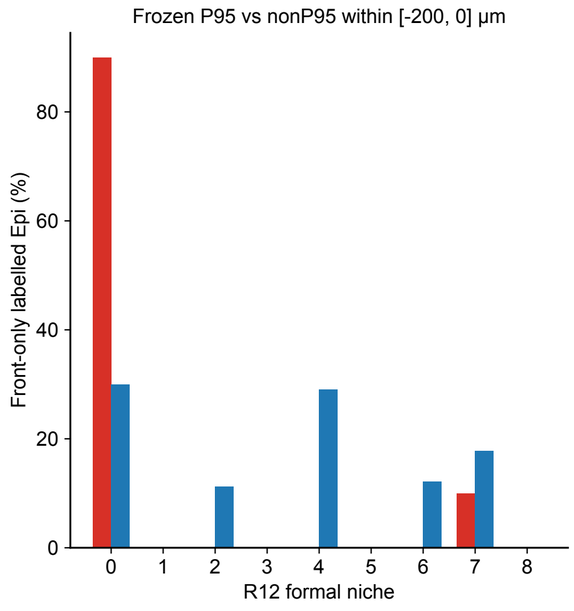










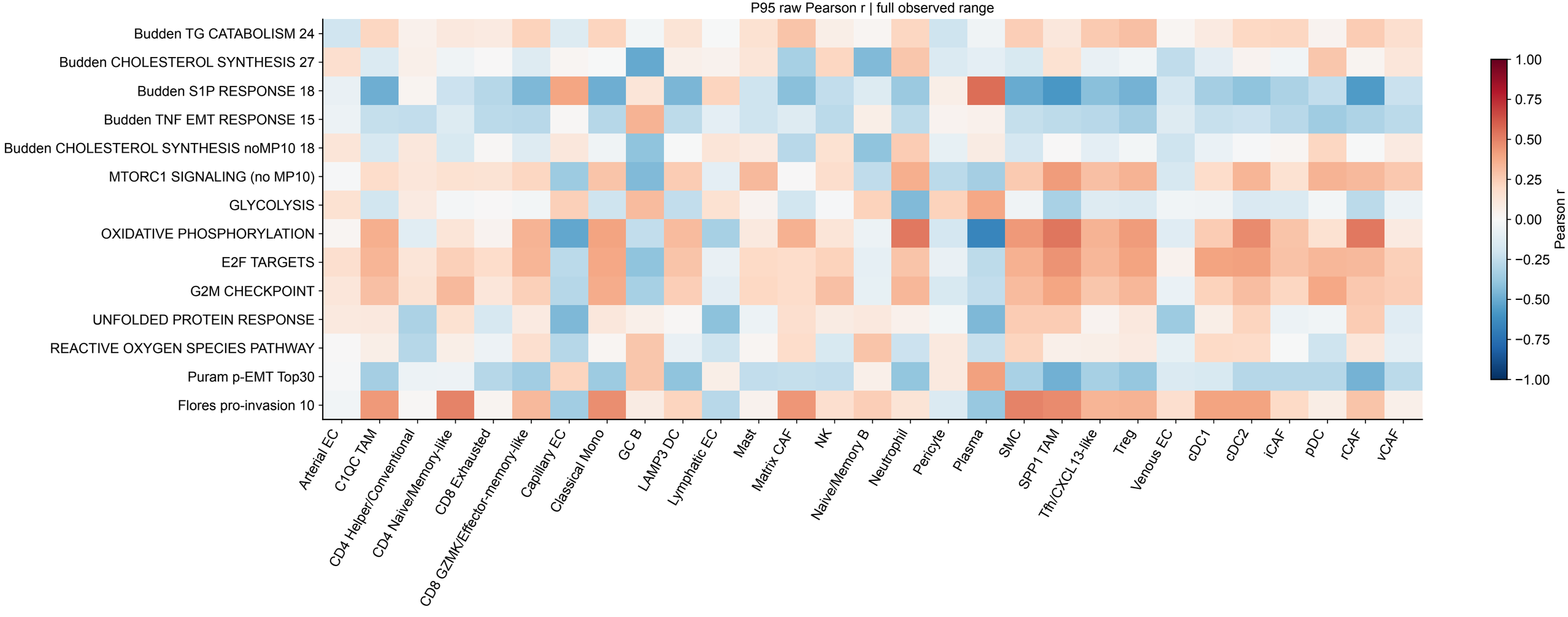


Tissue 2035 | Epi 1744 | internal 1287 | P95 66
Exact front in Epi 436 | R12 labelled 1341

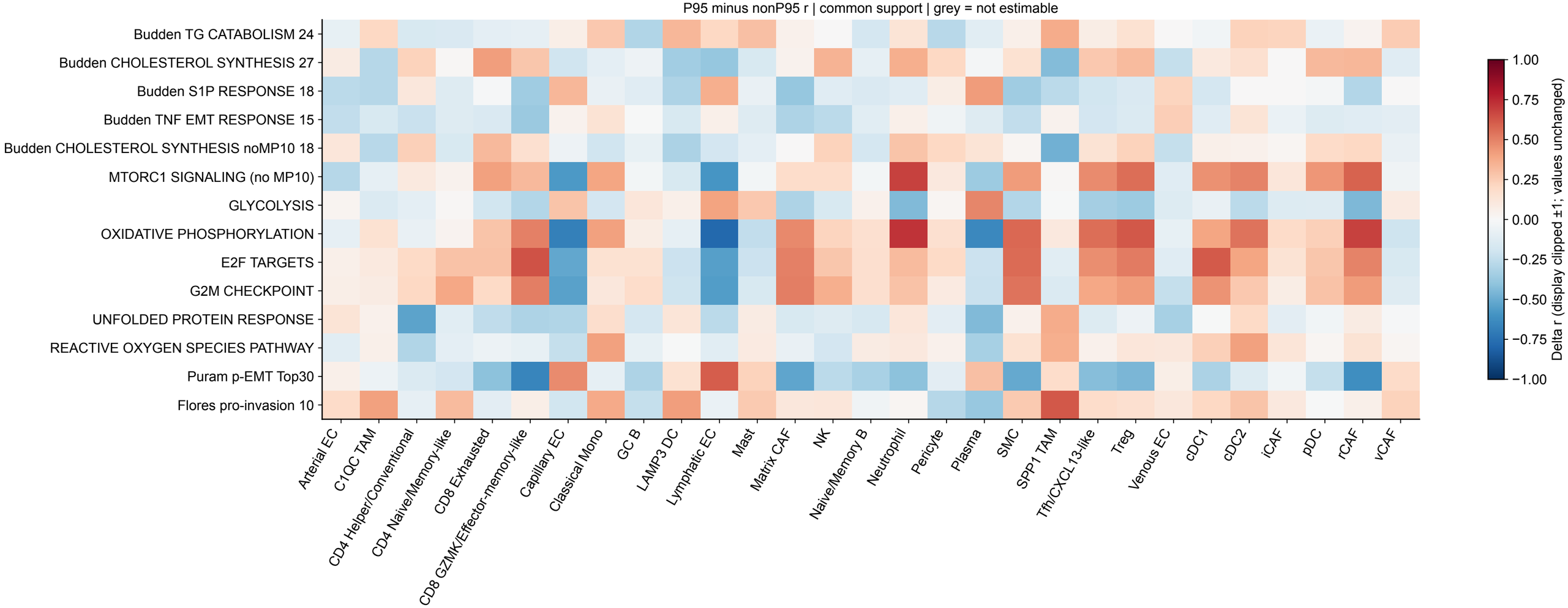
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC02 | all 29 candidates; fixed core programs

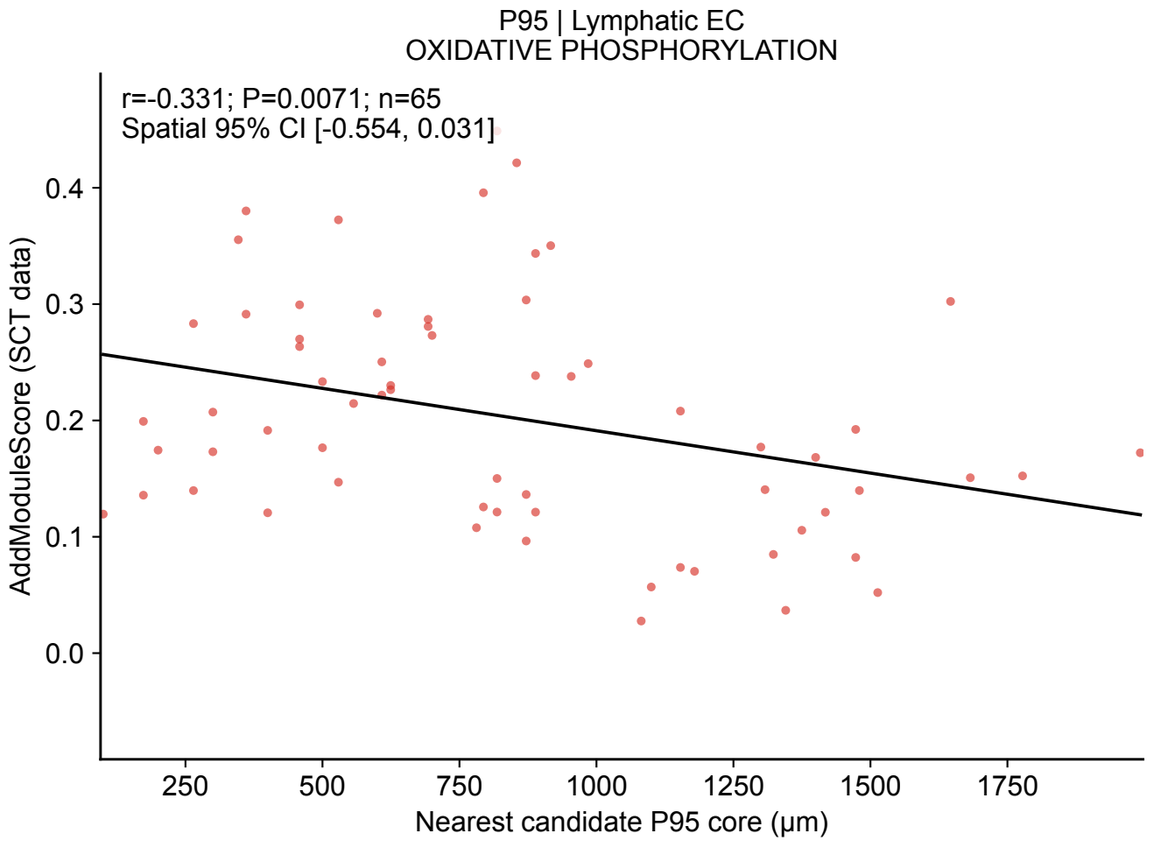
P95 raw Pearson r | full observed range

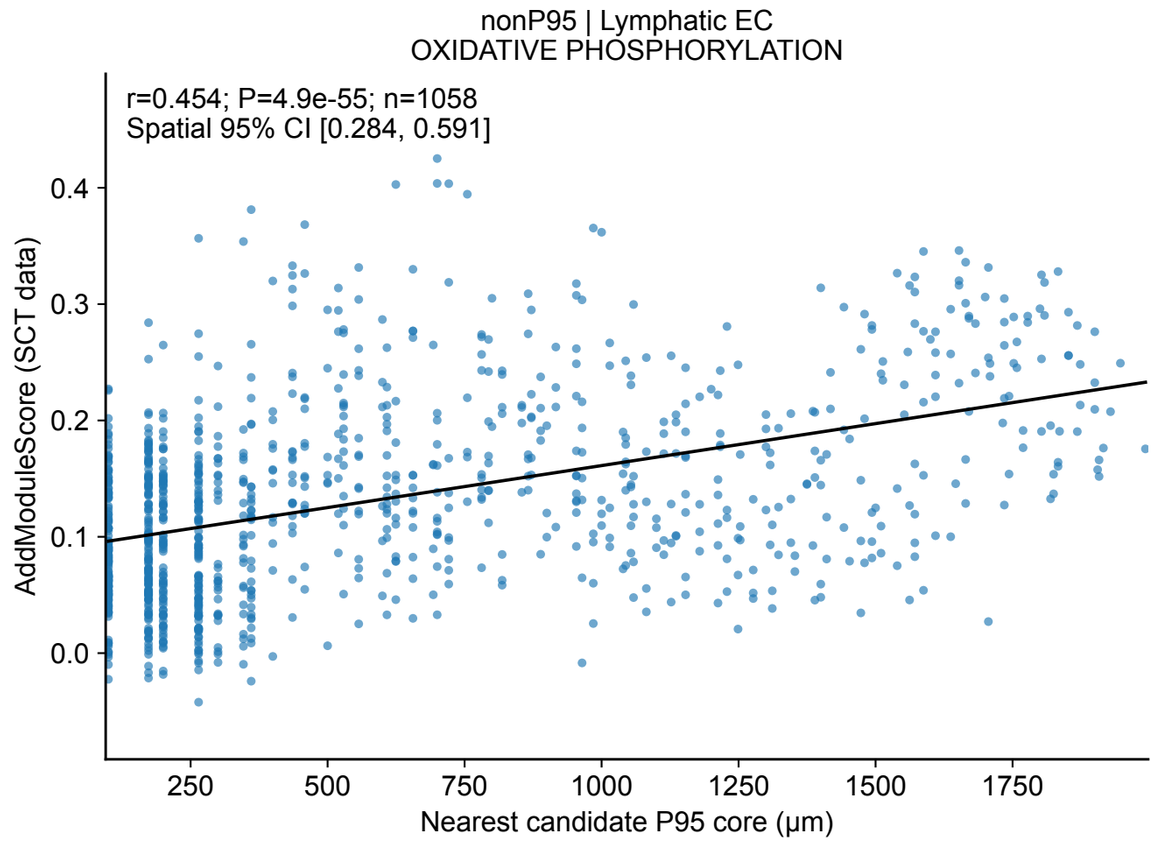


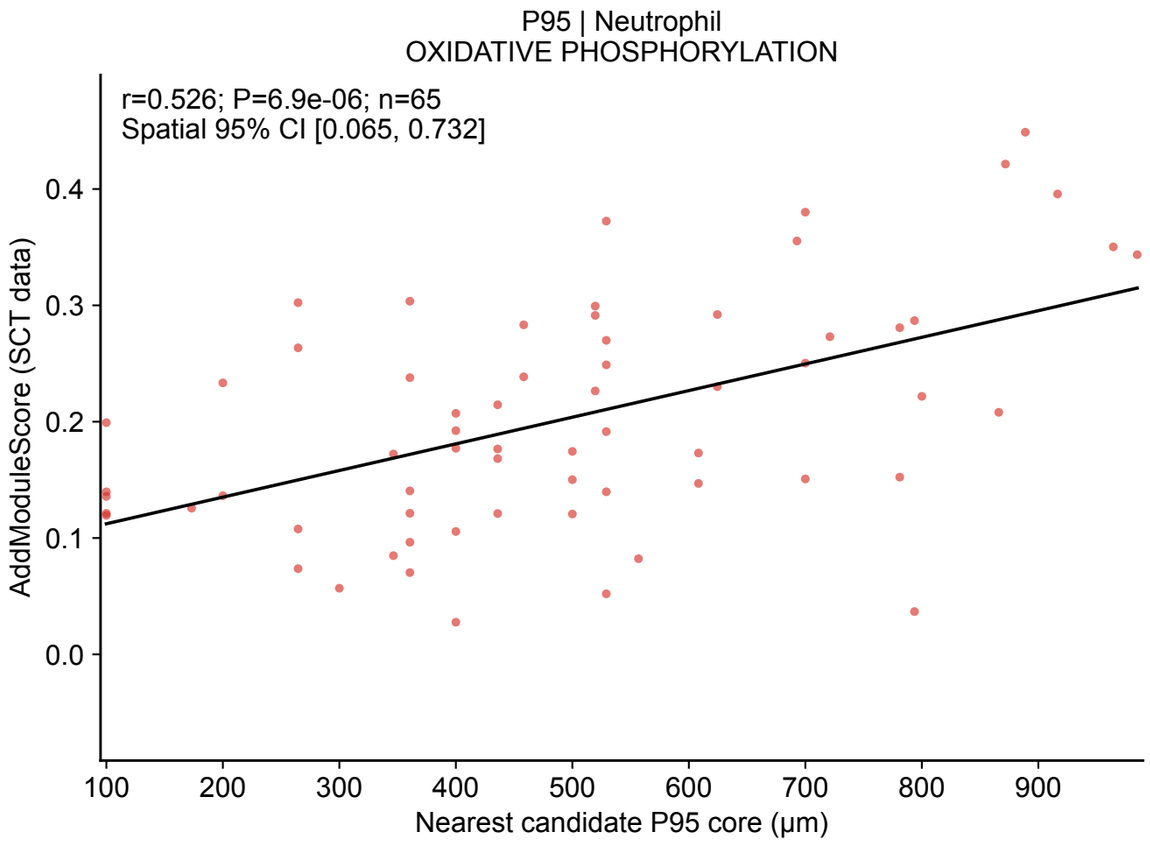
P95 minus nonP95 r | common support | grey = not estimable

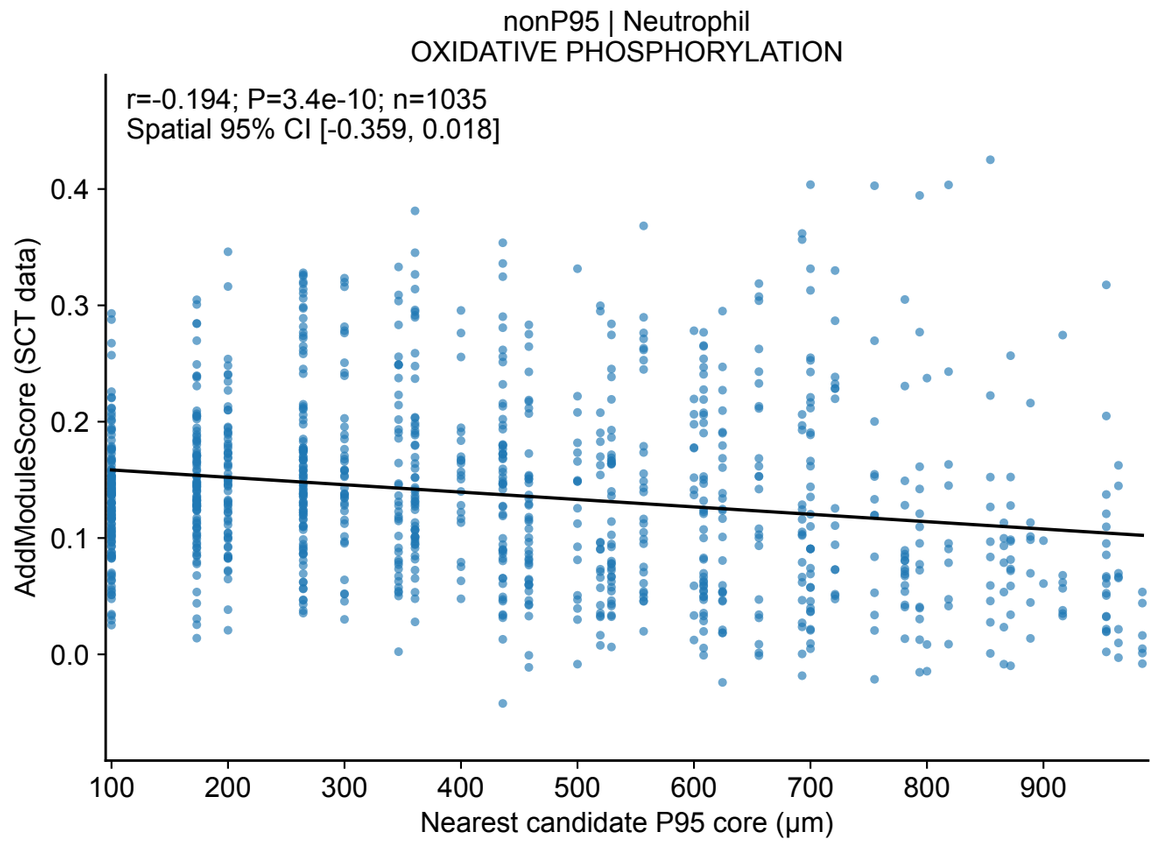


GSE319536_BC02 | exploratory scatter display selected by coverage and |delta r|, not P







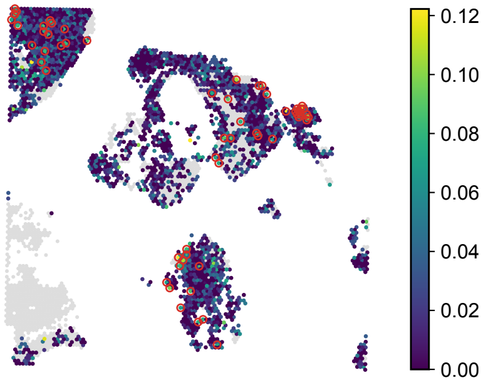


Lymphatic EC / OXIDATIVE PHOSPHORYLATION: delta r=-0.786 [-1.004, -0.440]; common 100–1992 µm; spatial P=5.5e-08
Neutrophil / OXIDATIVE PHOSPHORYLATION: delta r=0.719 [0.338, 0.925]; common 100–985 µm; spatial P=2.3e-05
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

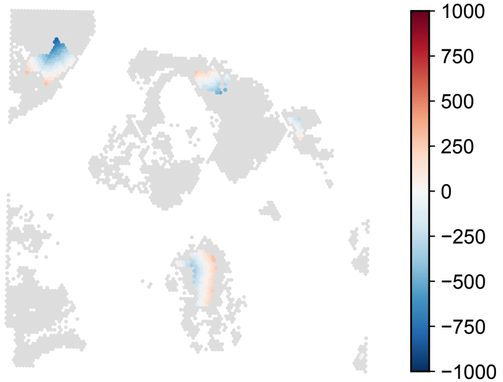
P95 nonP95 distance 0

GSE319536_BC04 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

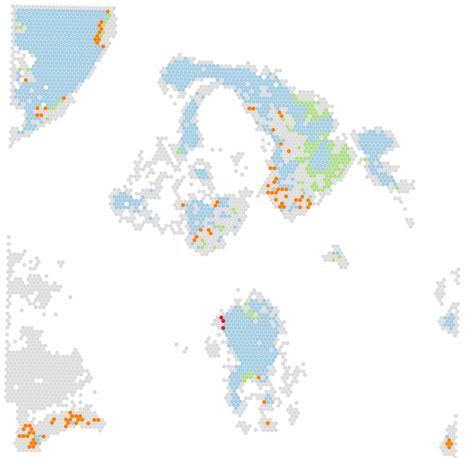
MP10 AUCell and frozen P95



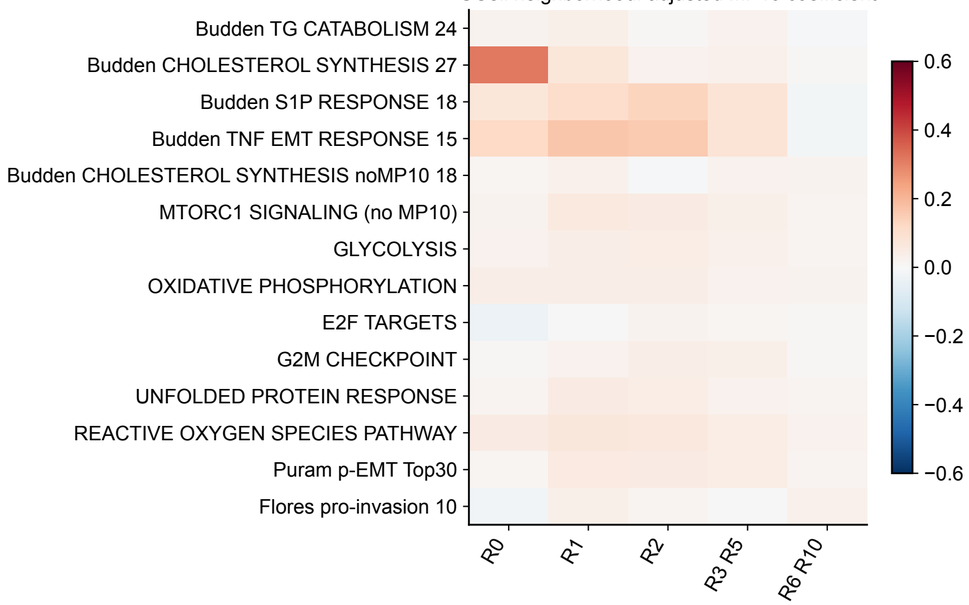
Strict v3.4 signed distance (µm)



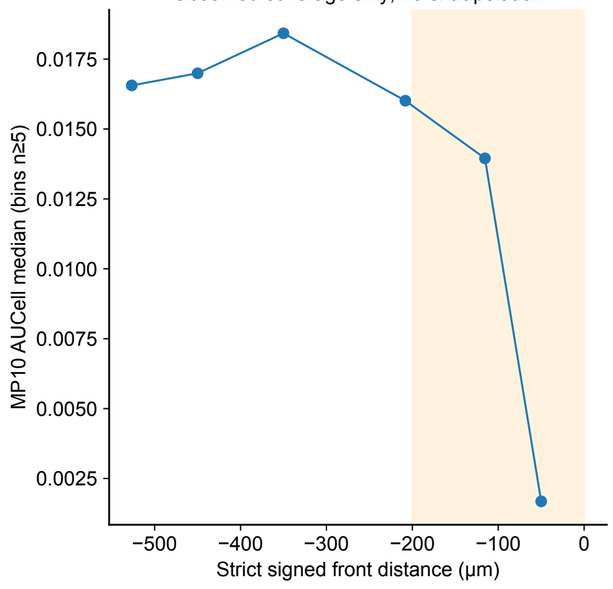
Formal R12 0–8; grey = unlabelled



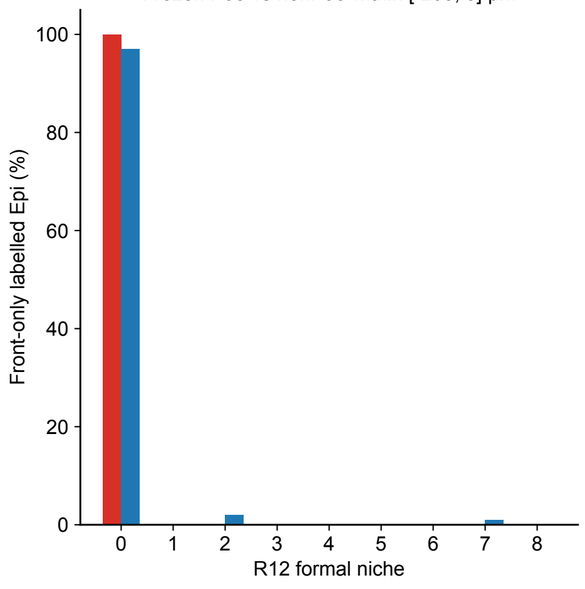
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

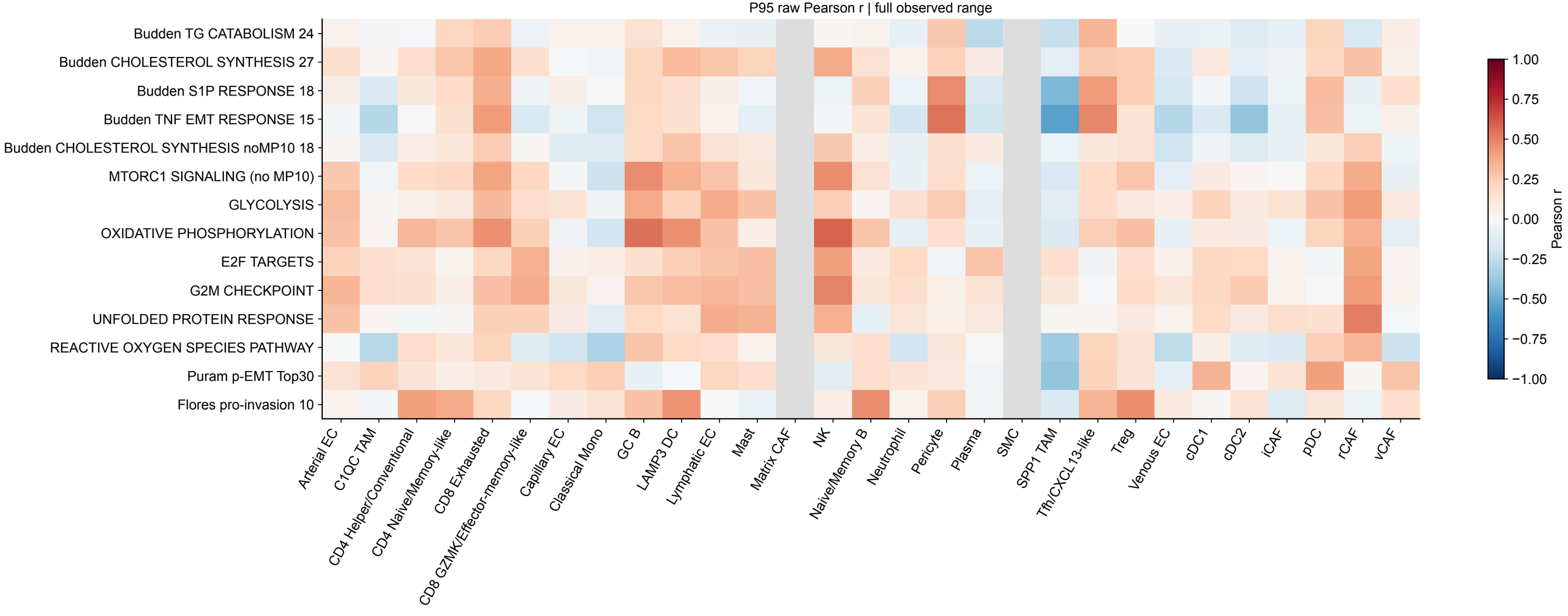


Tissue 3484 | Epi 2365 | internal 1154 | P95 58
Exact front in Epi 292 | R12 labelled 1580

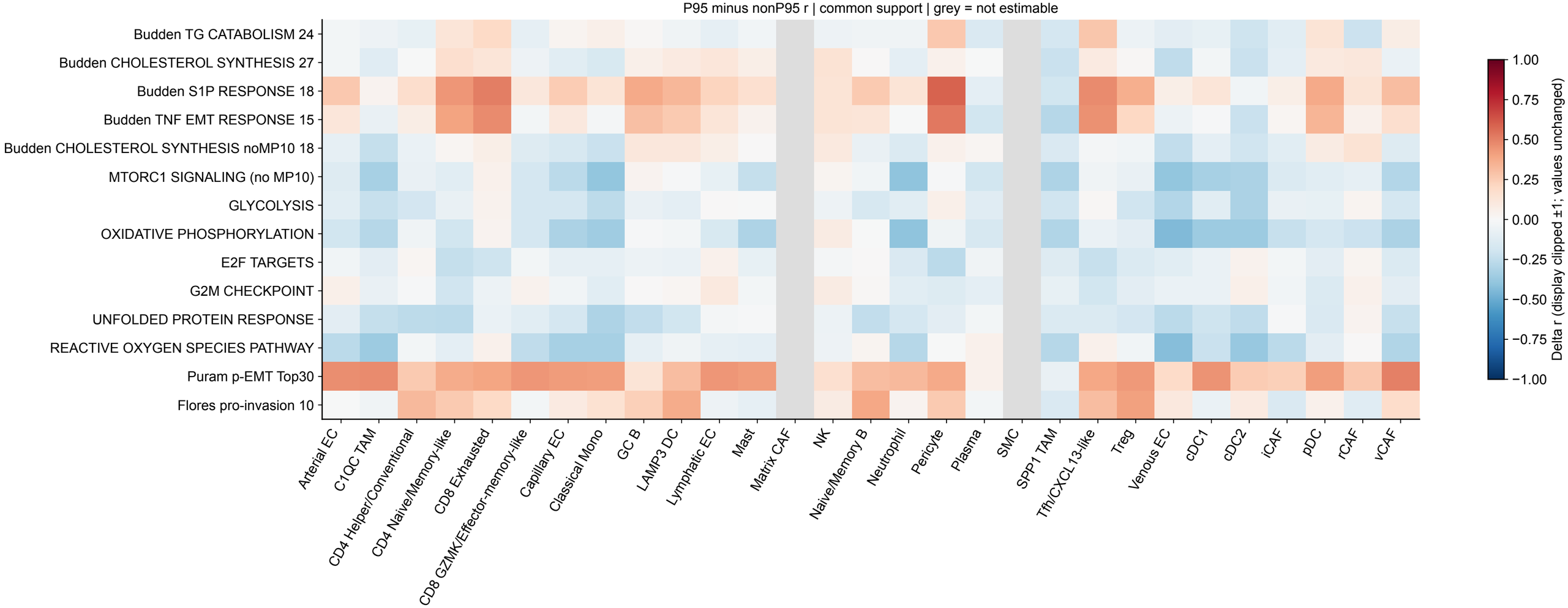
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC04 | all 29 candidates; fixed core programs

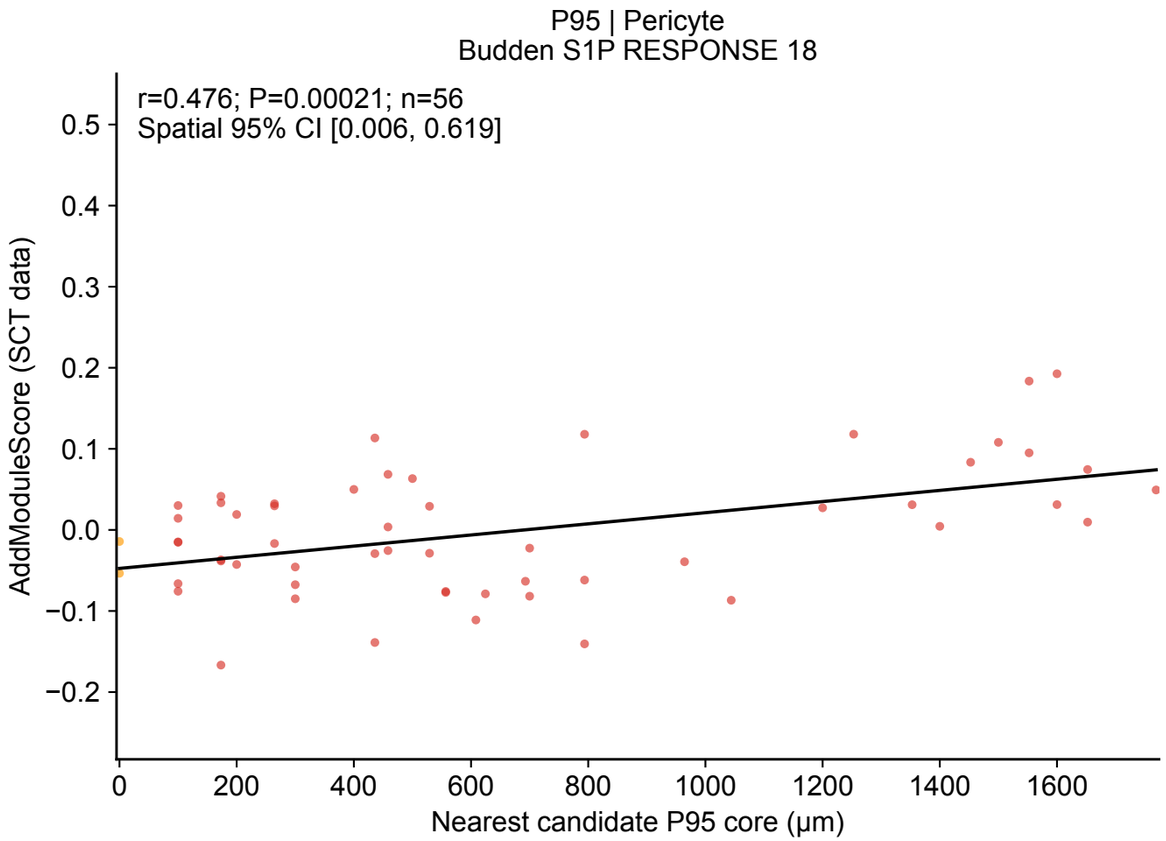
P95 raw Pearson r | full observed range

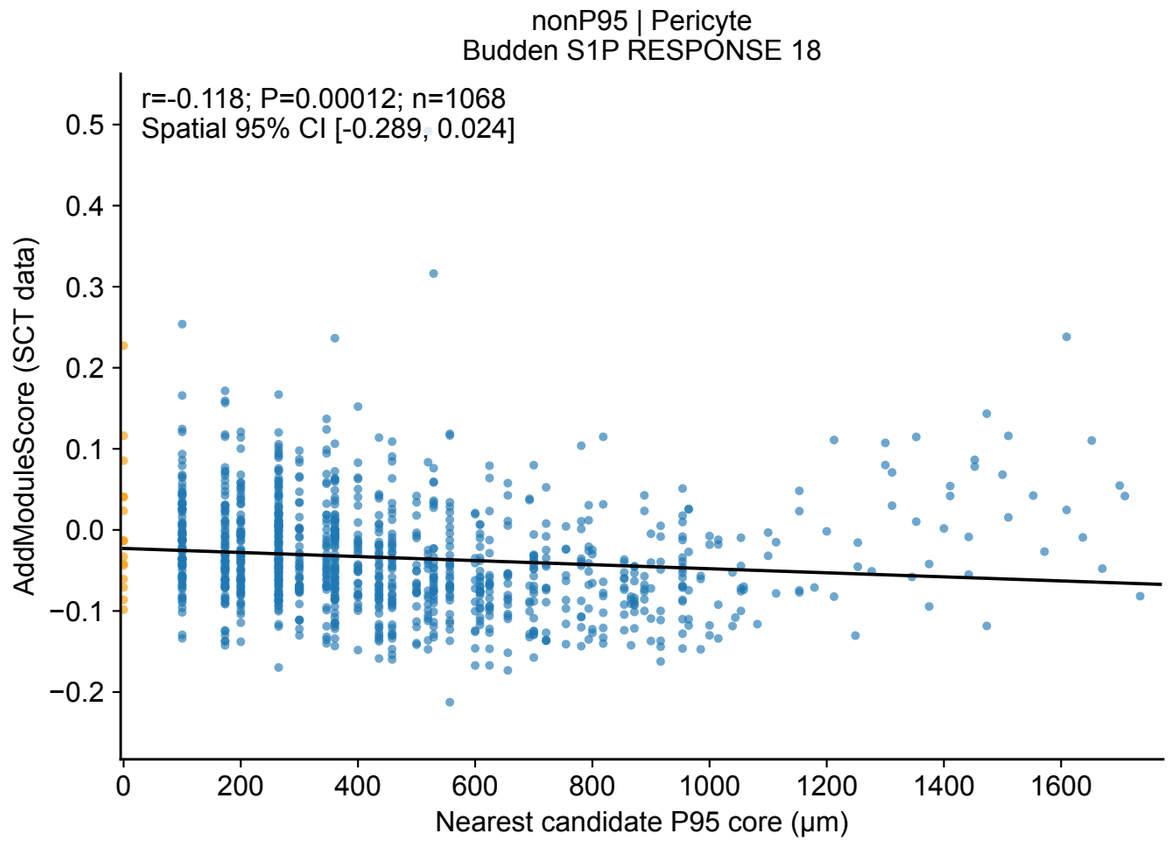


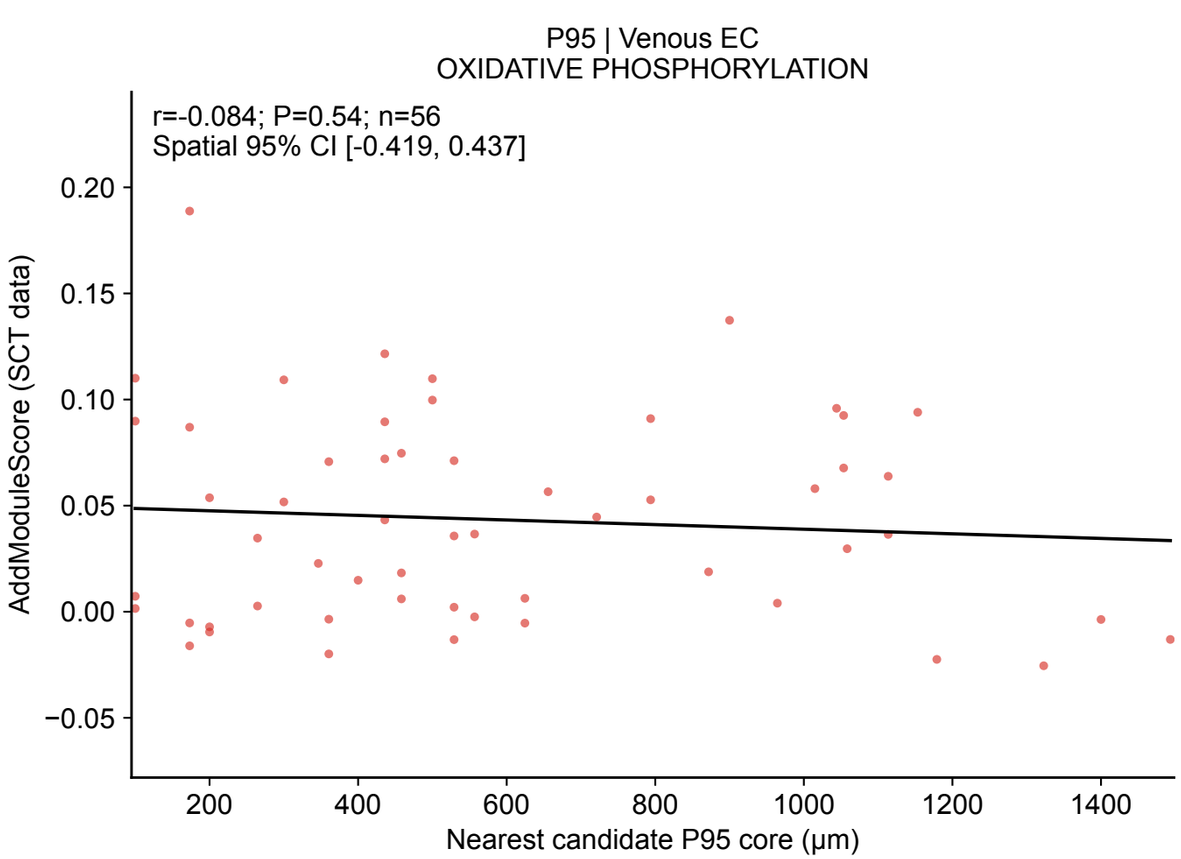
P95 minus nonP95 r | common support | grey = not estimable

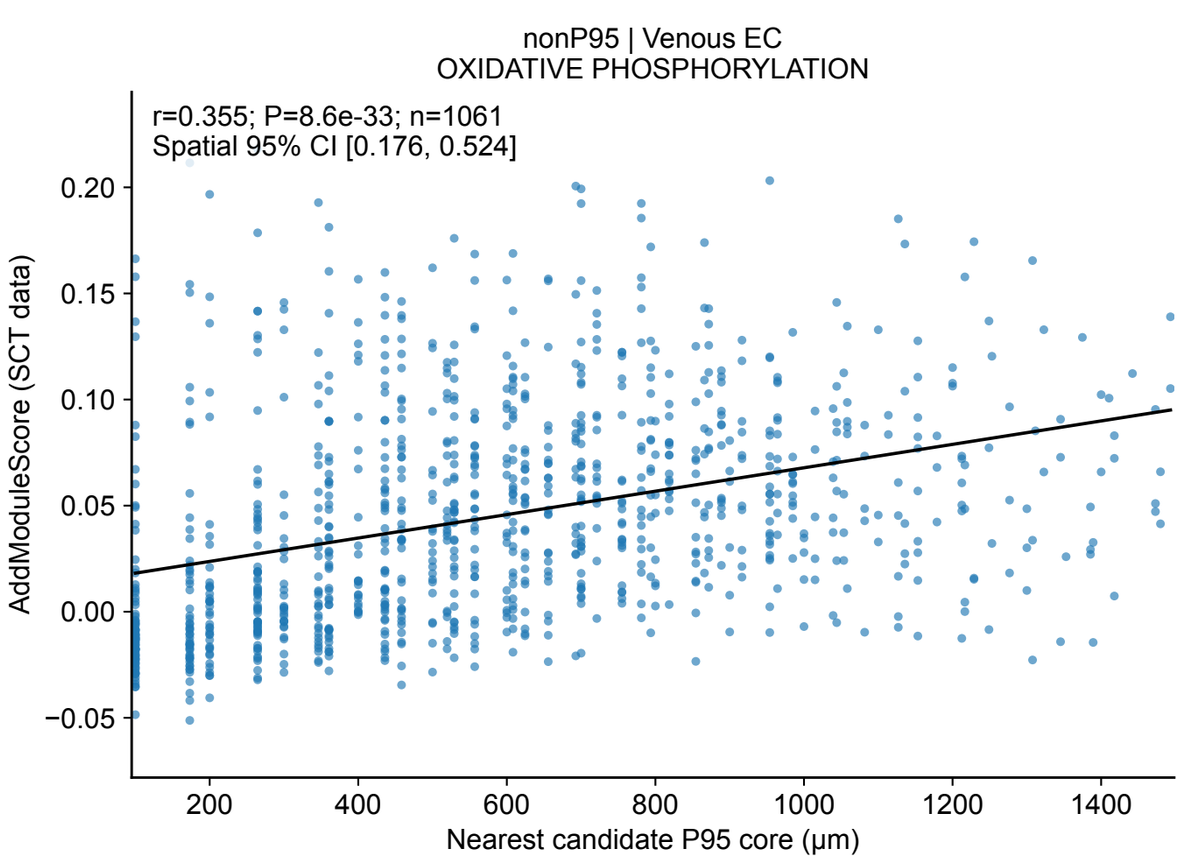


GSE319536_BC04 | exploratory scatter display selected by coverage and |delta r|, not P





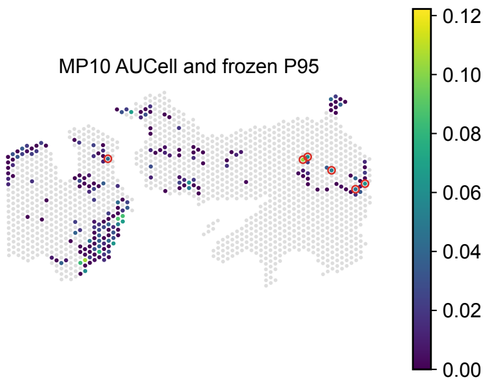


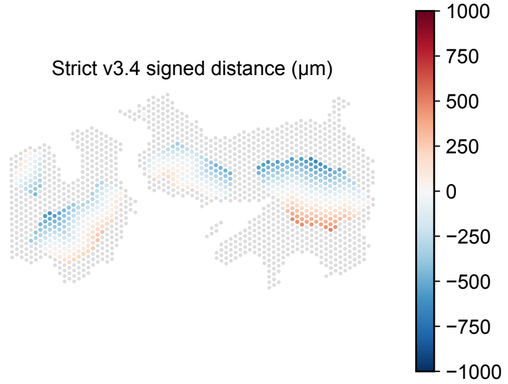


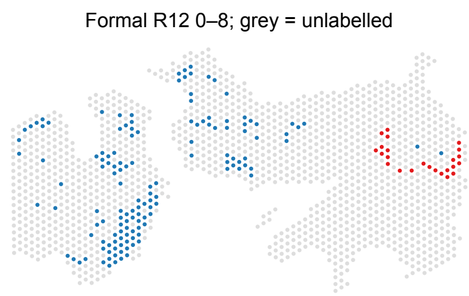
Pericyte / Budden S1P RESPONSE 18: delta r=0.593 [0.288, 0.719]; common 0–1769 µm; spatial P=1e-09
Venous EC / OXIDATIVE PHOSPHORYLATION: delta r=-0.439 [-0.692, -0.014]; common 100–1493 µm; spatial P=0.0092
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

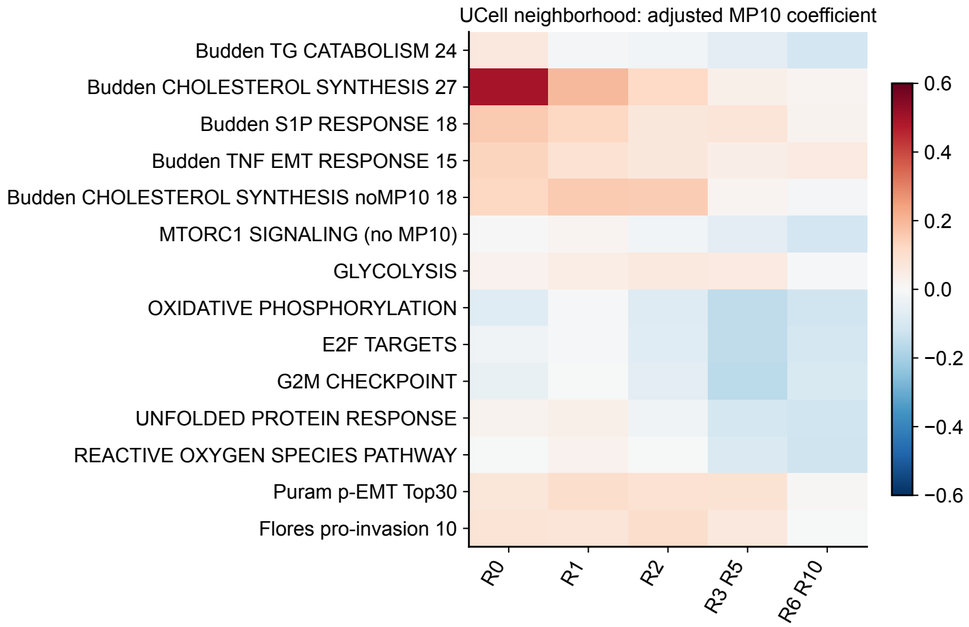
P95 nonP95 distance 0

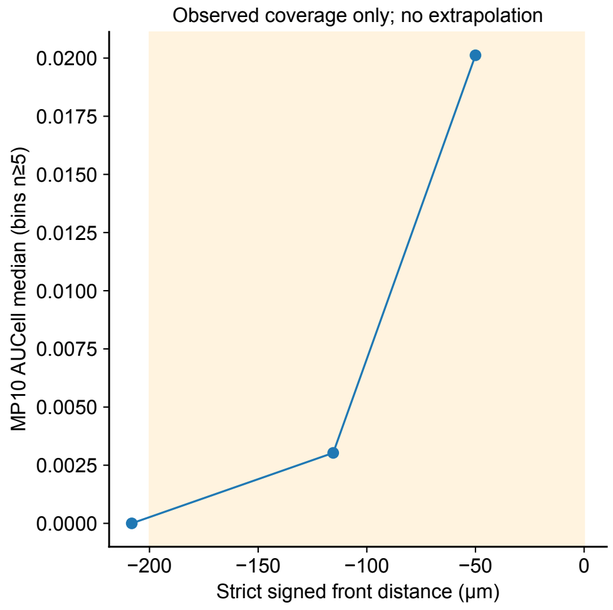
GSE319536_BC05 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

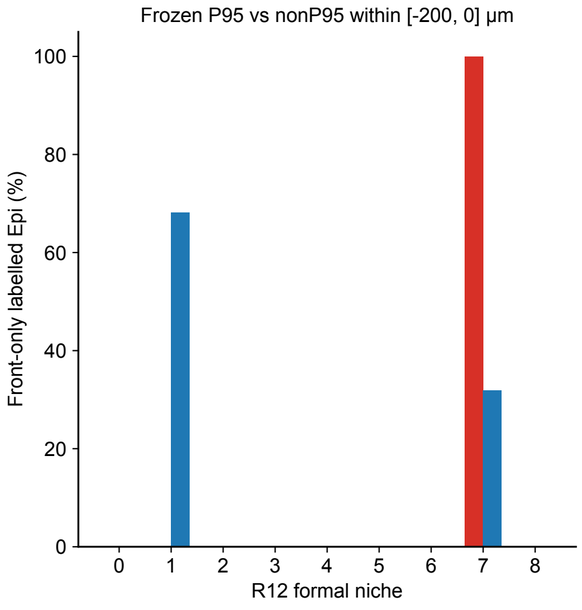








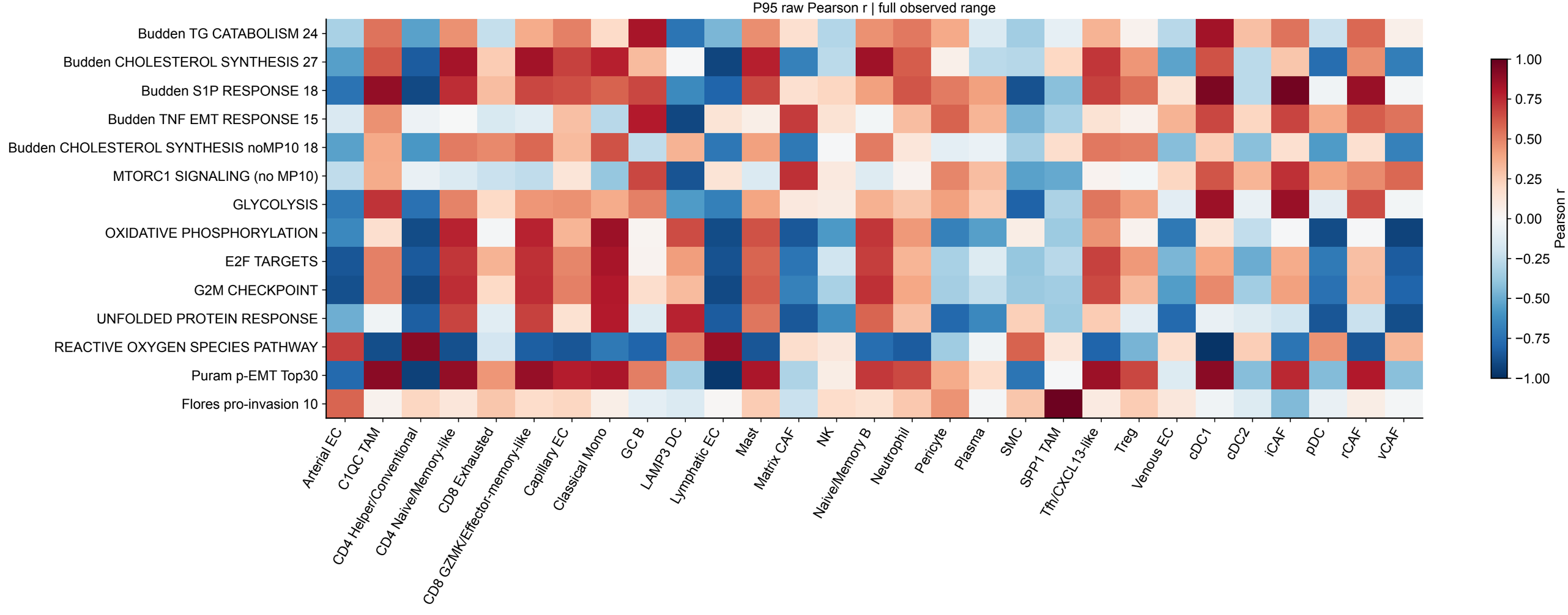




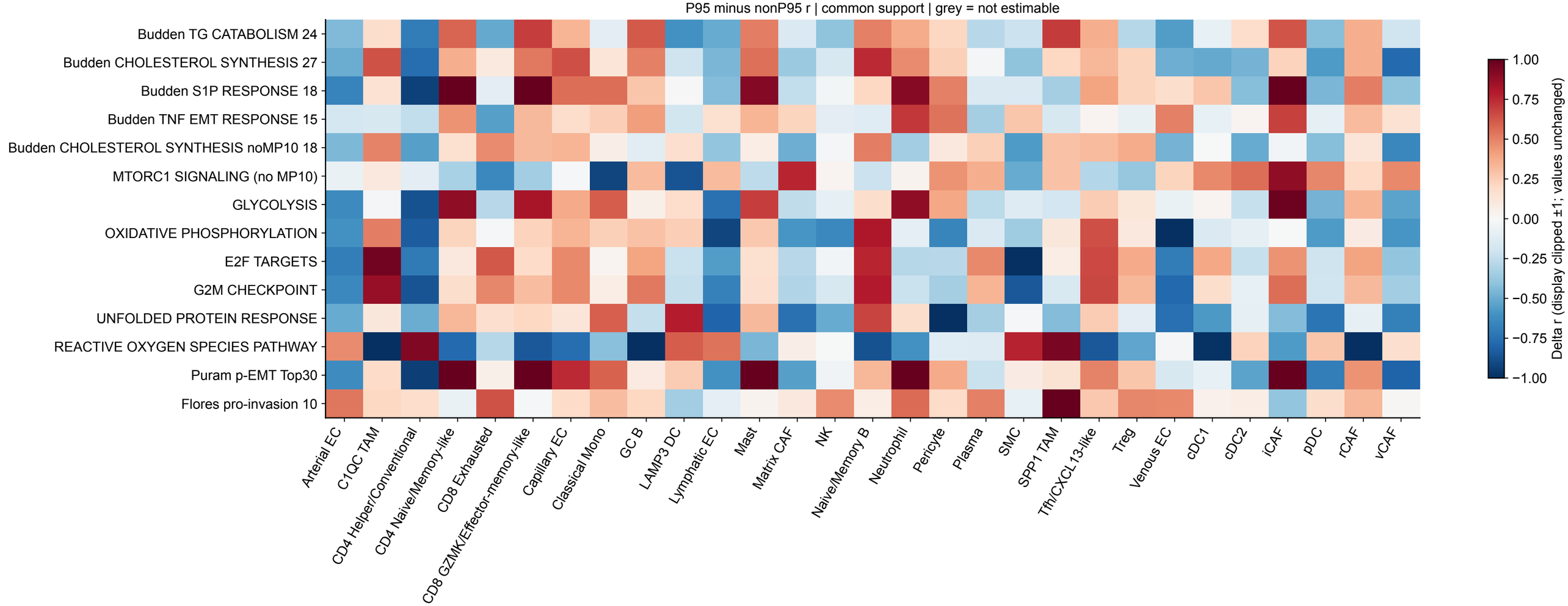
Tissue 1550 | Epi 210 | internal 102 | P95 6
Exact front in Epi 72 | R12 labelled 162

GSE319536_BC05 | all 29 candidates; fixed core programs

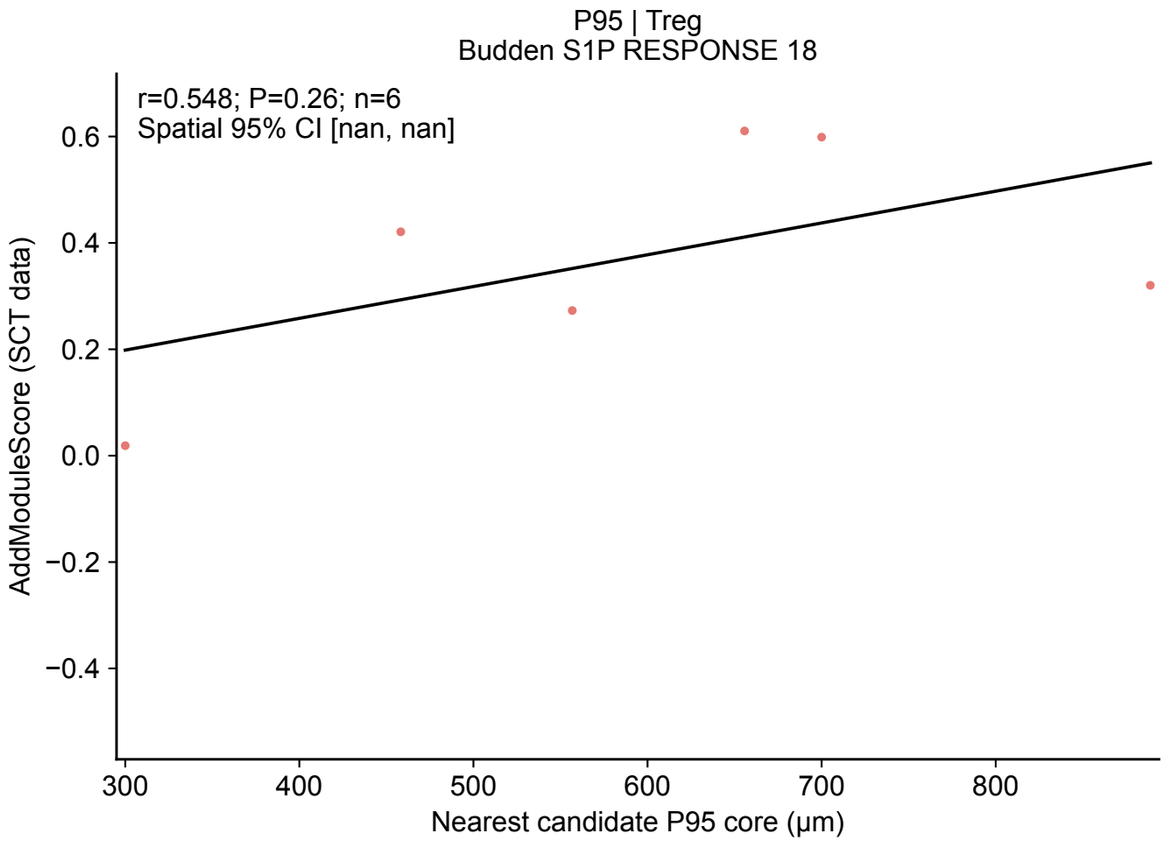
P95 raw Pearson r | full observed range

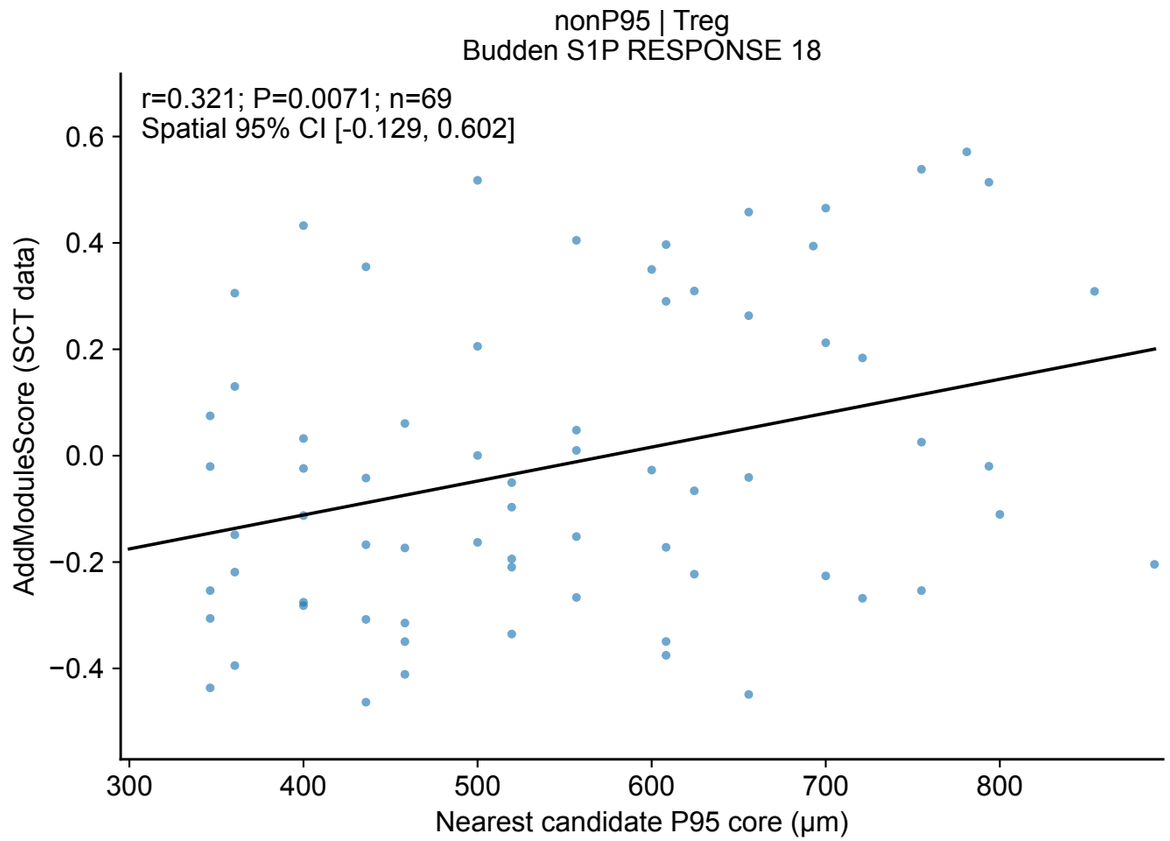


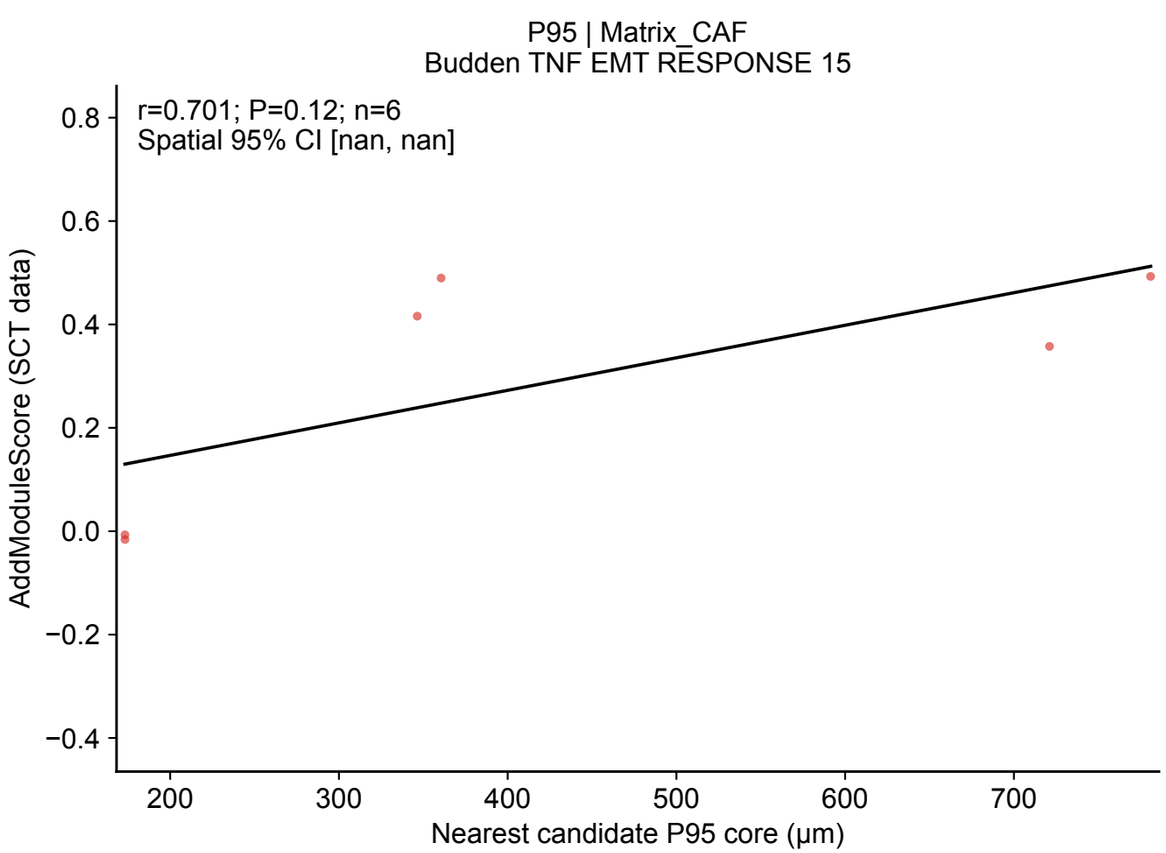
P95 minus nonP95 r | common support | grey = not estimable

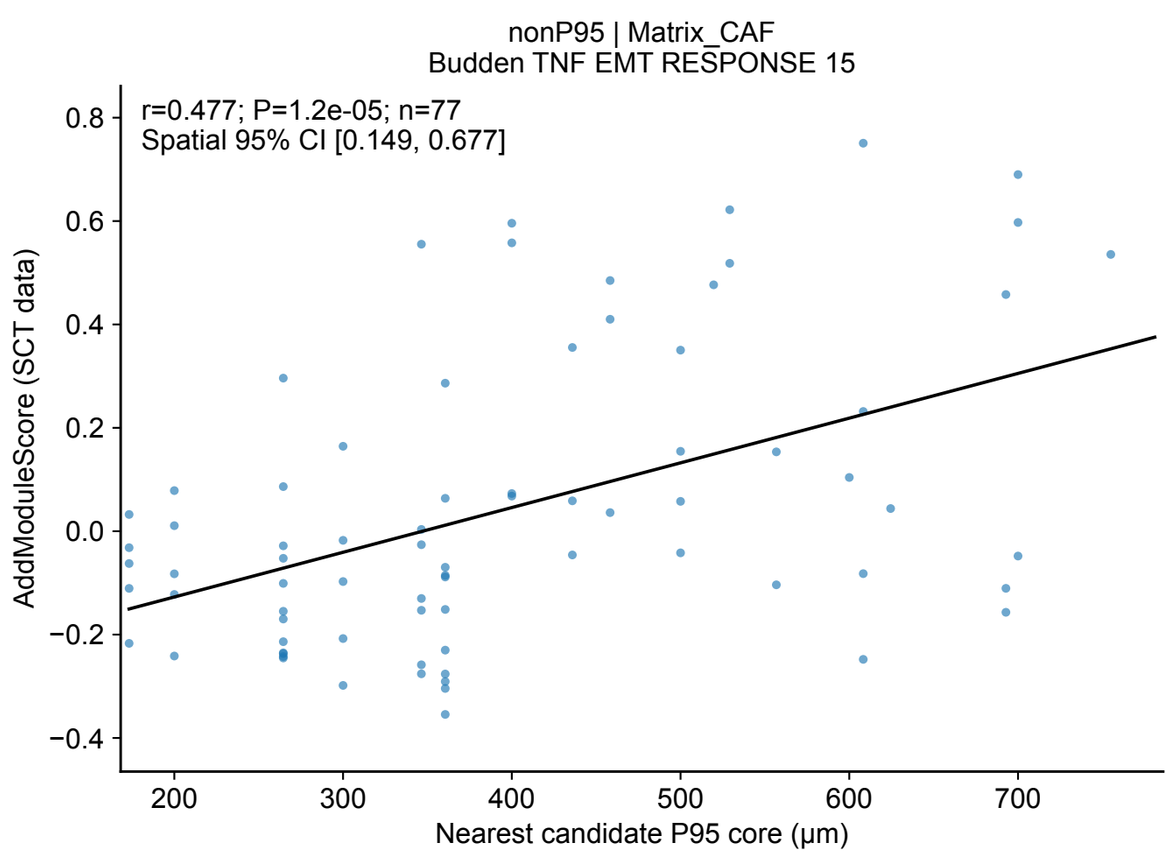


GSE319536_BC05 | exploratory scatter display selected by coverage and |delta r|, not P







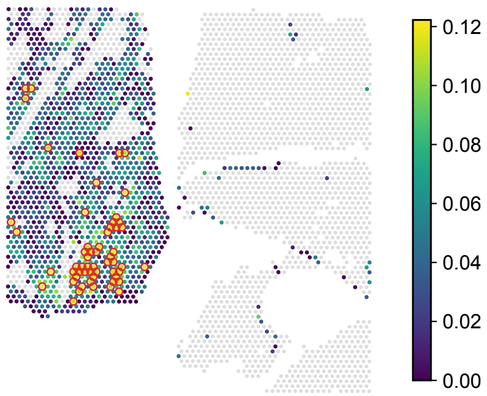


Treg / Budden S1P RESPONSE 18: delta r=0.226 [nan, nan]; common 300–889 µm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=0.224 [nan, nan]; common 173–781 µm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

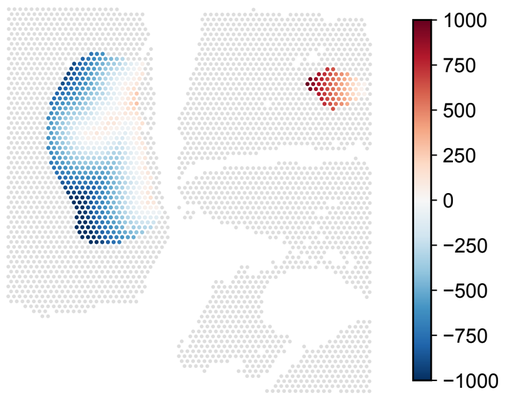
P95 nonP95 distance 0

GSE319536_BC08 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

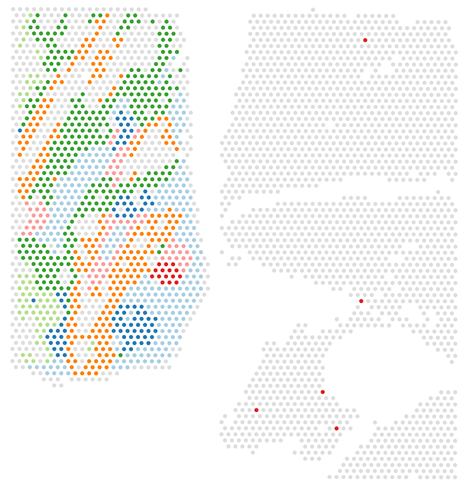
MP10 AUCell and frozen P95



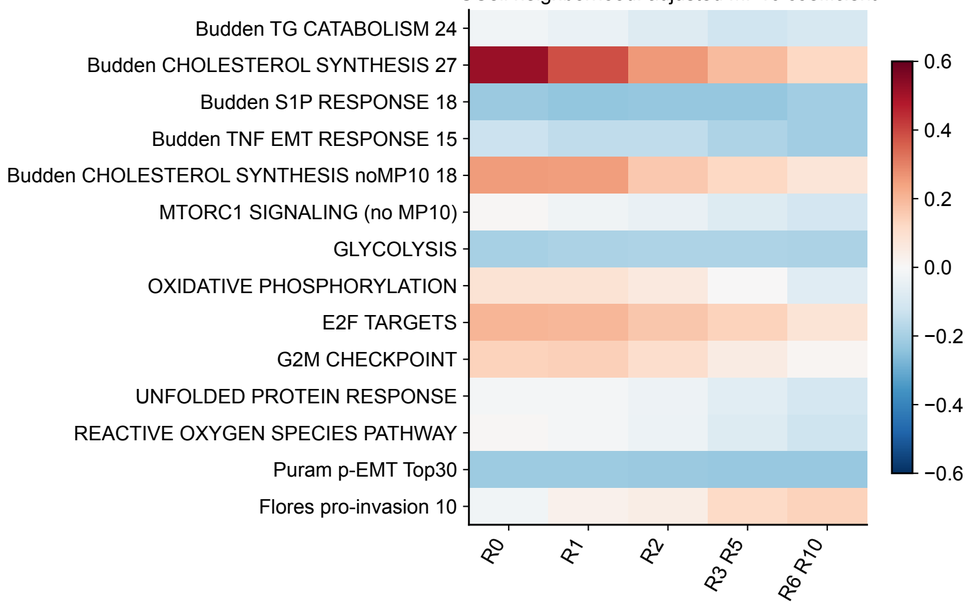
Strict v3.4 signed distance (µm)



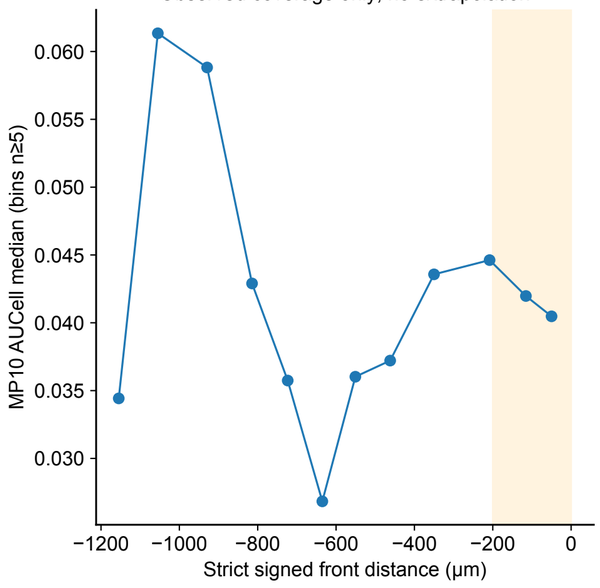
Formal R12 0–8; grey = unlabelled



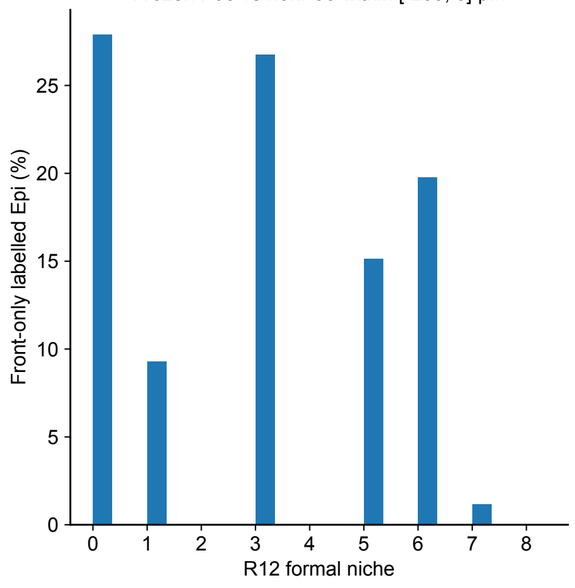
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

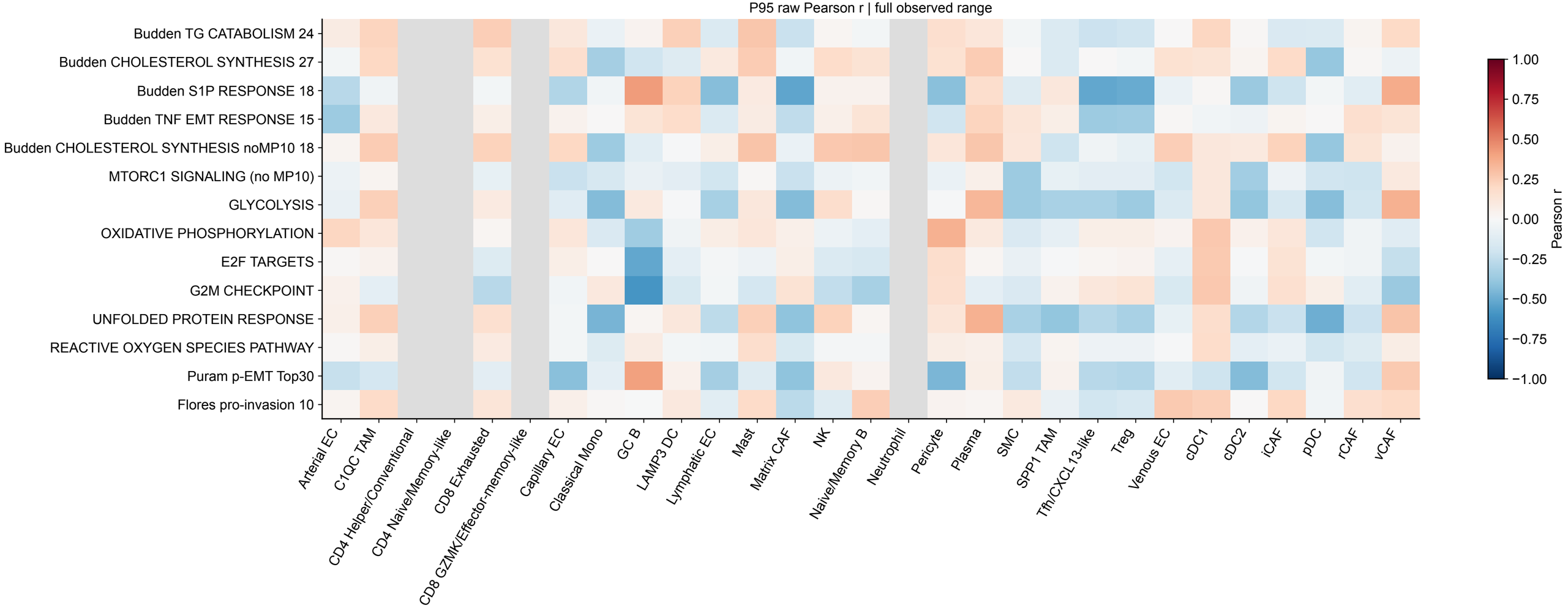


Tissue 3444 | Epi 1297 | internal 1121 | P95 57
Exact front in Epi 482 | R12 labelled 1088

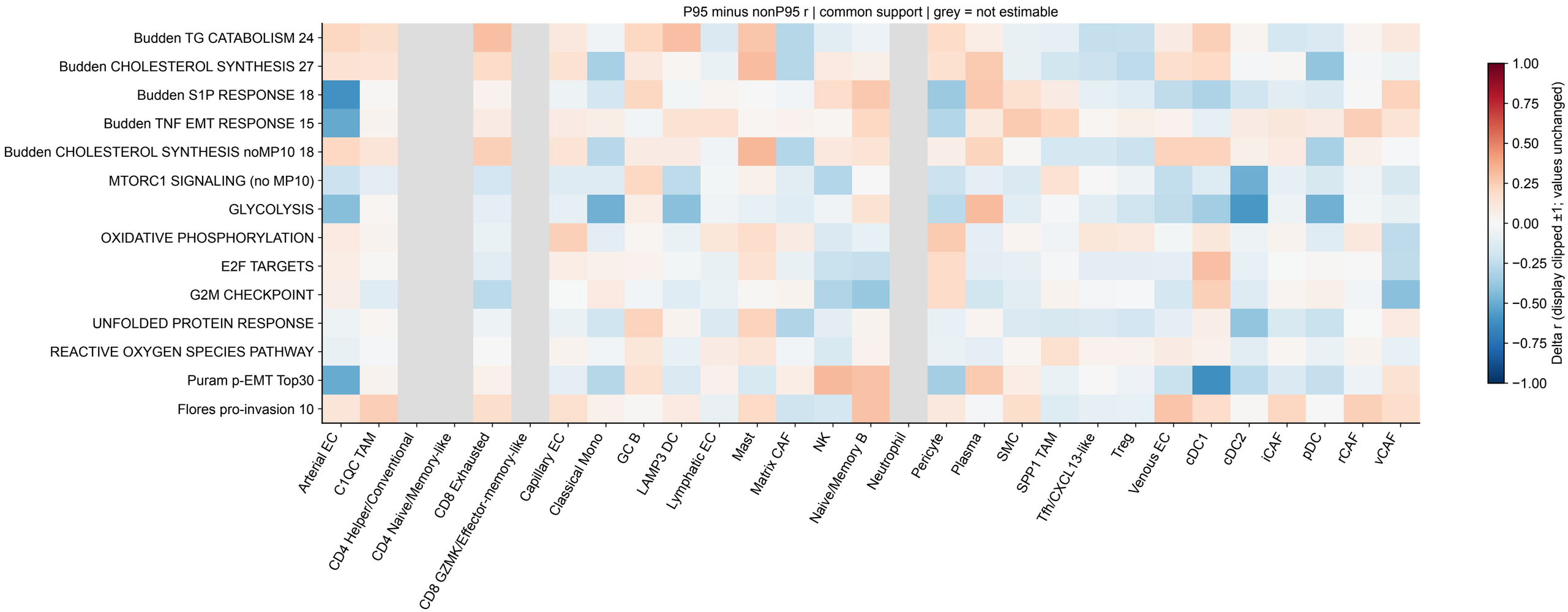
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC08 | all 29 candidates; fixed core programs

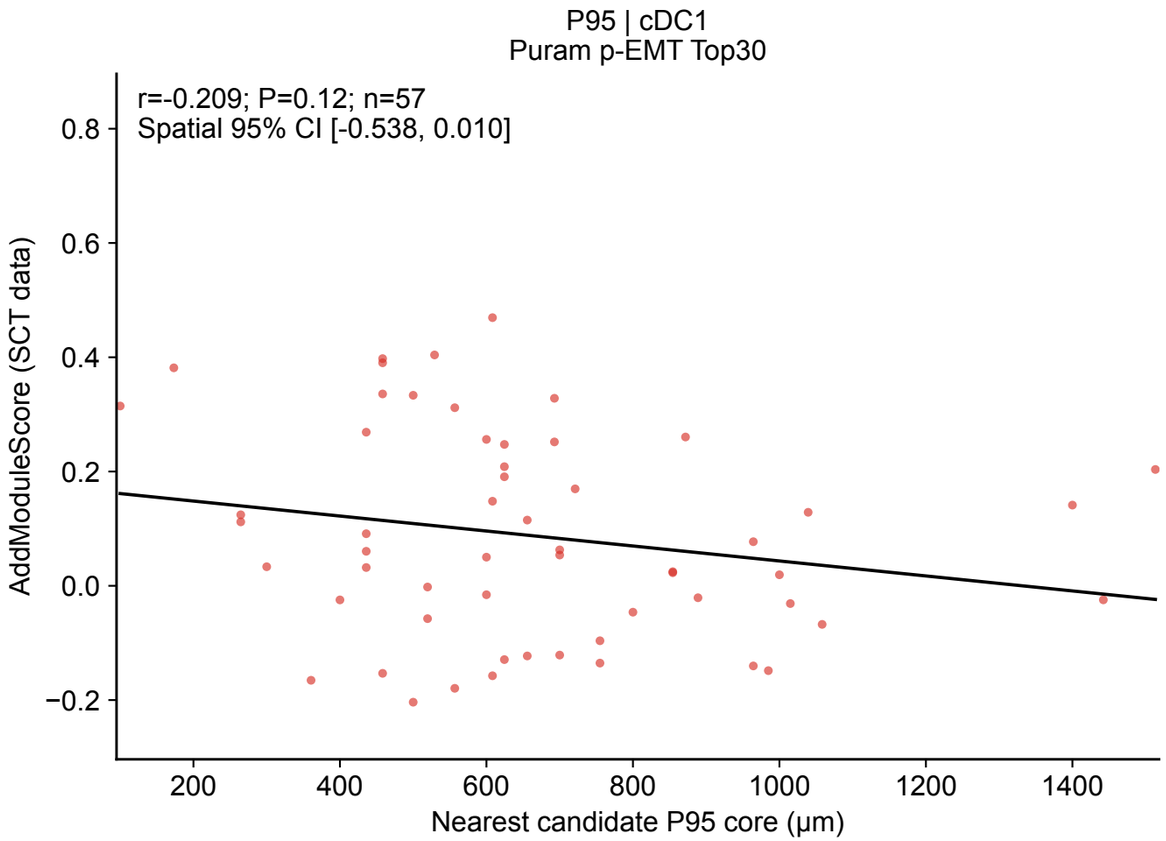
P95 raw Pearson r | full observed range

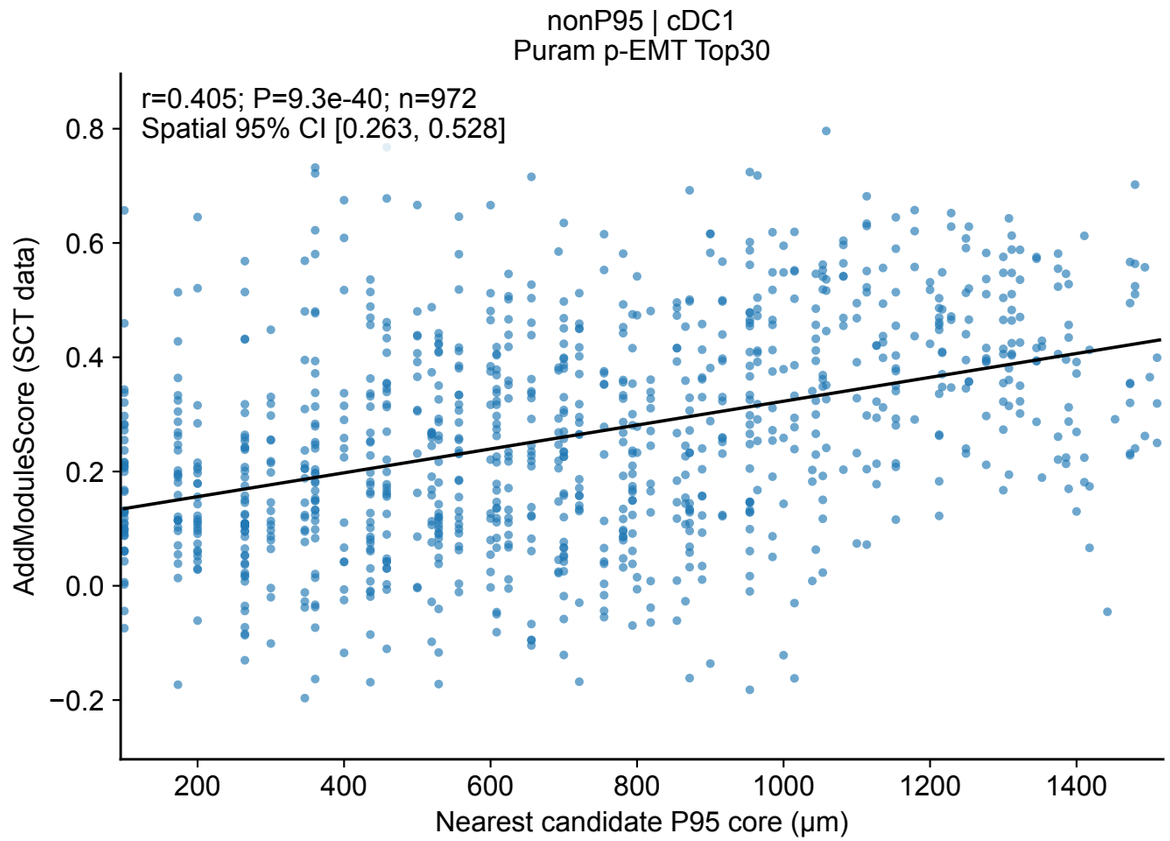


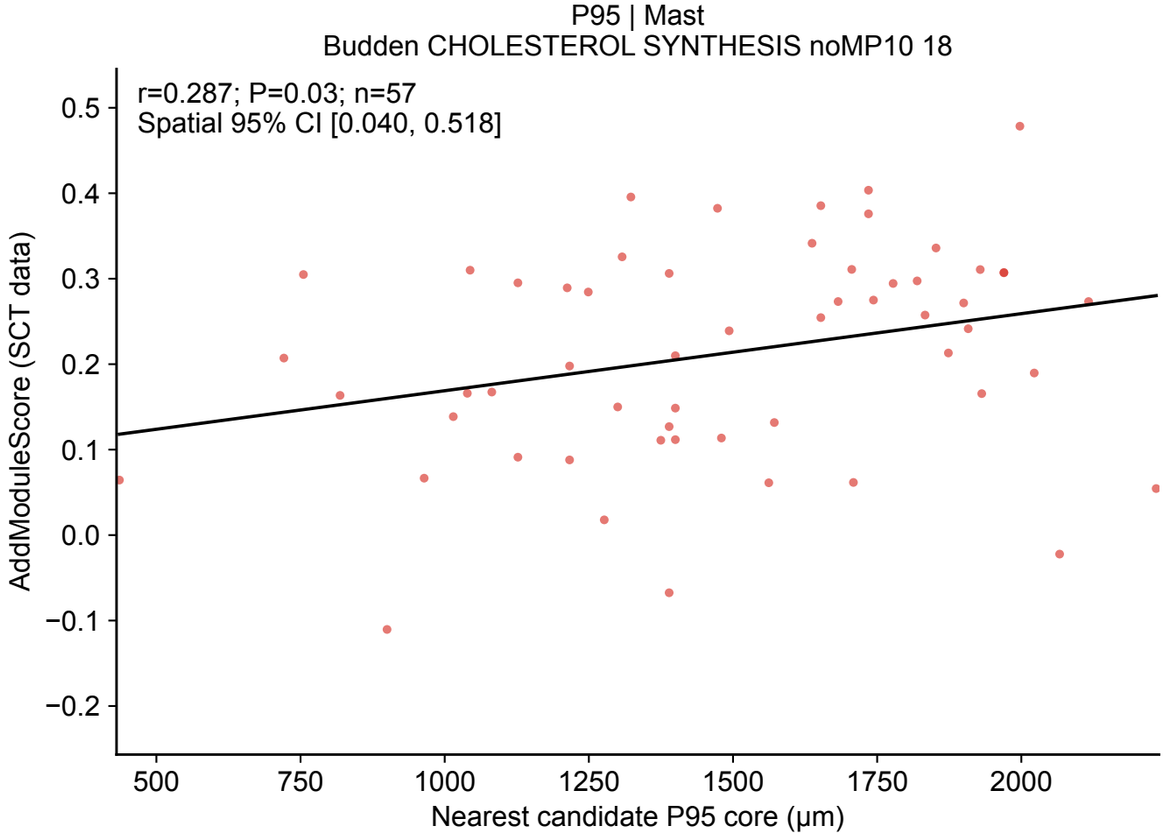
P95 minus nonP95 r | common support | grey = not estimable

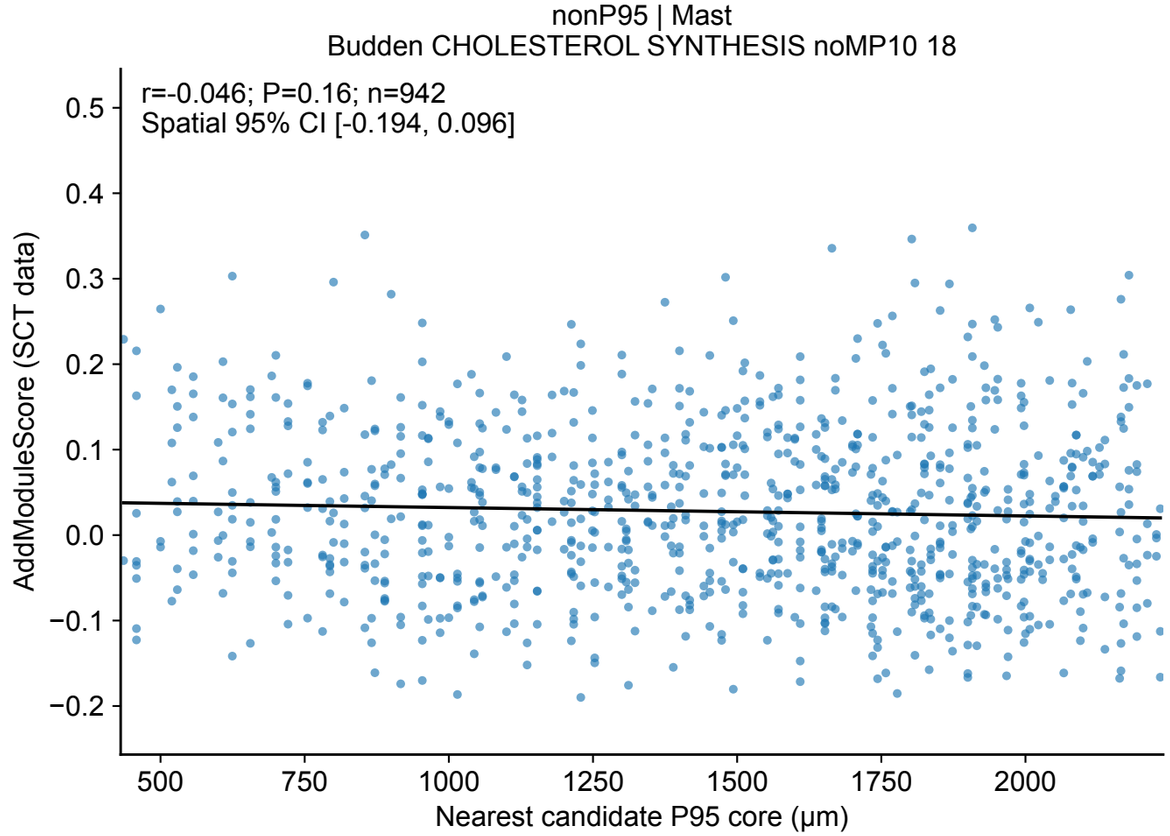


GSE319536_BC08 | exploratory scatter display selected by coverage and |delta r|, not P







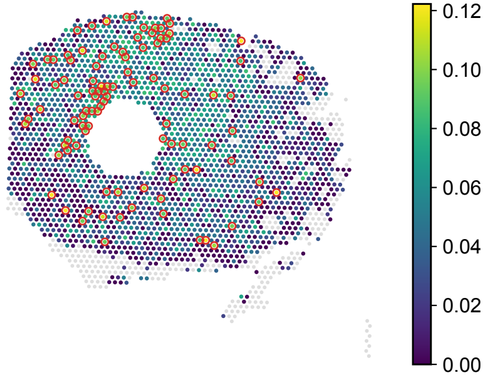


cDC1 / Puram p-EMT Top30: delta r=-0.614 [-0.943, -0.386]; common 100–1513 µm; spatial P=1.4e-05
Mast / Budden CHOLESTEROL SYNTHESIS noMP10 18: delta r=0.334 [0.055, 0.599]; common 436–2234 µm; spatial P=0.013
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

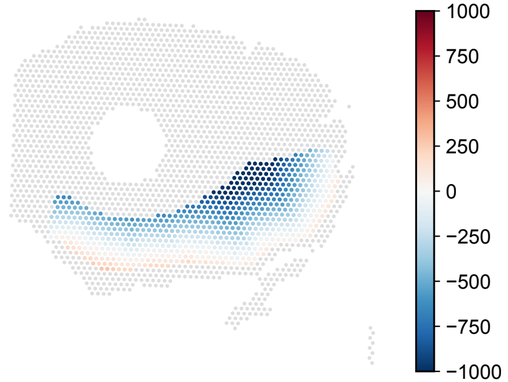
P95 nonP95 distance 0

GSE319536_BC10 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

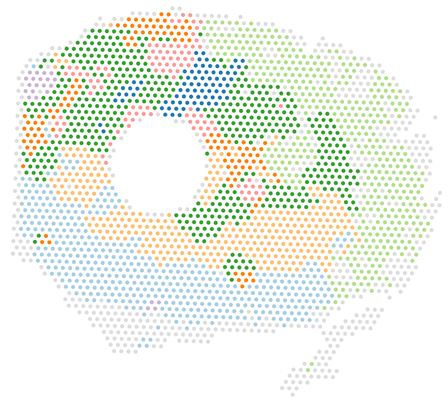
MP10 AUCell and frozen P95



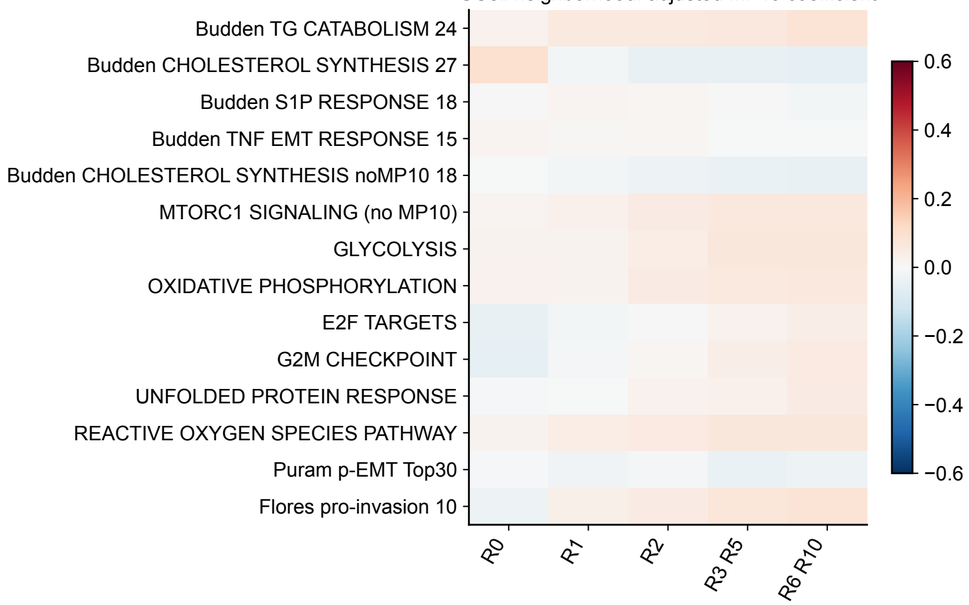
Strict v3.4 signed distance (µm)



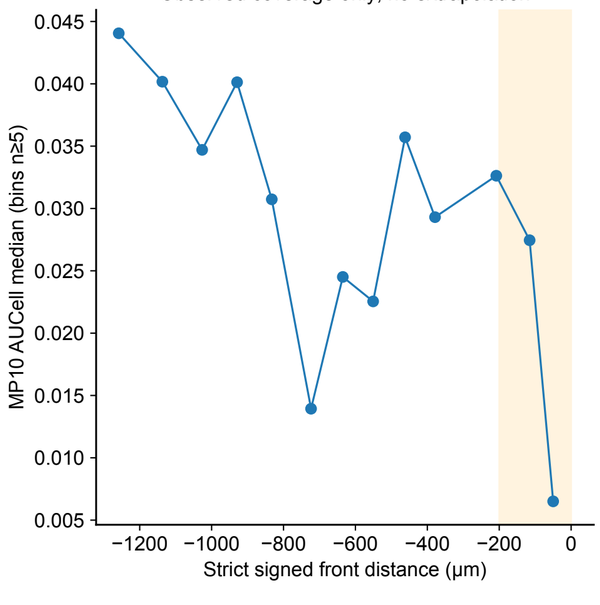
Formal R12 0–8; grey = unlabelled



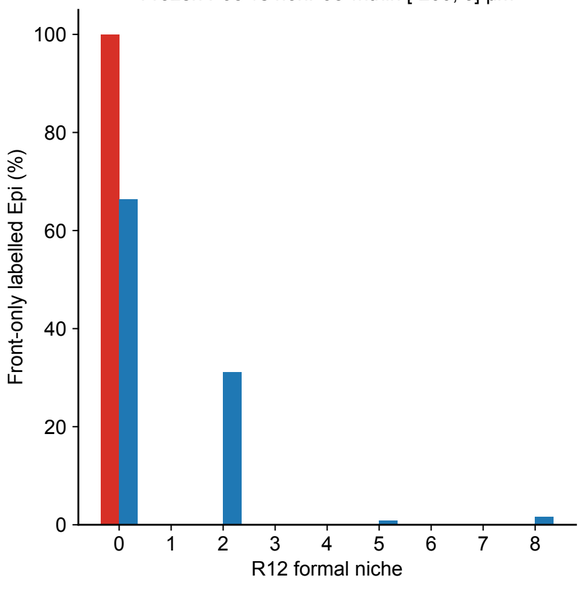
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

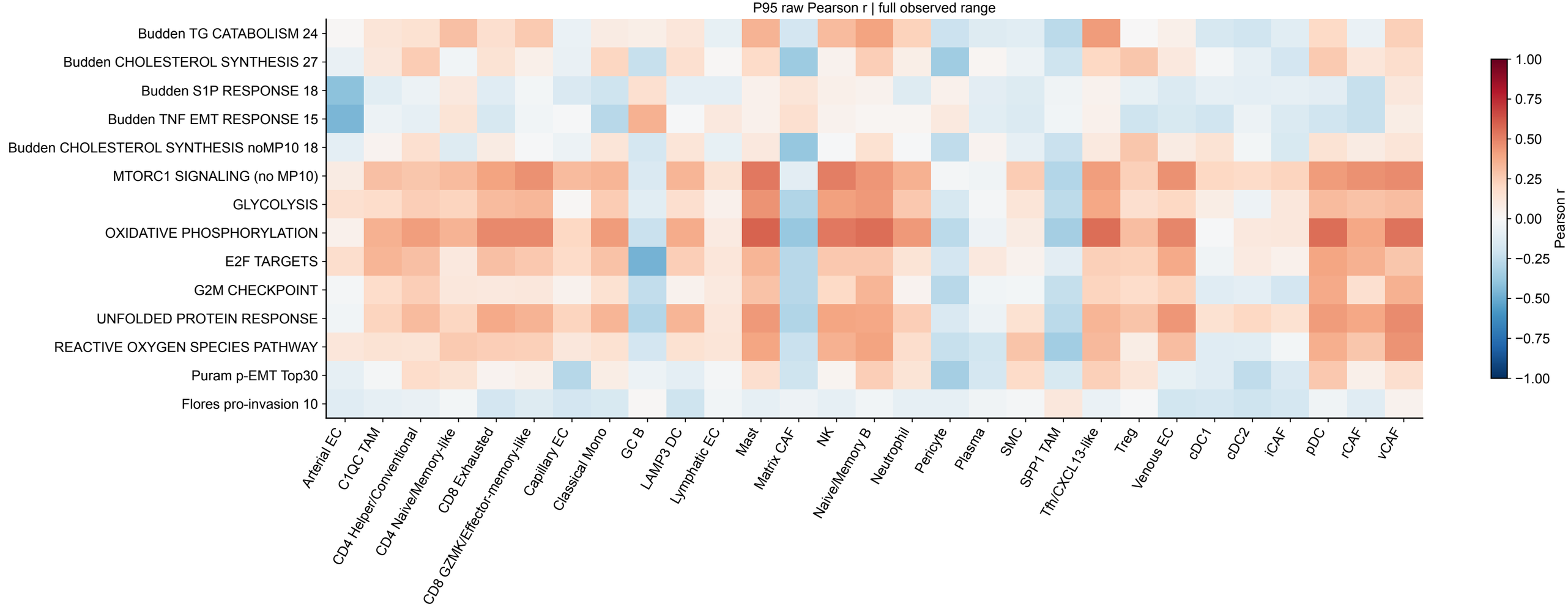


Tissue 2499 | Epi 2174 | internal 1998 | P95 100
Exact front in Epi 628 | R12 labelled 2002

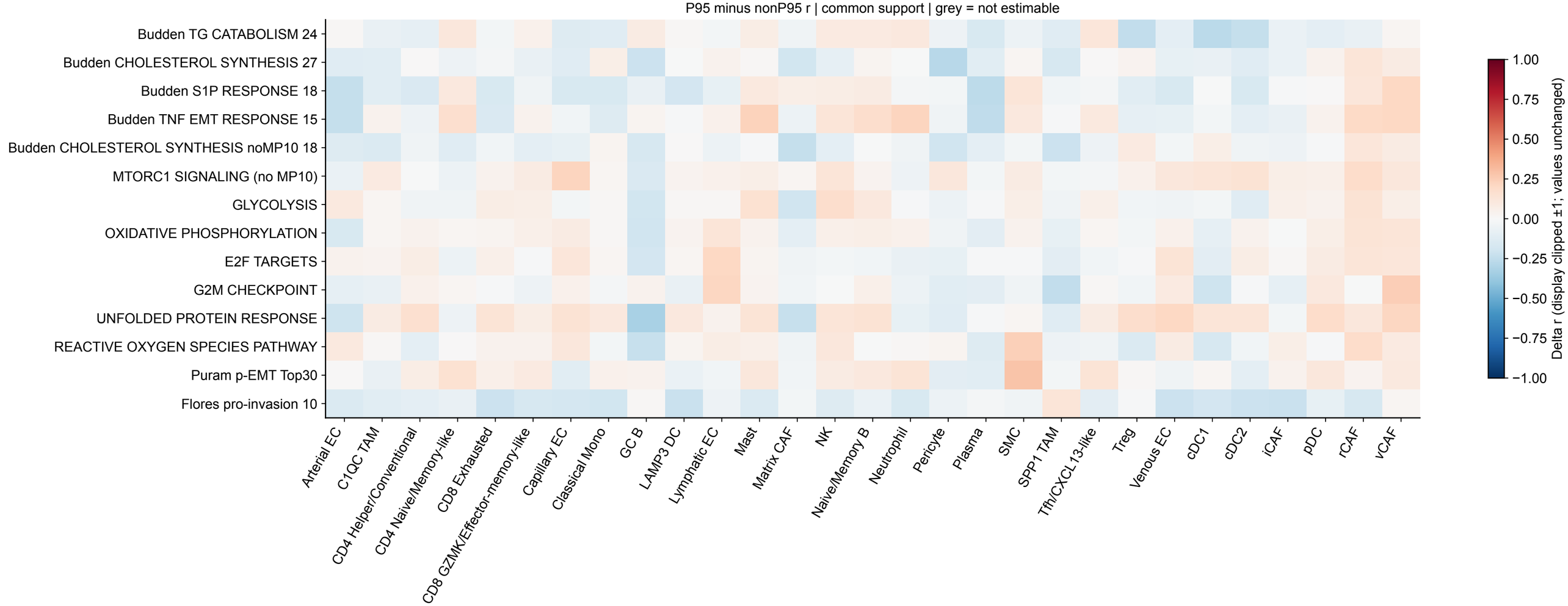
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC10 | all 29 candidates; fixed core programs

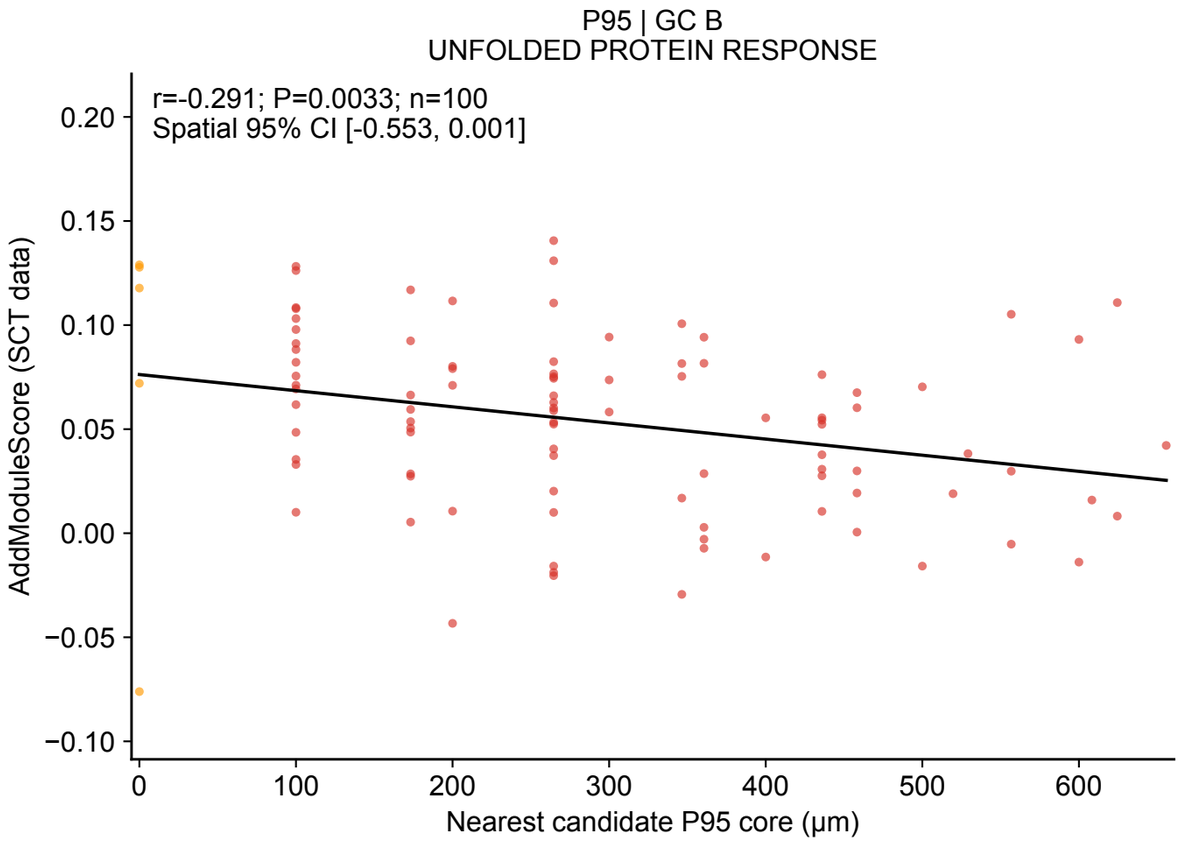
P95 raw Pearson r | full observed range

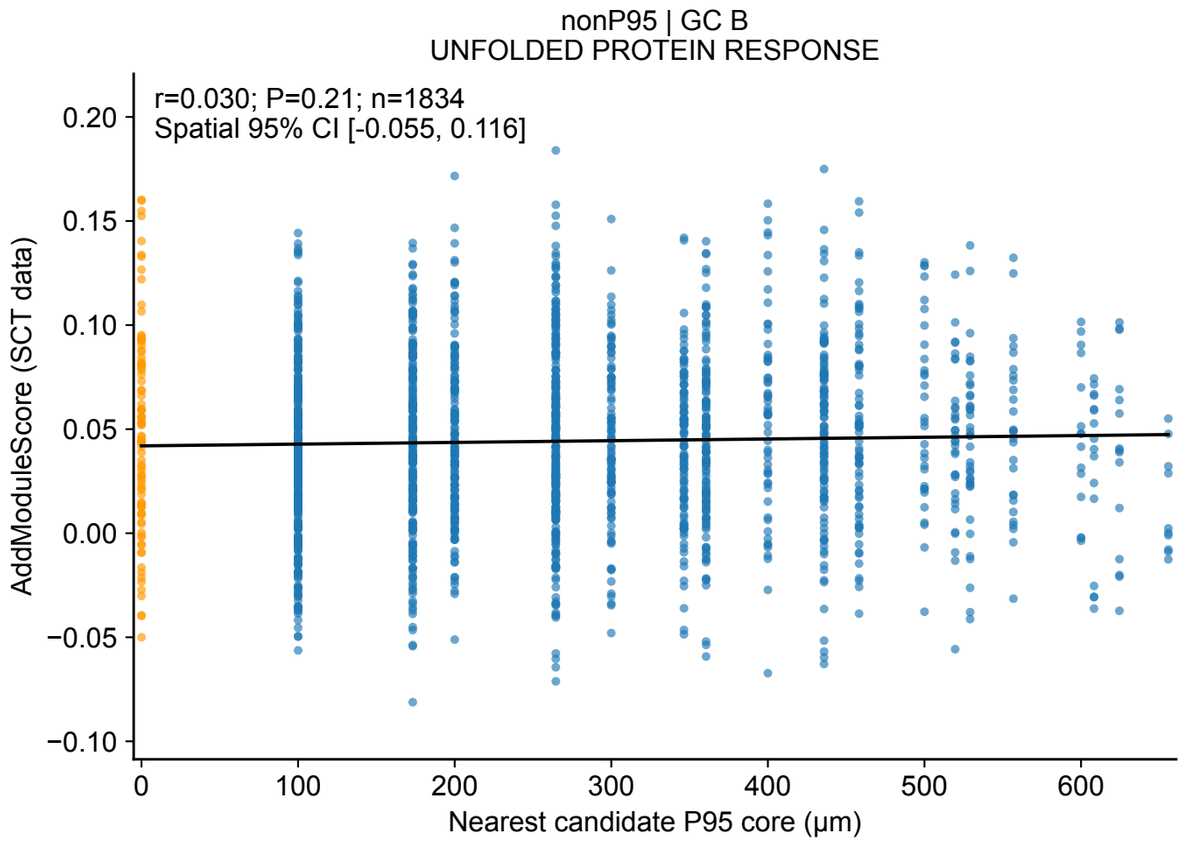


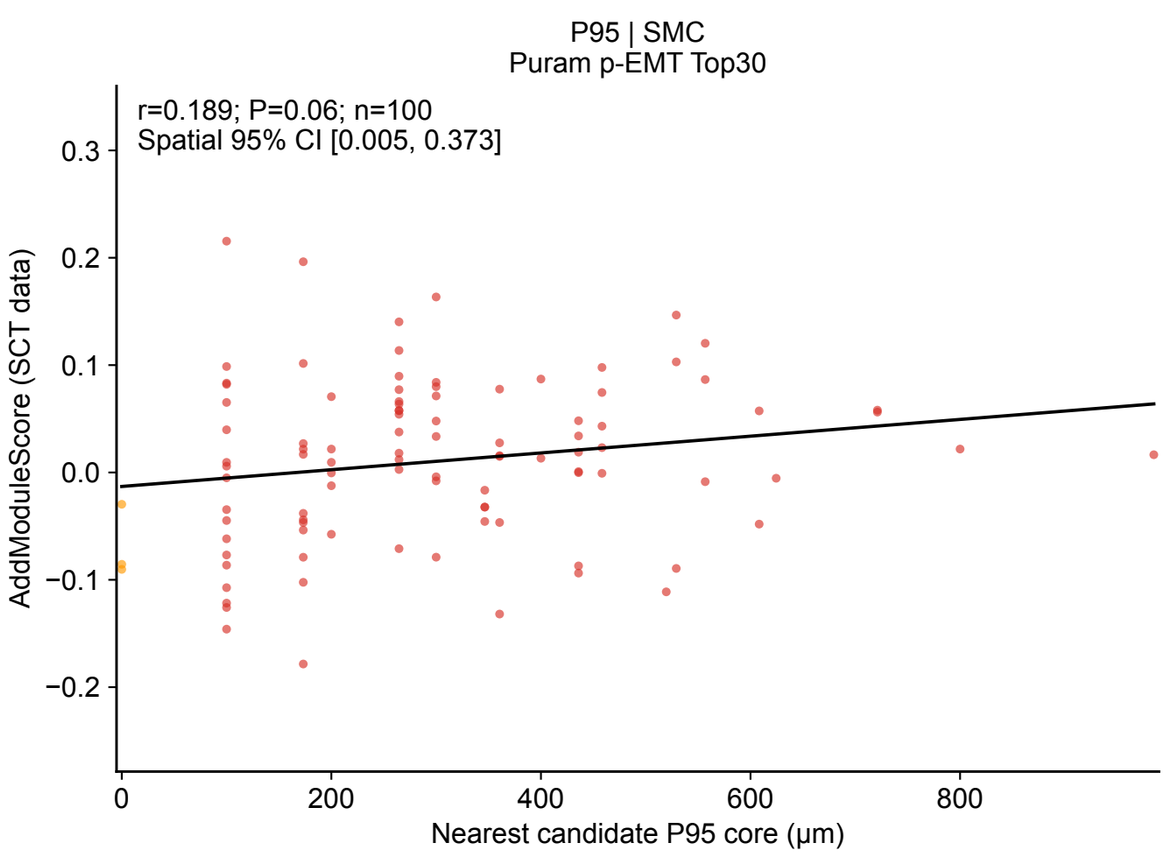
P95 minus nonP95 r | common support | grey = not estimable

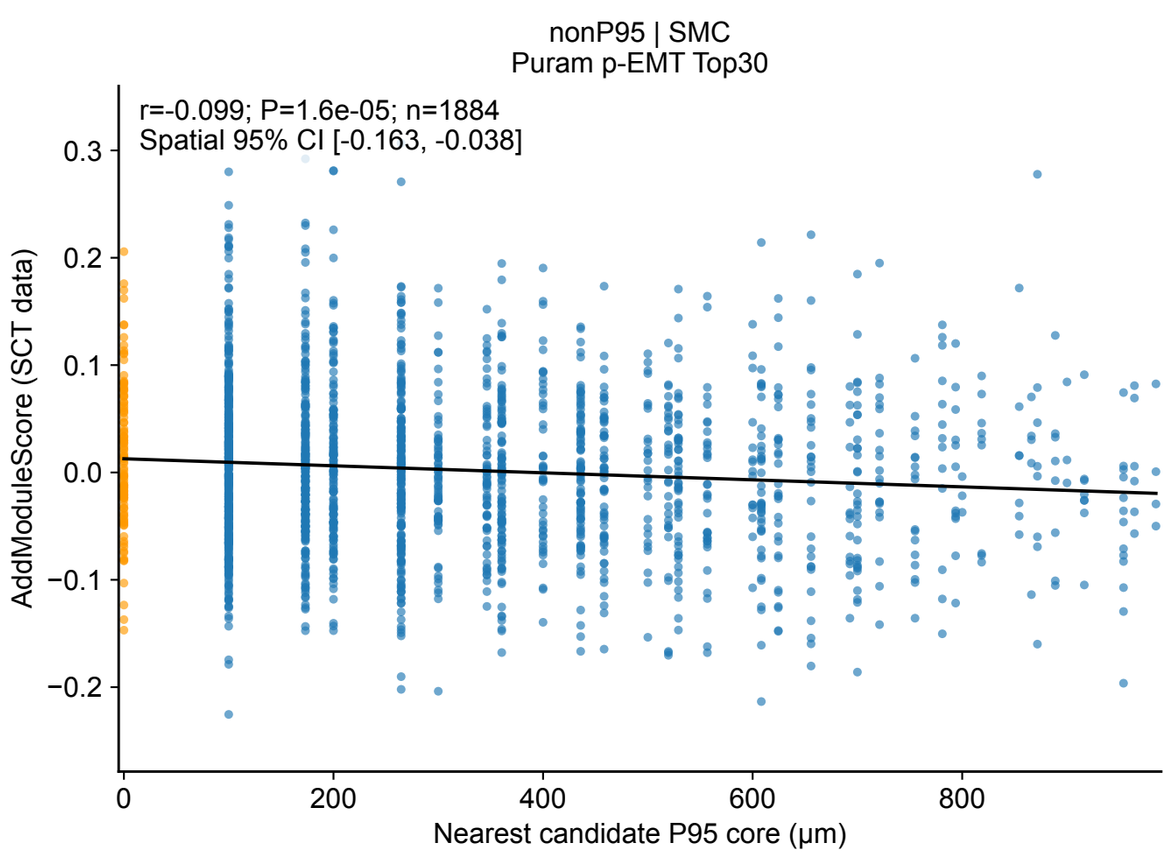


GSE319536_BC10 | exploratory scatter display selected by coverage and |delta r|, not P







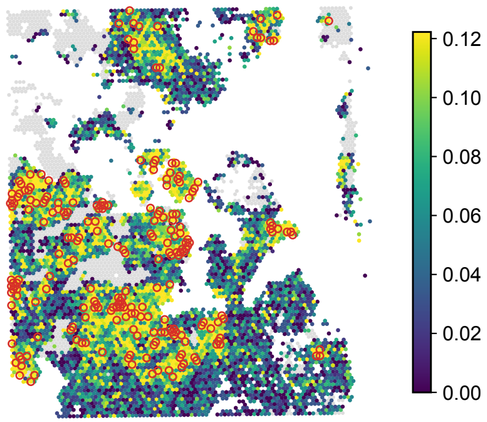


GC B / UNFOLDED PROTEIN RESPONSE: delta r=-0.320 [-0.580, -0.034]; common 0–656 µm; spatial P=0.036
SMC / Puram p-EMT Top30: delta r=0.288 [0.104, 0.479]; common 0–985 µm; spatial P=0.0039
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

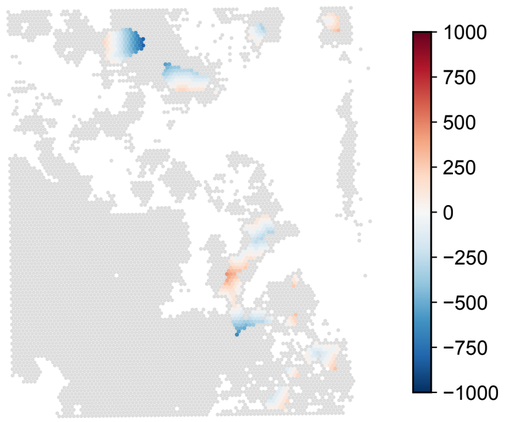
P95 nonP95 distance 0

GSE319536_BC11 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

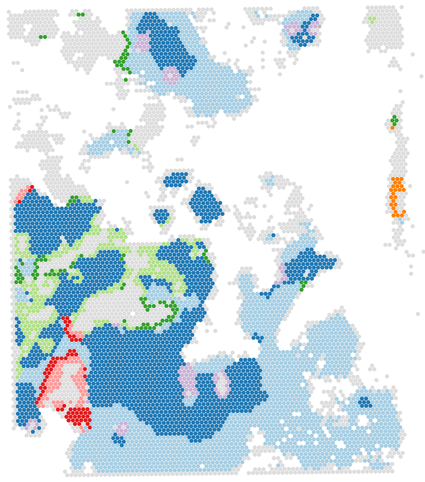
MP10 AUCell and frozen P95



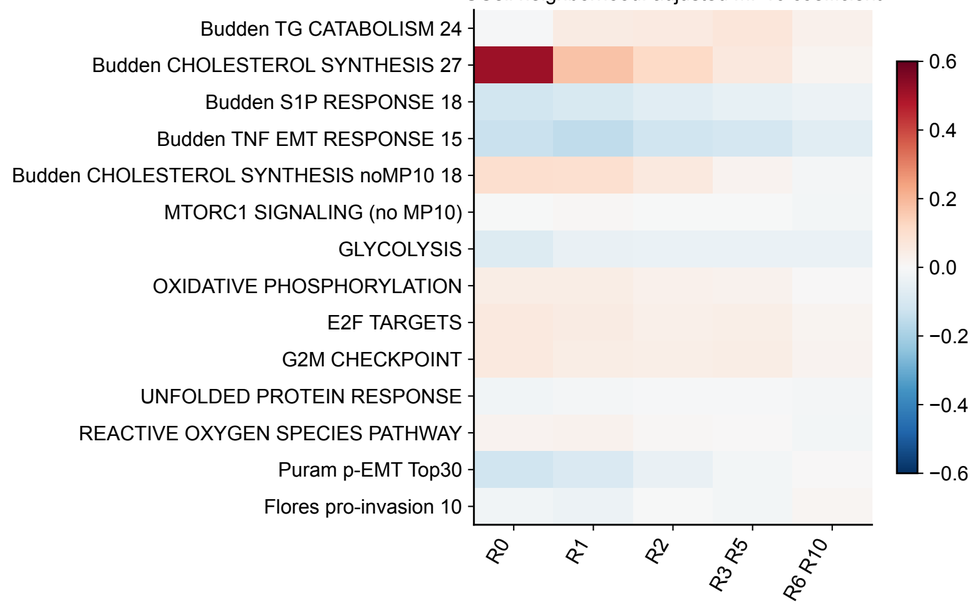
Strict v3.4 signed distance (µm)



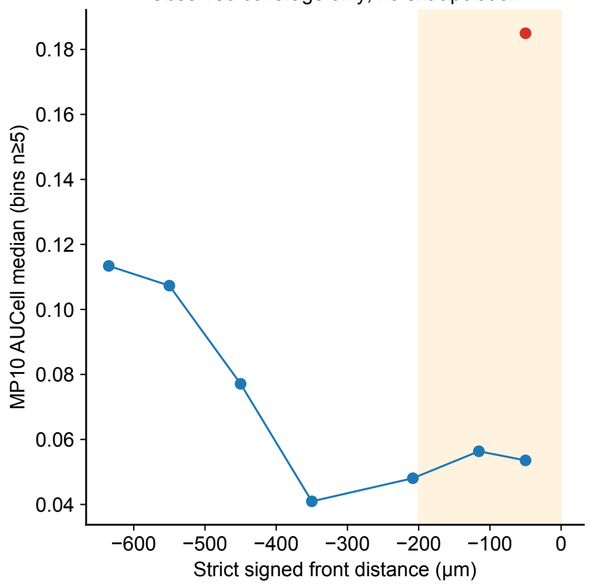
Formal R12 0–8; grey = unlabelled



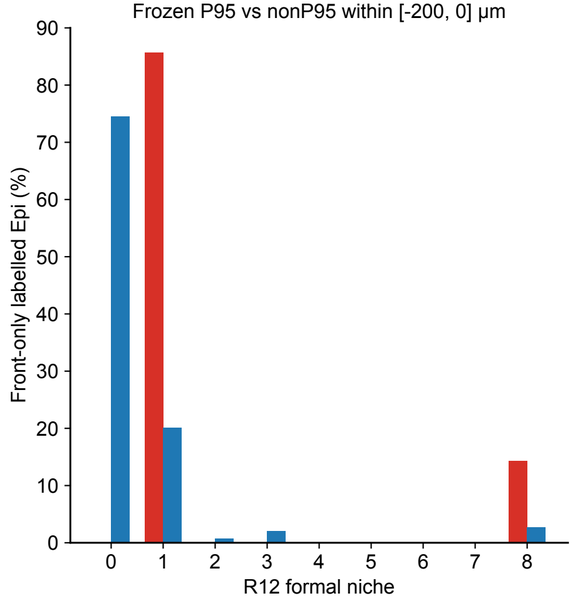
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

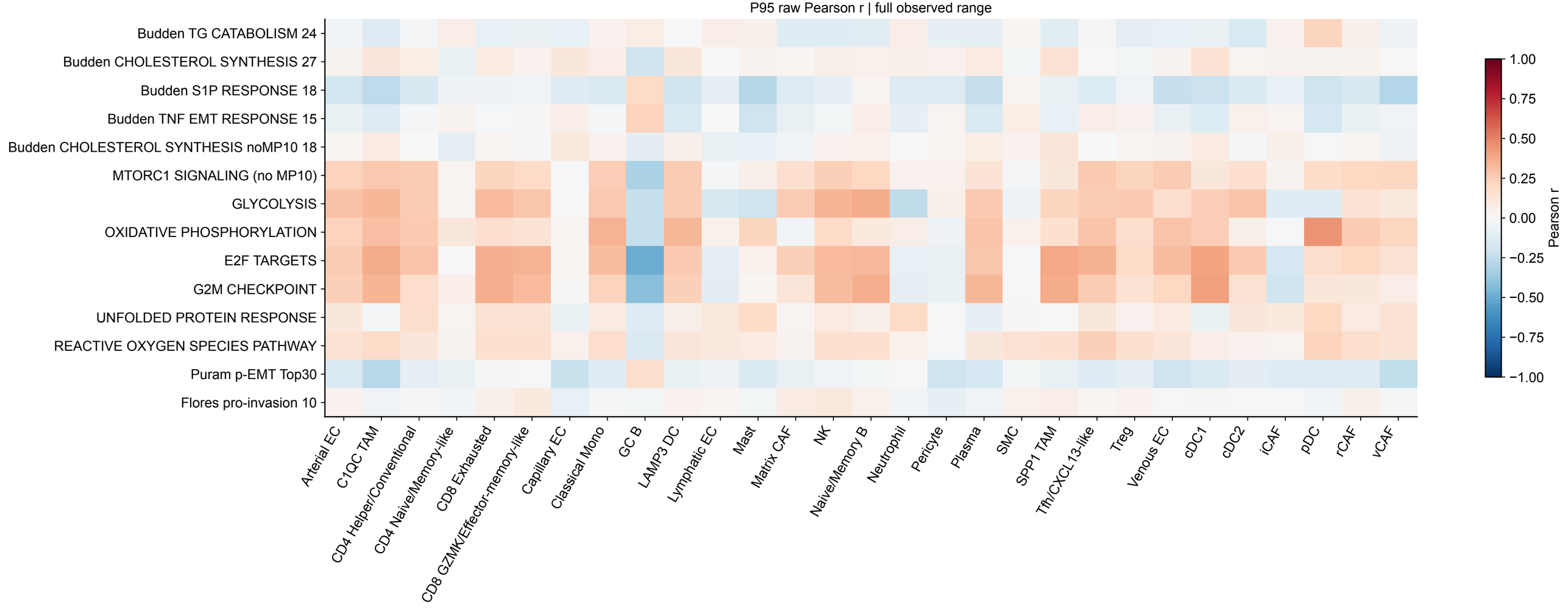


Tissue 6859 | Epi 5719 | internal 4430 | P95 222
Exact front in Epi 476 | R12 labelled 4702

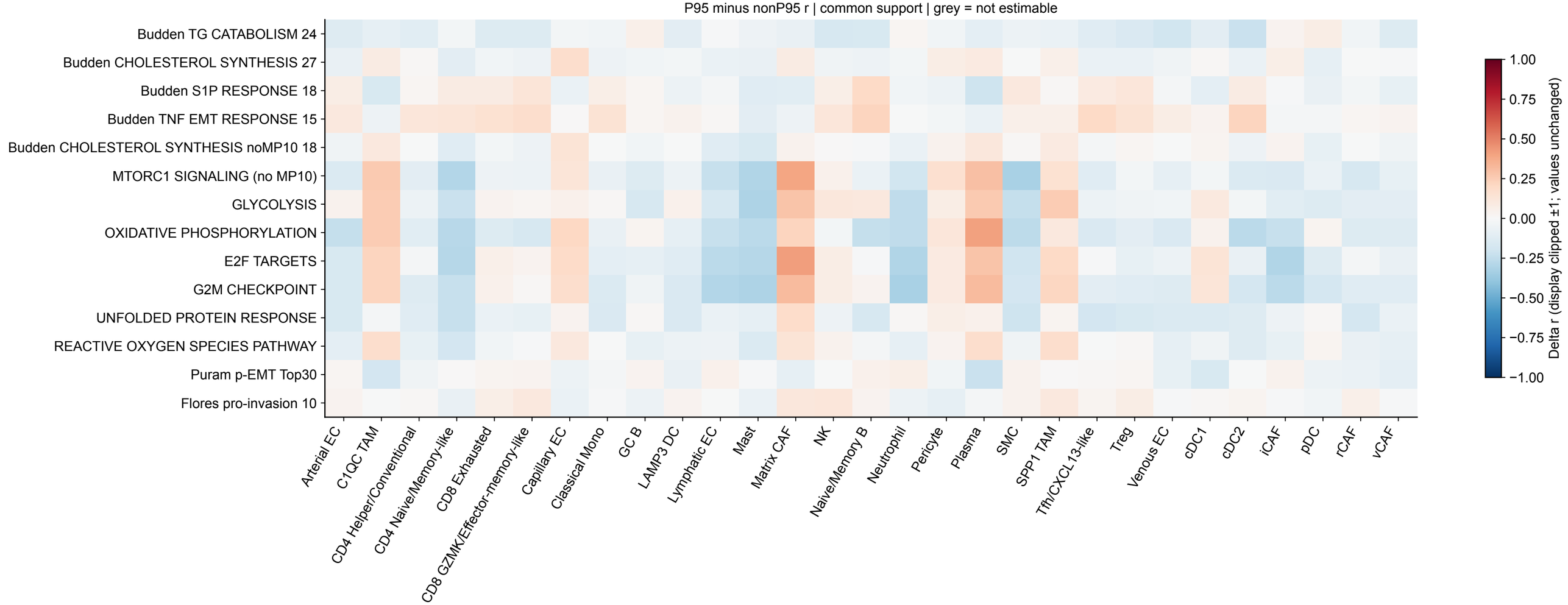
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC11 | all 29 candidates; fixed core programs

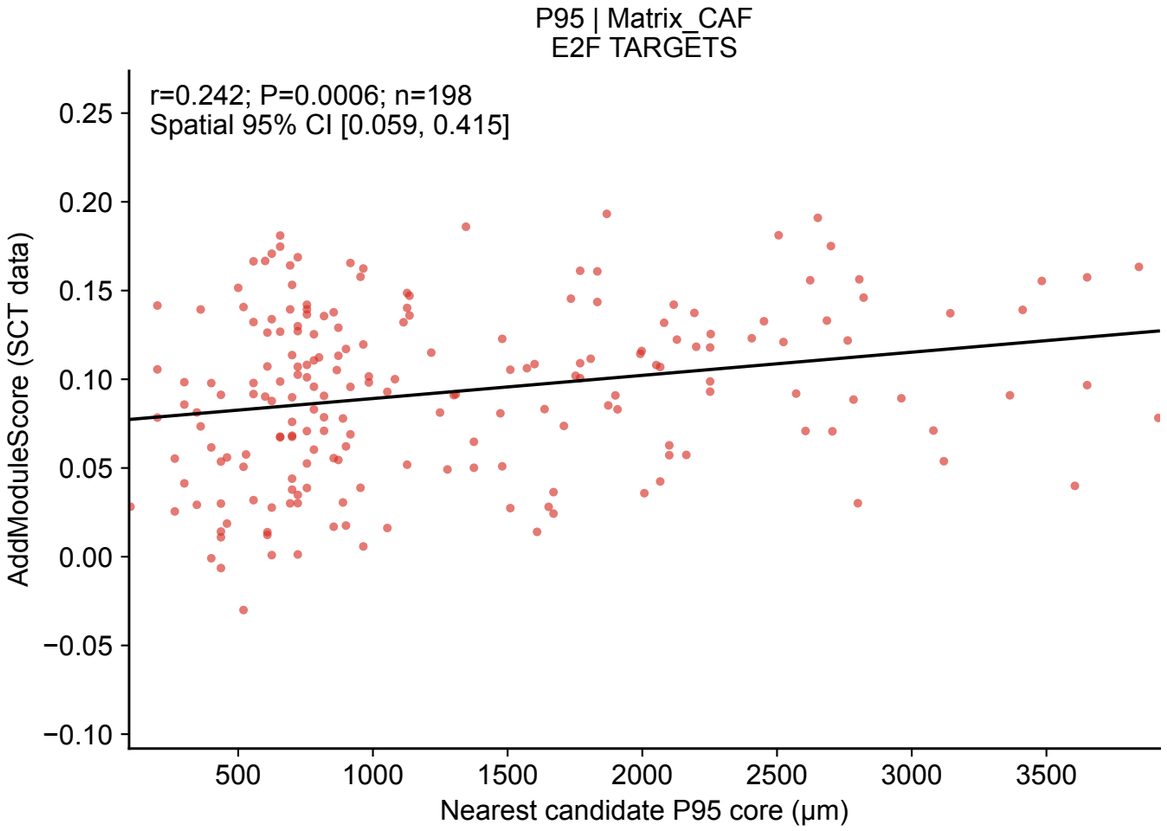
P95 raw Pearson r | full observed range

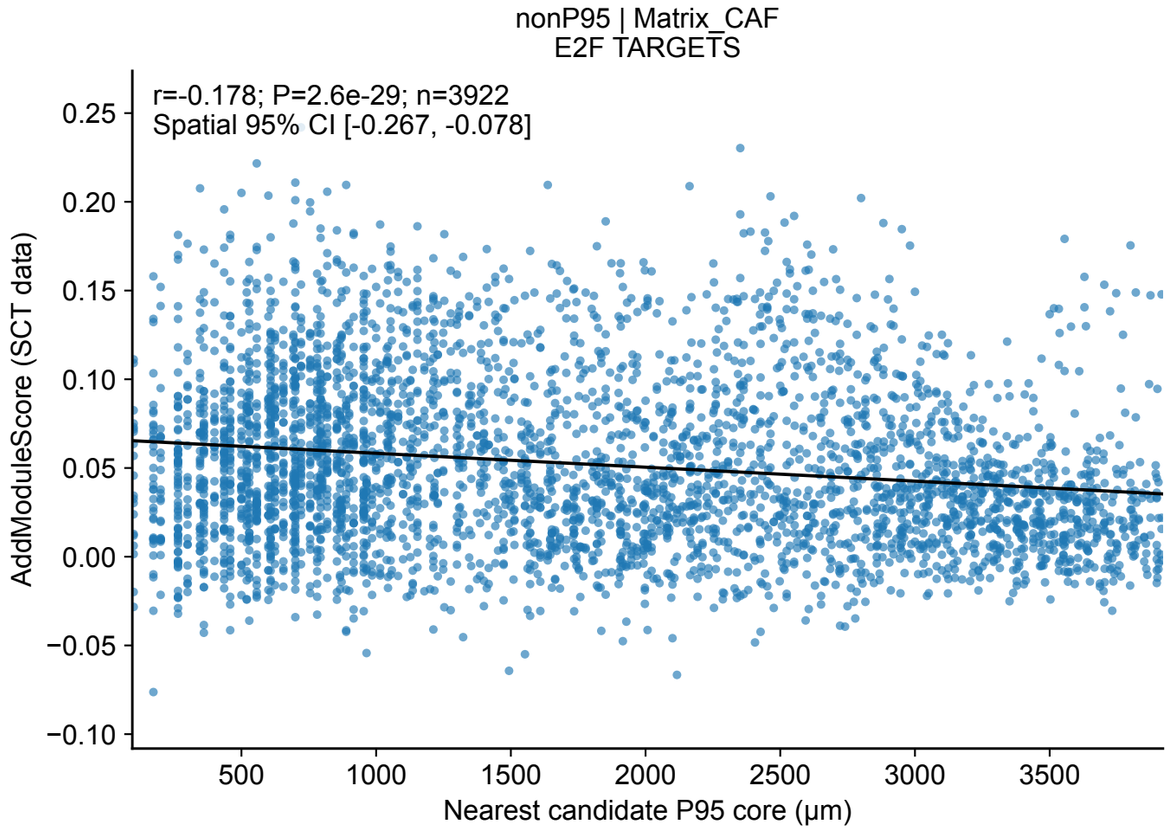


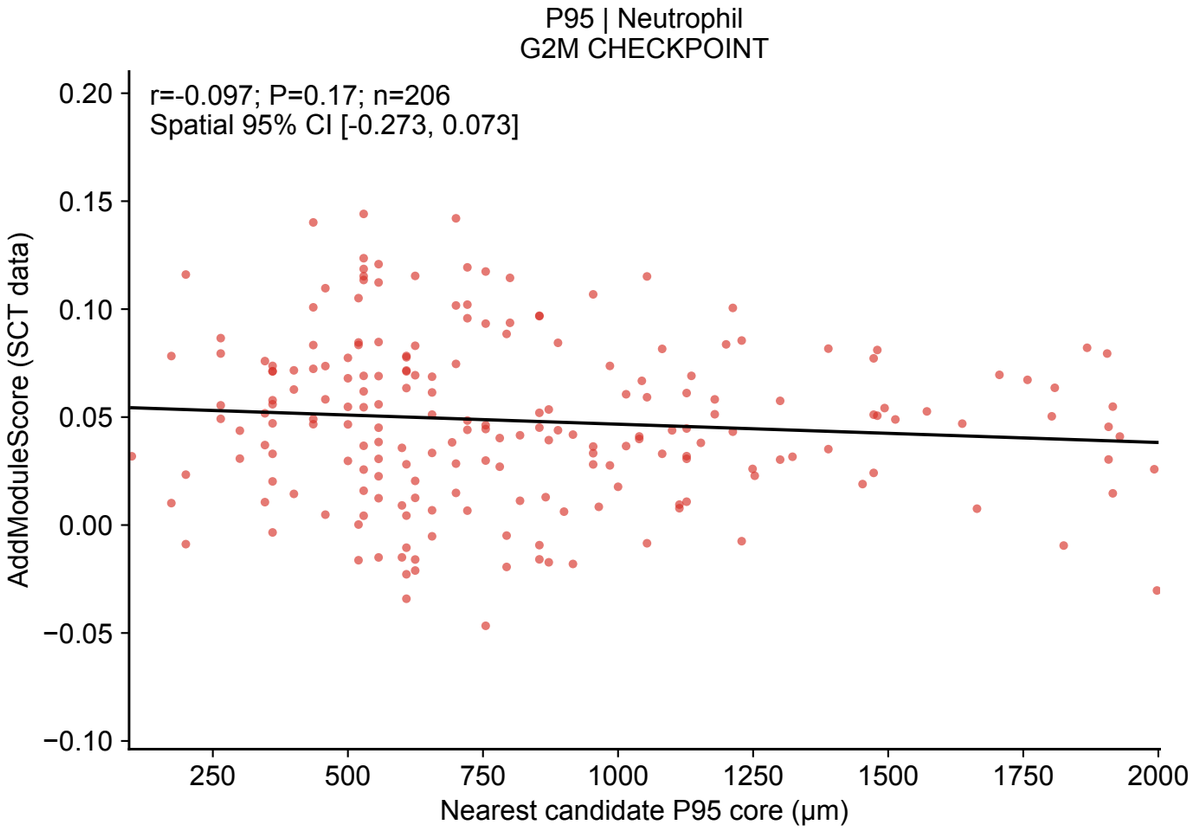
P95 minus nonP95 r | common support | grey = not estimable

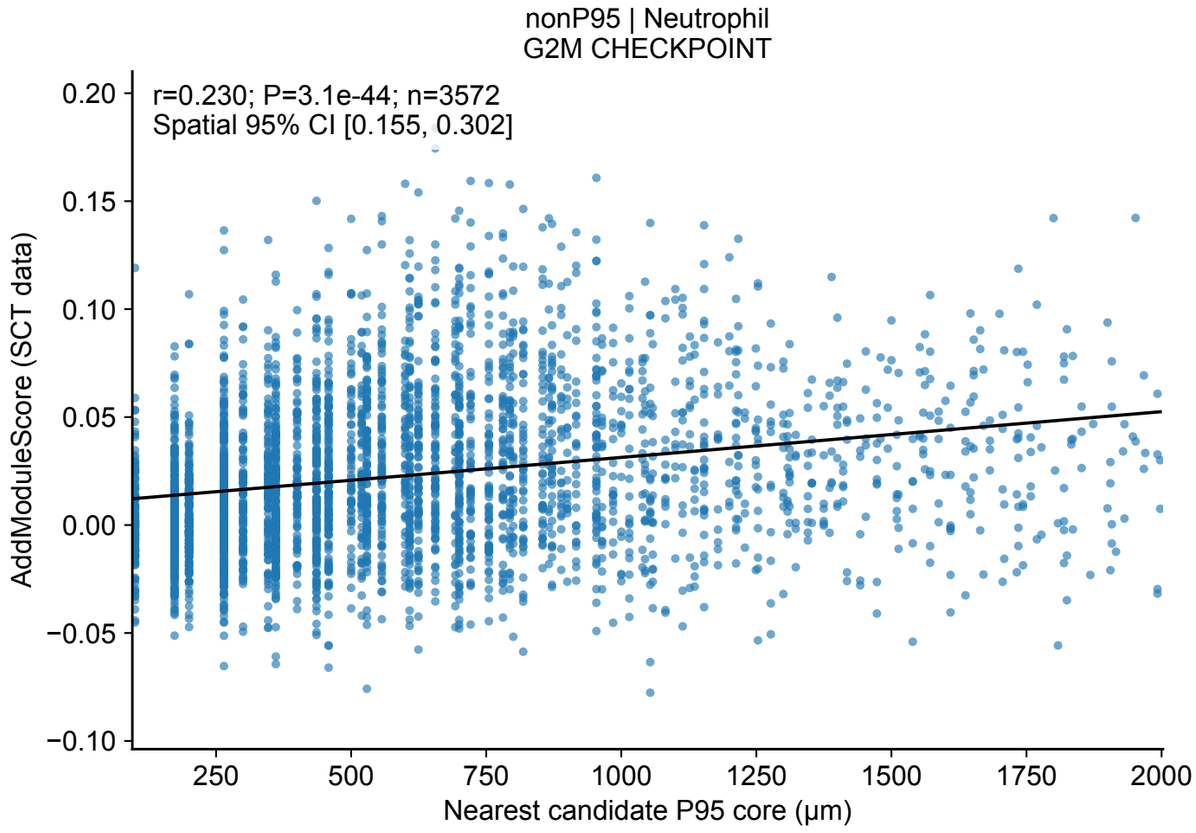


GSE319536_BC11 | exploratory scatter display selected by coverage and |delta r|, not P







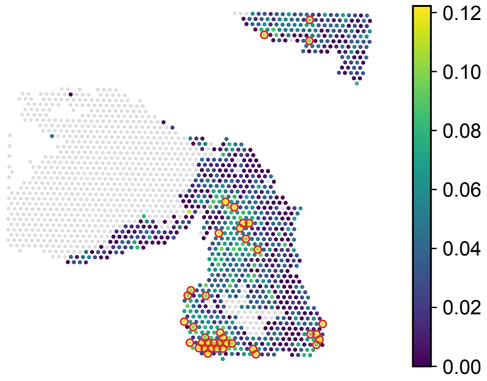


Matrix_CAF / E2F TARGETS: delta r=0.420 [0.224, 0.598]; common 100–3915 μm; spatial P=2.2e-05
Neutrophil / G2M CHECKPOINT: delta r=-0.327 [-0.502, -0.164]; common 100–1997 μm; spatial P=0.00013
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

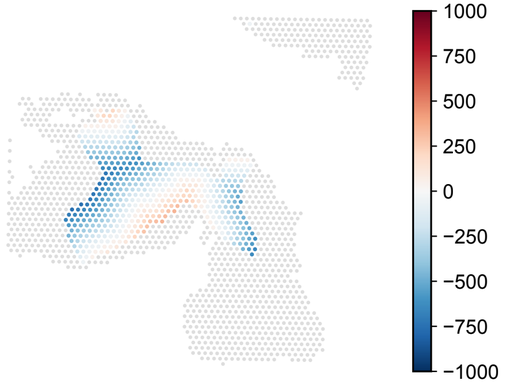
P95 nonP95 distance 0

GSE319536_BC12 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

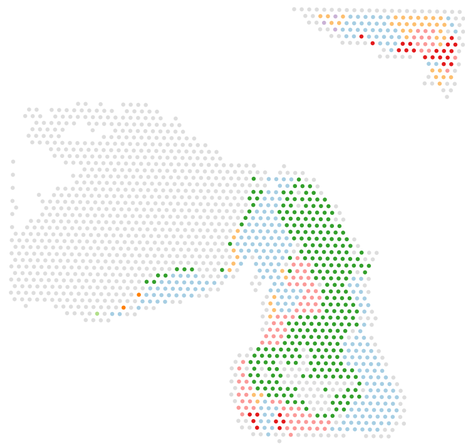
MP10 AUCell and frozen P95



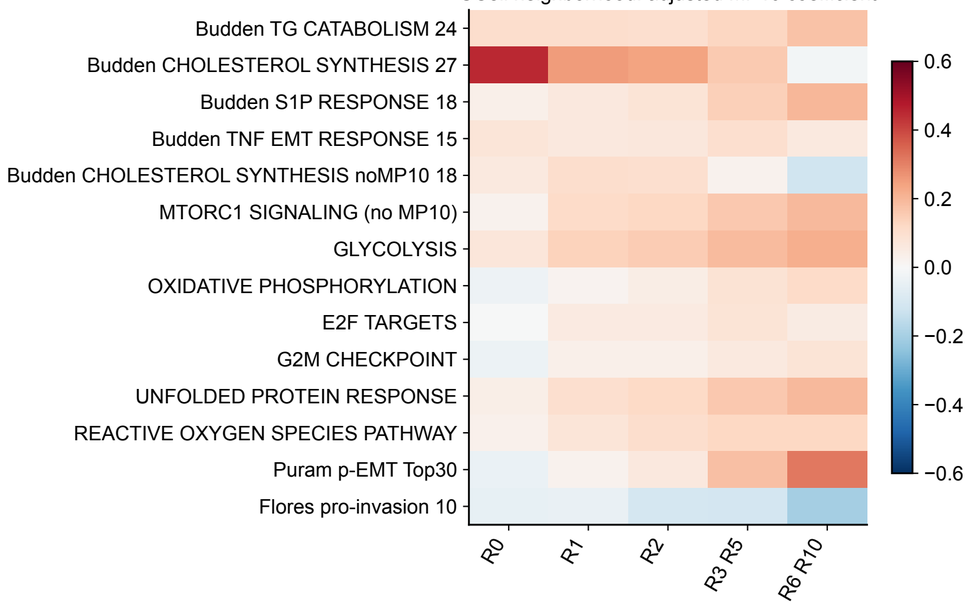
Strict v3.4 signed distance (µm)



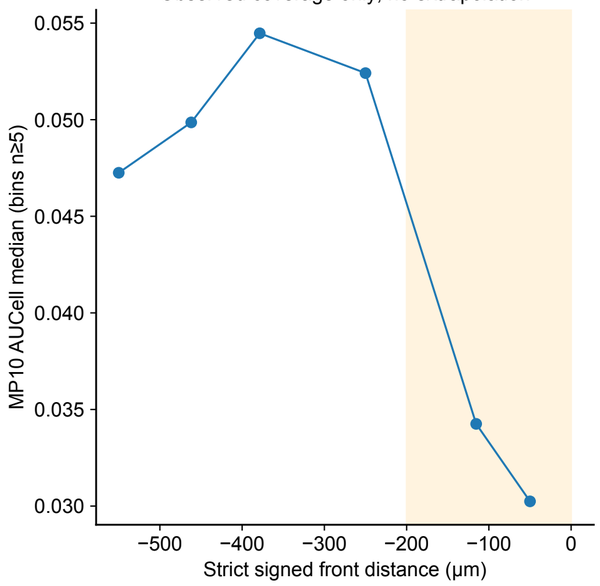
Formal R12 0–8; grey = unlabelled



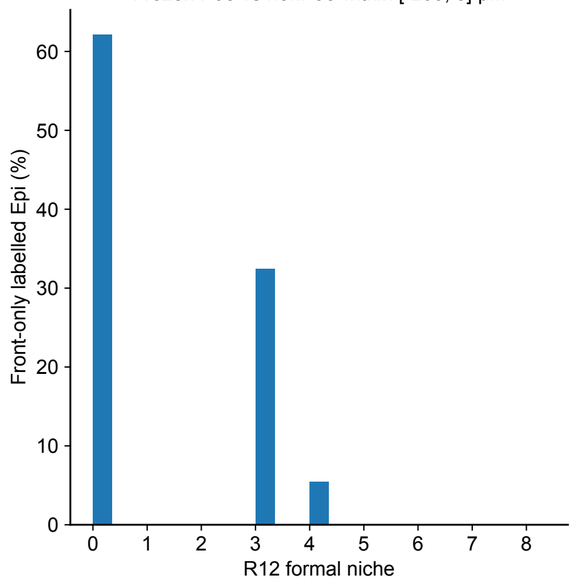
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

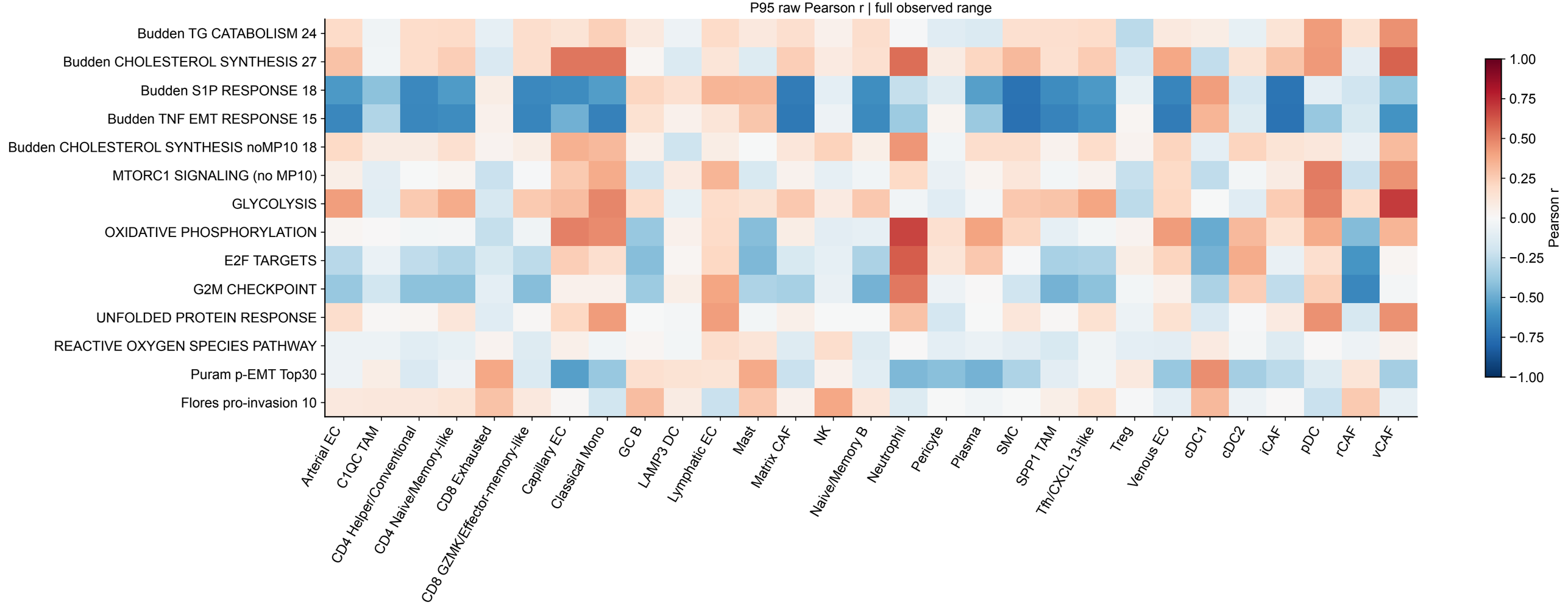


Tissue 1654 | Epi 884 | internal 764 | P95 39
Exact front in Epi 142 | R12 labelled 698

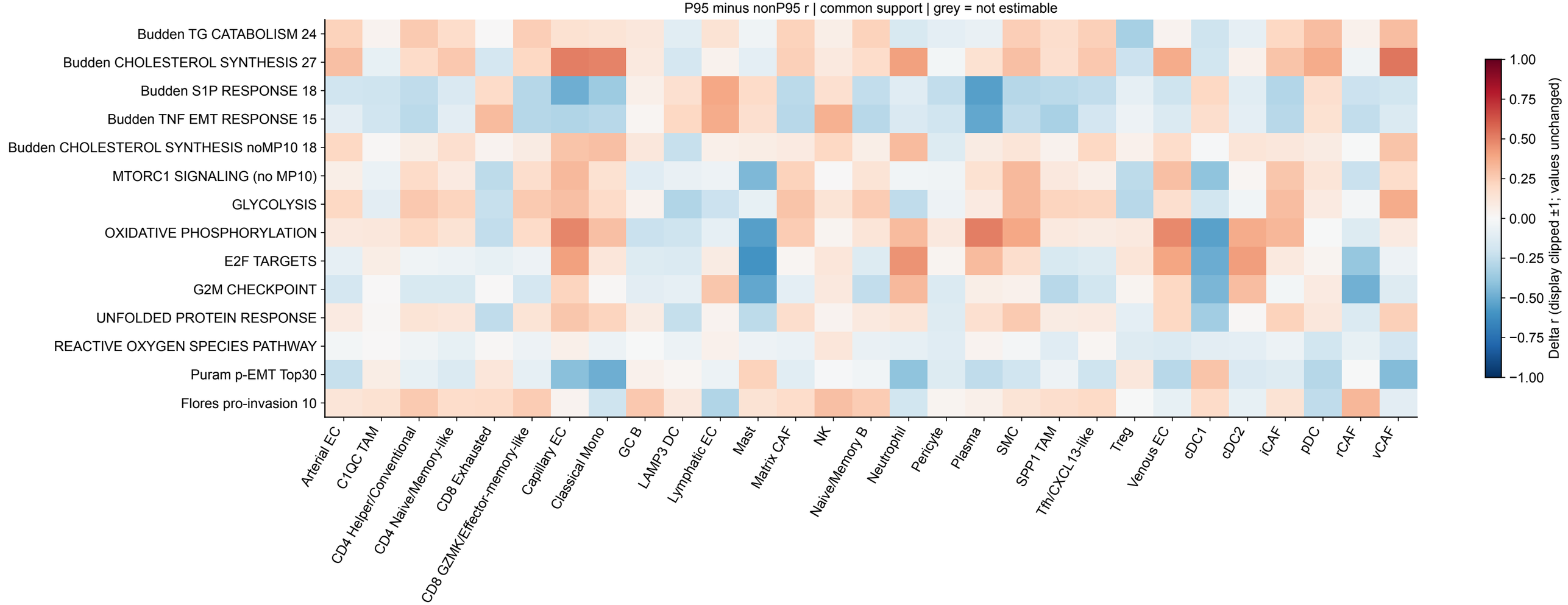
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC12 | all 29 candidates; fixed core programs

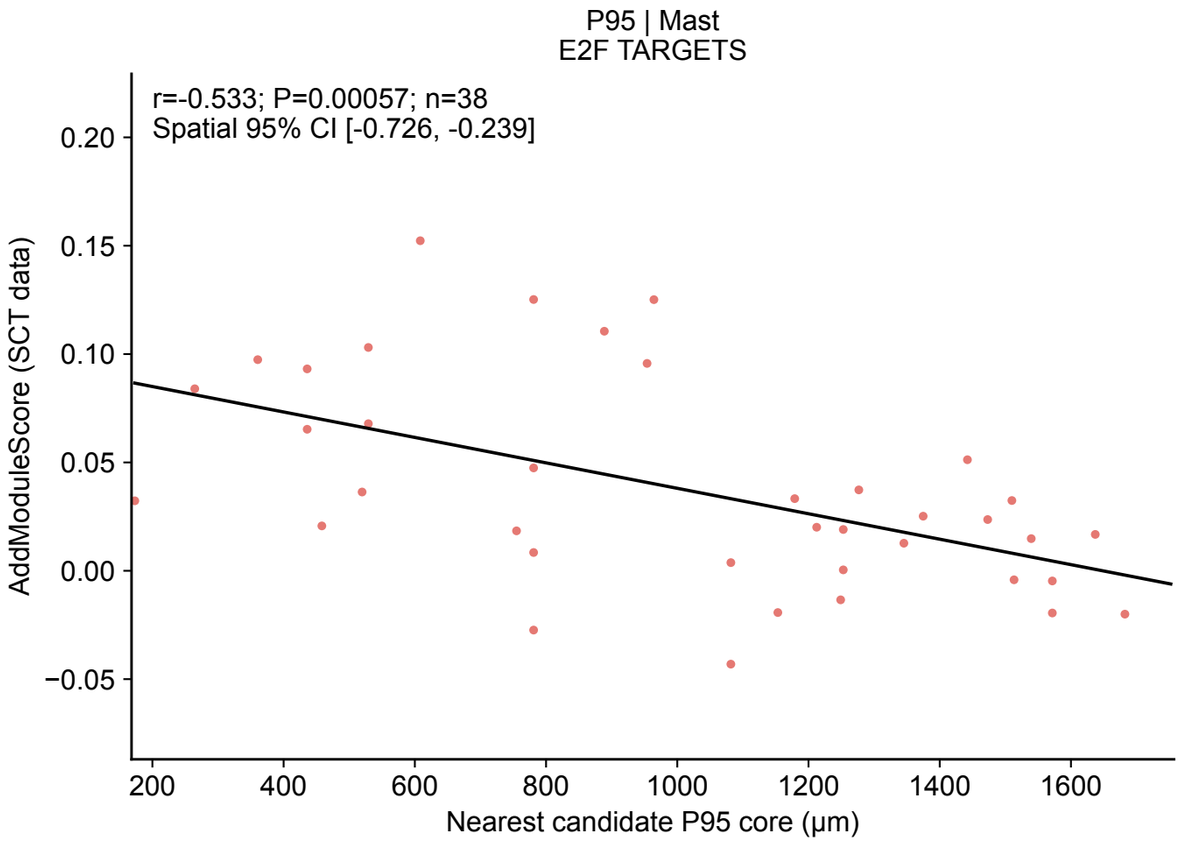
P95 raw Pearson r | full observed range

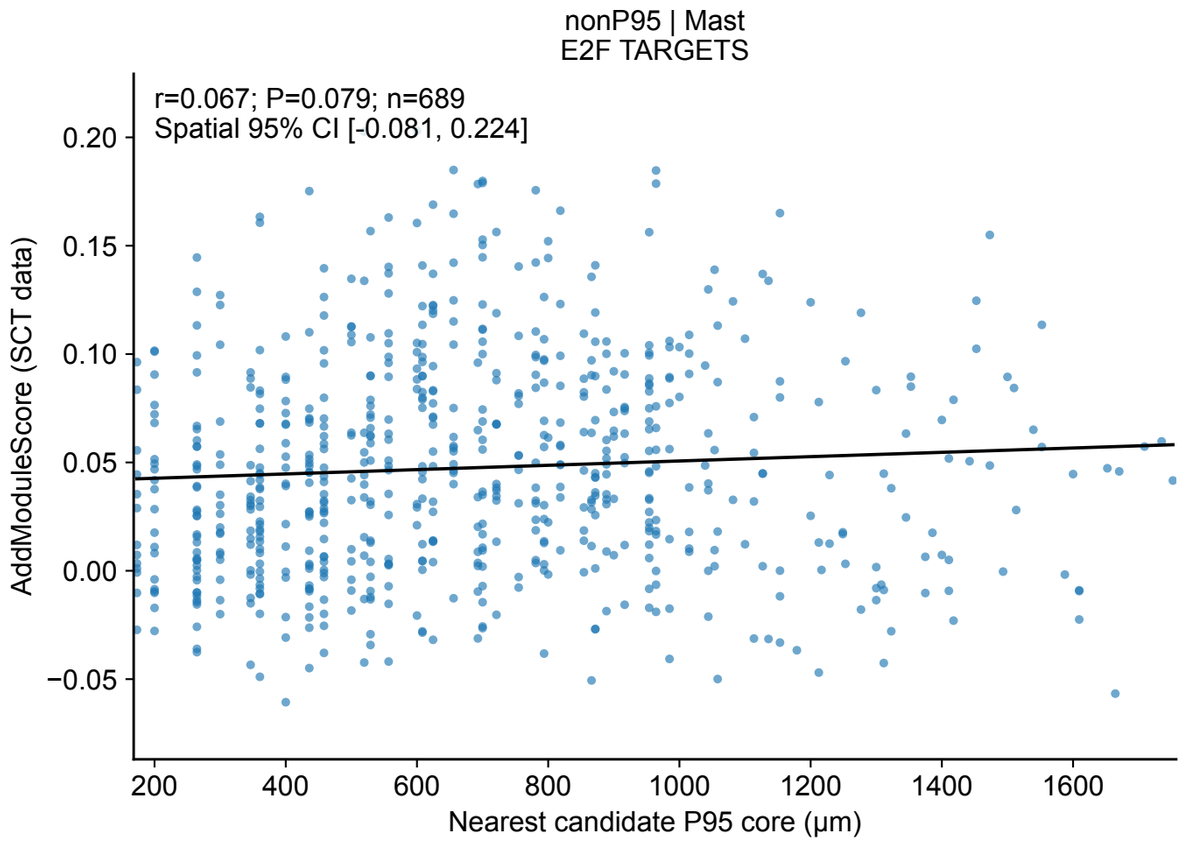


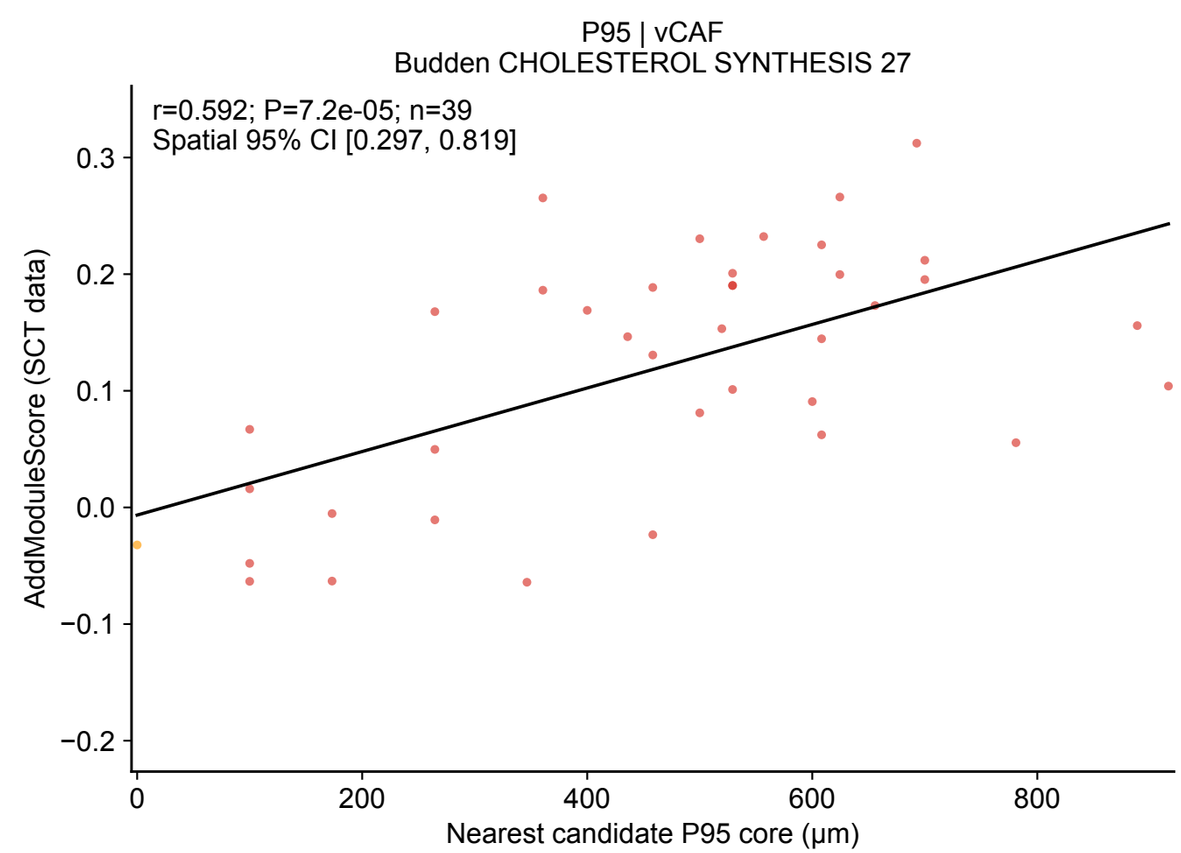
P95 minus nonP95 r | common support | grey = not estimable

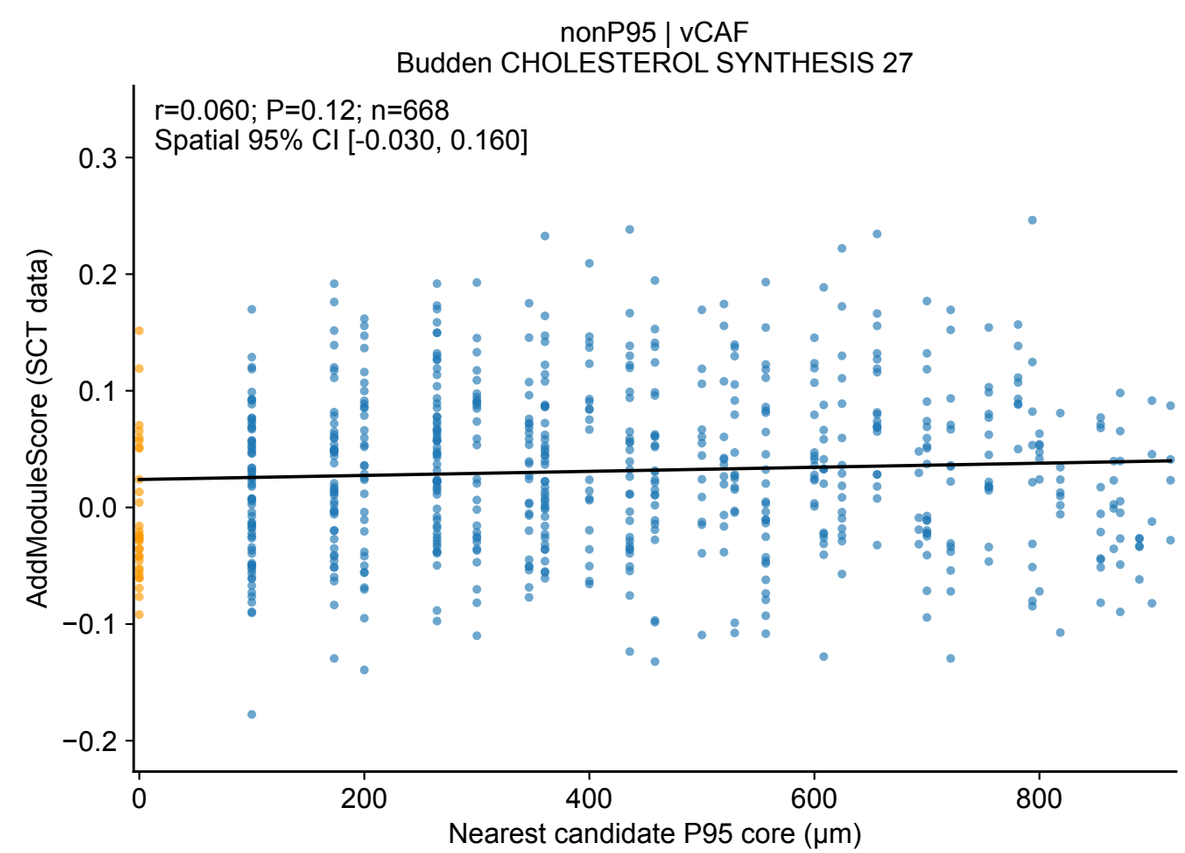


GSE319536_BC12 | exploratory scatter display selected by coverage and |delta r|, not P







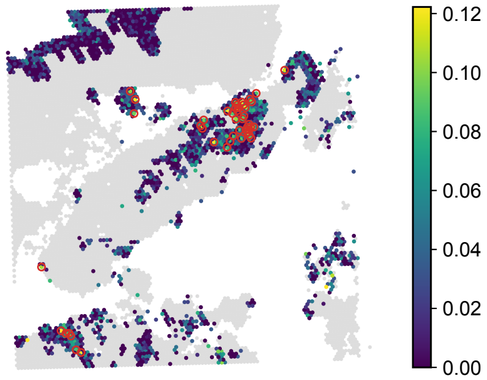


Mast / E2F TARGETS: delta r=-0.600 [-0.840, -0.318]; common 173–1752 µm; spatial P=1.2e-05
vCAF / Budden CHOLESTEROL SYNTHESIS 27: delta r=0.532 [0.232, 0.761]; common 0–917 µm; spatial P=0.0013
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

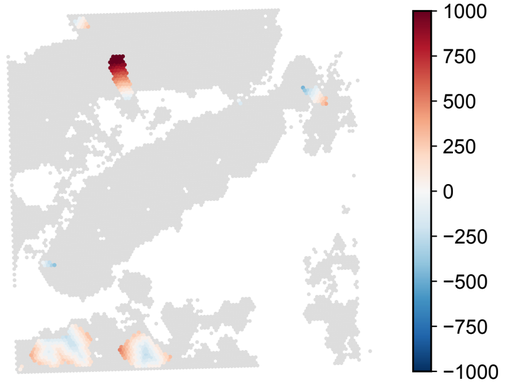
P95 nonP95 distance 0

GSE319536_BC13 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

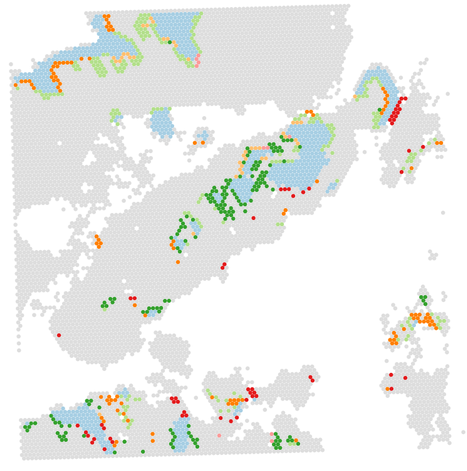
MP10 AUCell and frozen P95



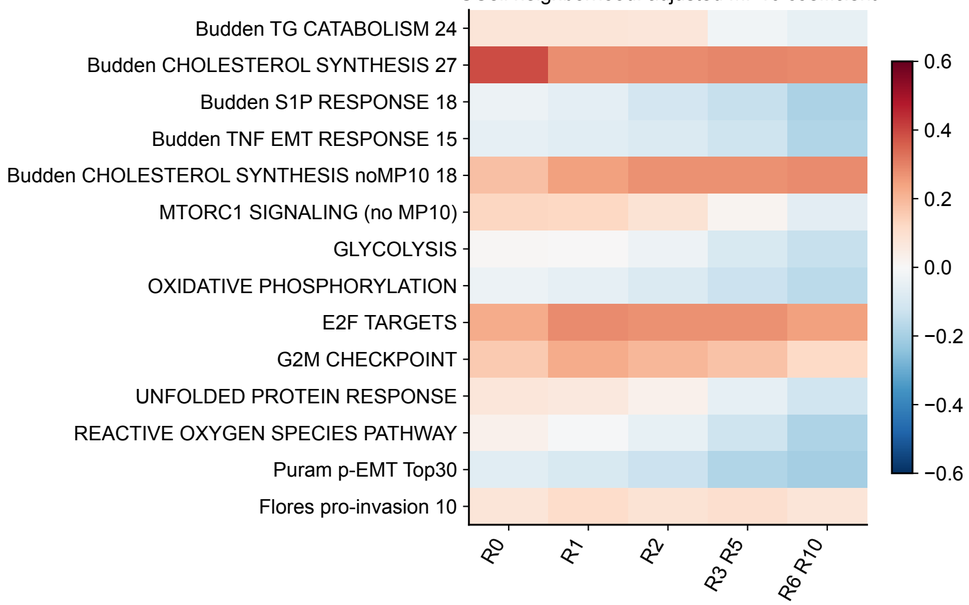
Strict v3.4 signed distance (µm)



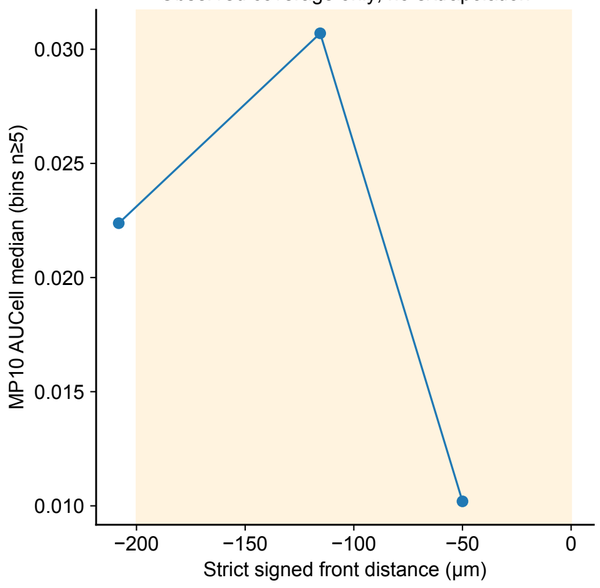
Formal R12 0–8; grey = unlabelled



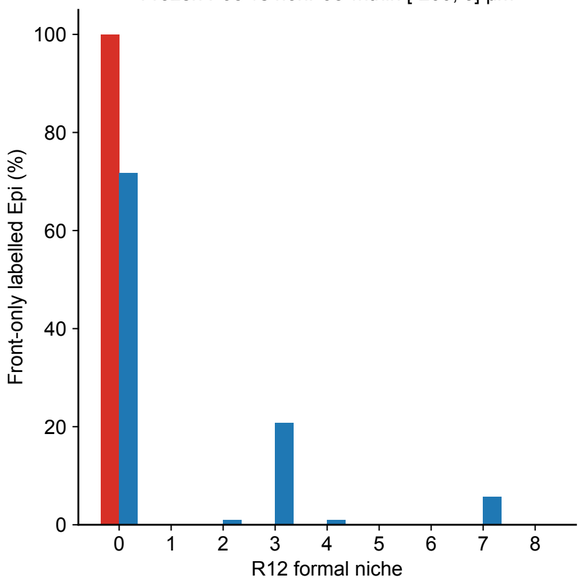
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

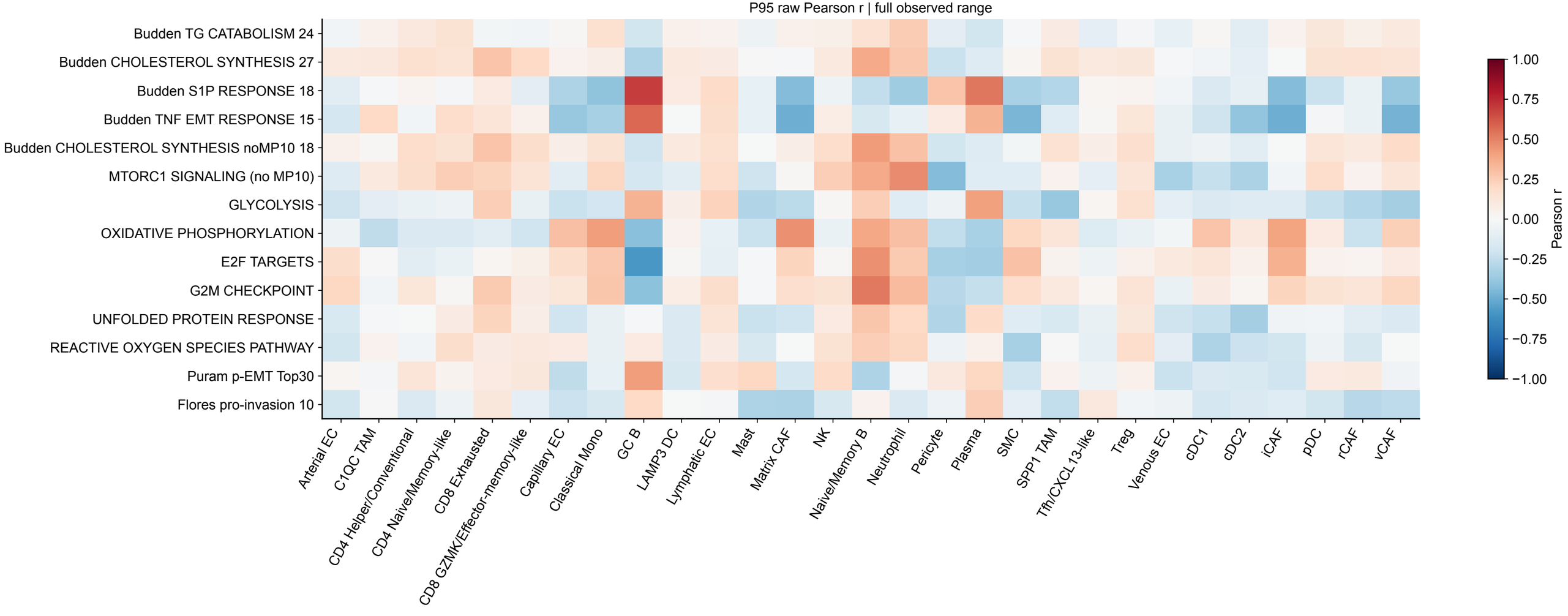


Tissue 7758 | Epi 1759 | internal 881 | P95 45
Exact front in Epi 184 | R12 labelled 1351

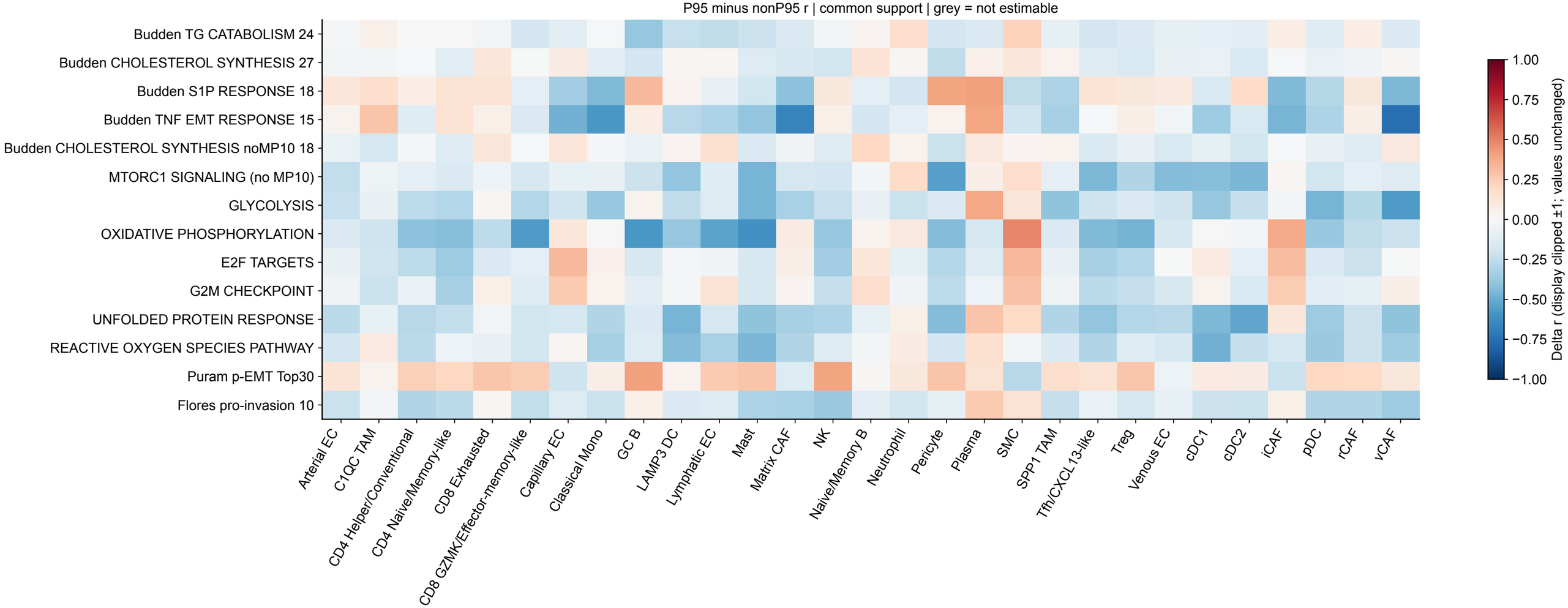
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC13 | all 29 candidates; fixed core programs

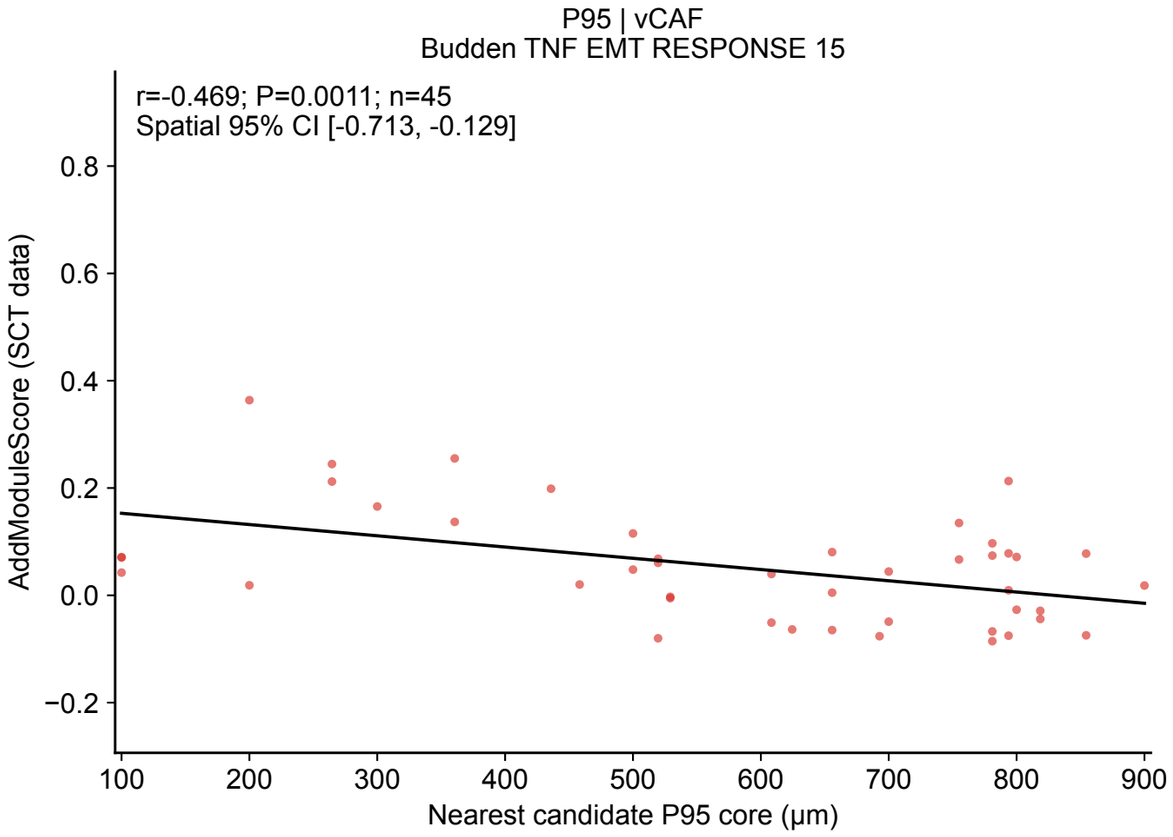
P95 raw Pearson r | full observed range

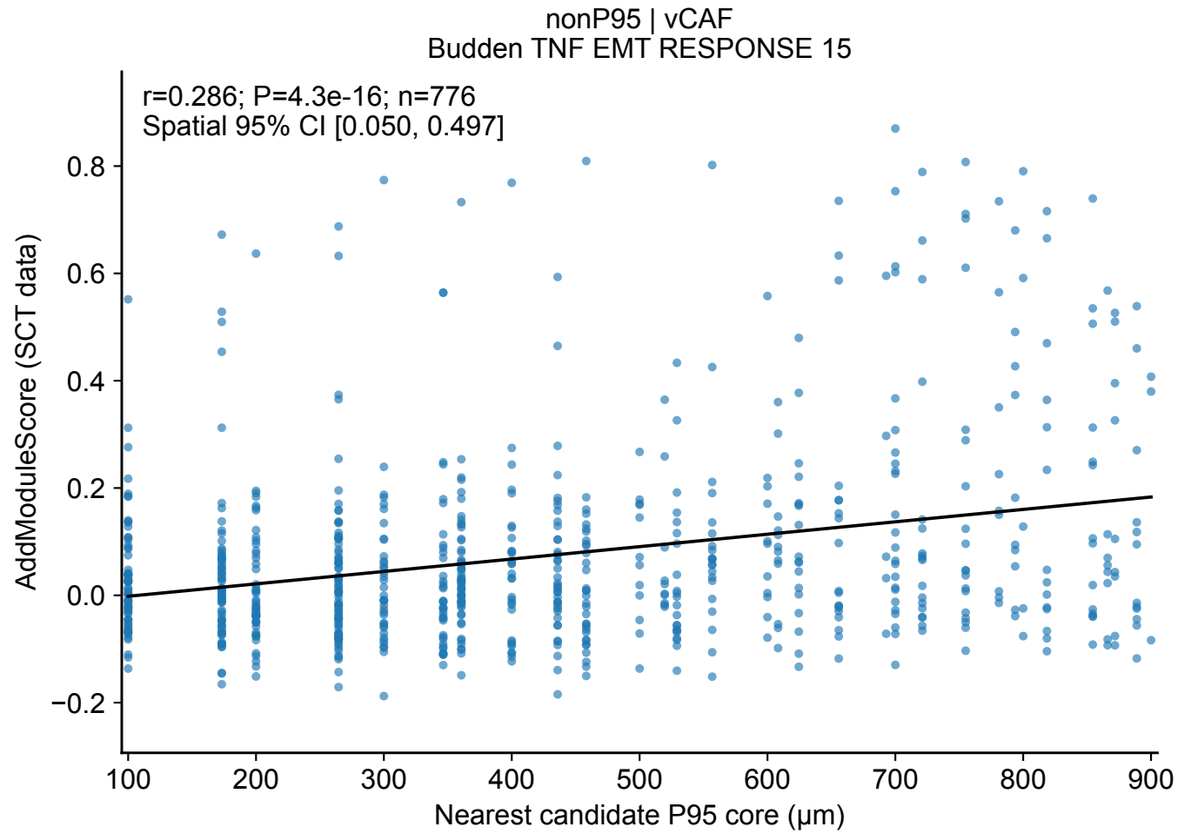


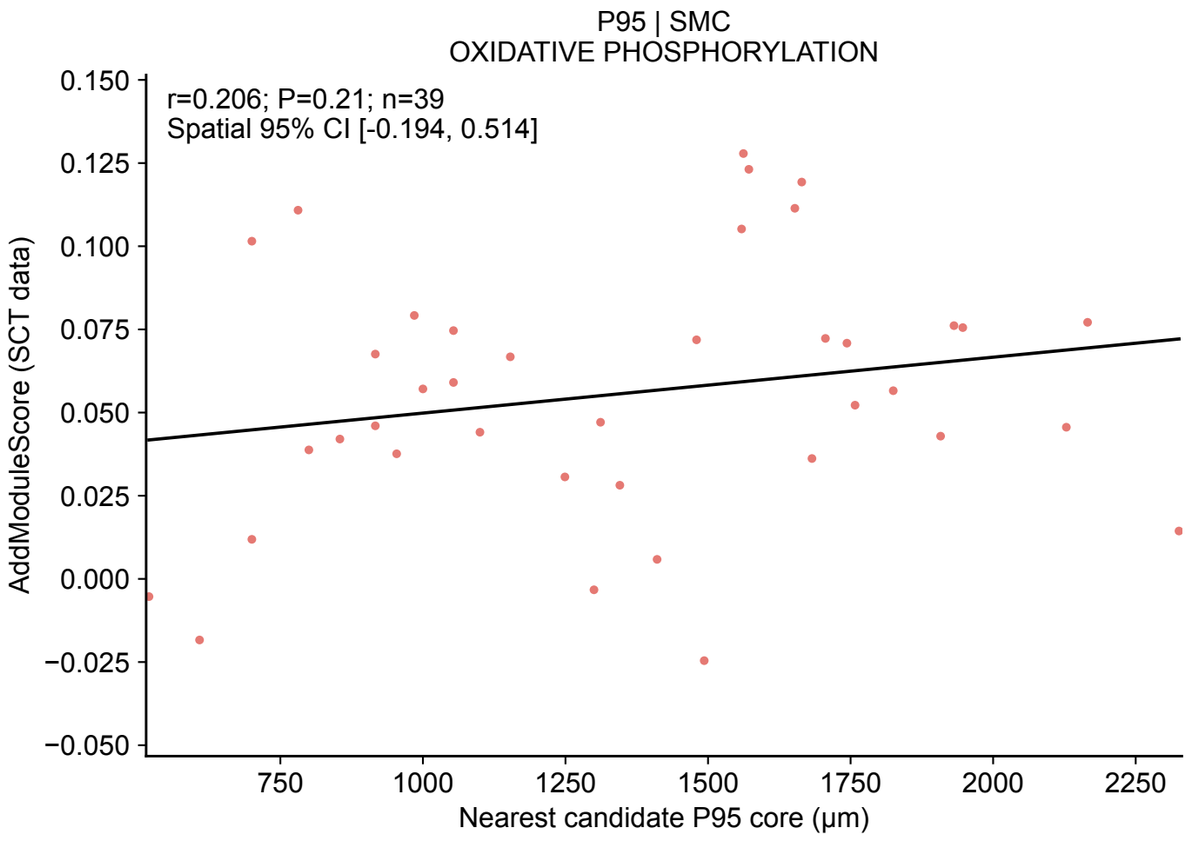
P95 minus nonP95 r | common support | grey = not estimable

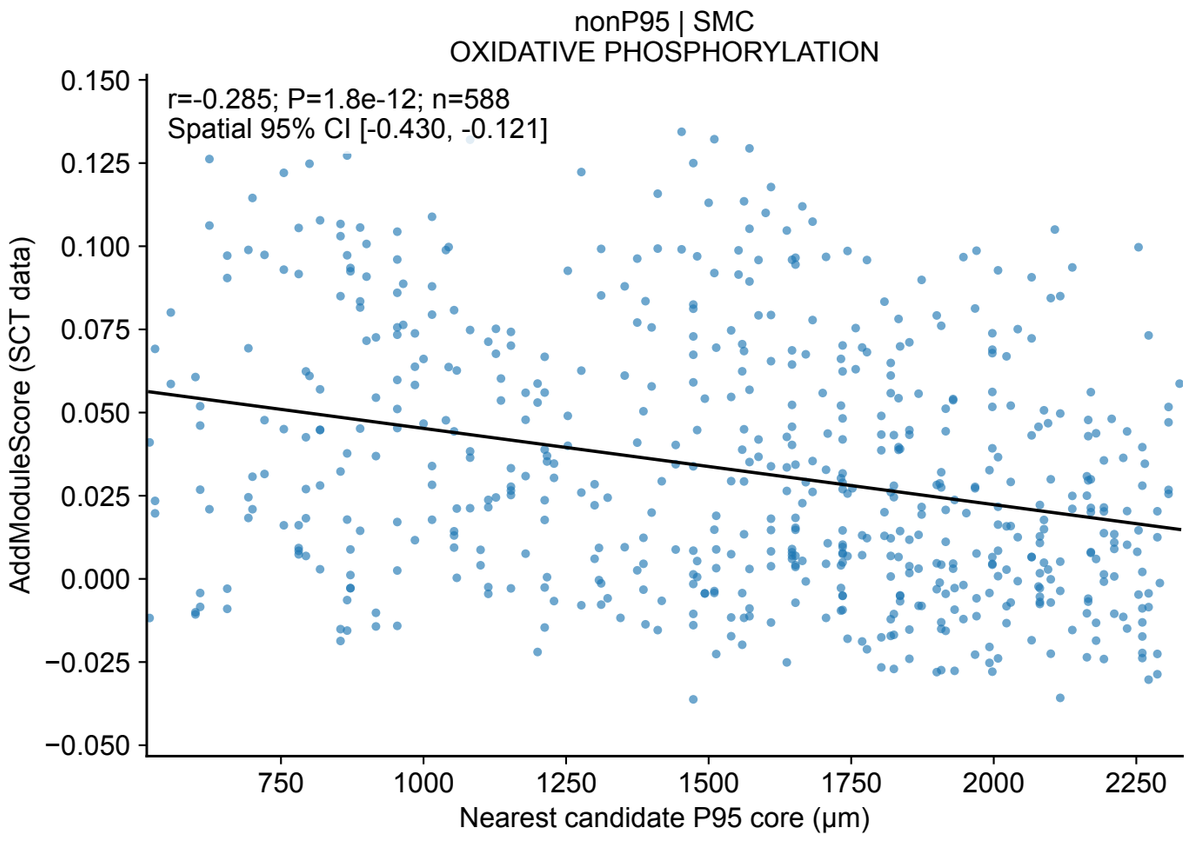


GSE319536_BC13 | exploratory scatter display selected by coverage and |delta r|, not P







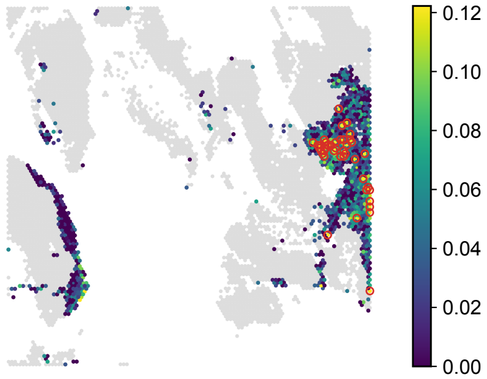


vCAF / Budden TNF EMT RESPONSE 15: delta r=-0.755 [-1.043, -0.380]; common 100–900 µm; spatial P=3.5e-05
SMC / OXIDATIVE PHOSPHORYLATION: delta r=0.491 [0.083, 0.835]; common 520–2326 µm; spatial P=0.0093
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

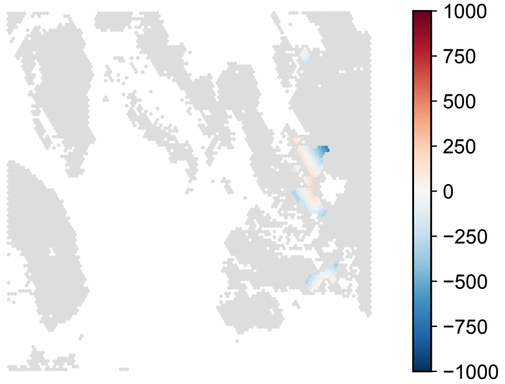
P95 nonP95 distance 0

GSE319536_BC14 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

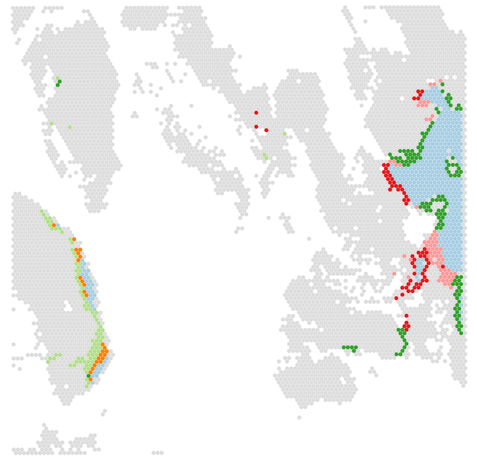
MP10 AUCell and frozen P95



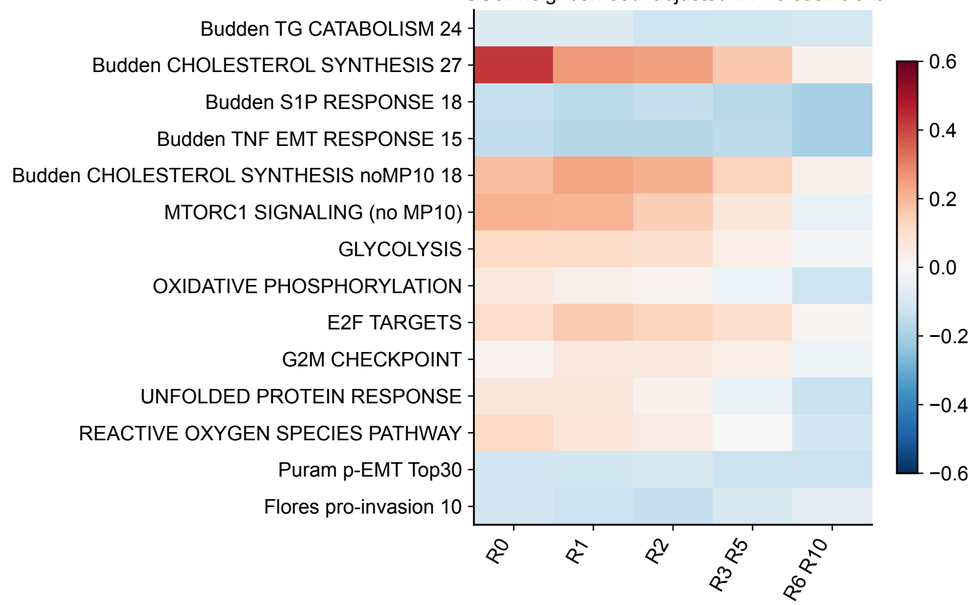
Strict v3.4 signed distance (µm)



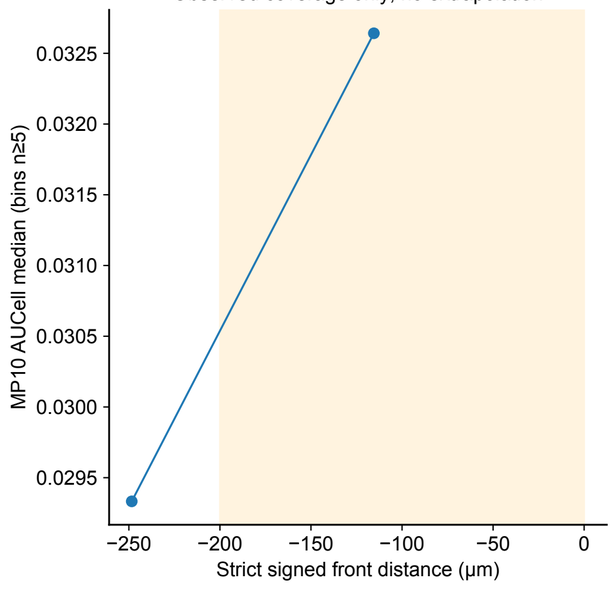
Formal R12 0–8; grey = unlabelled



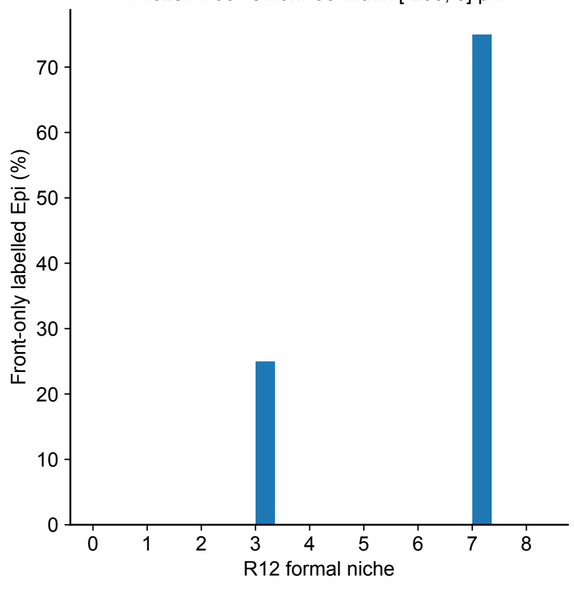
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

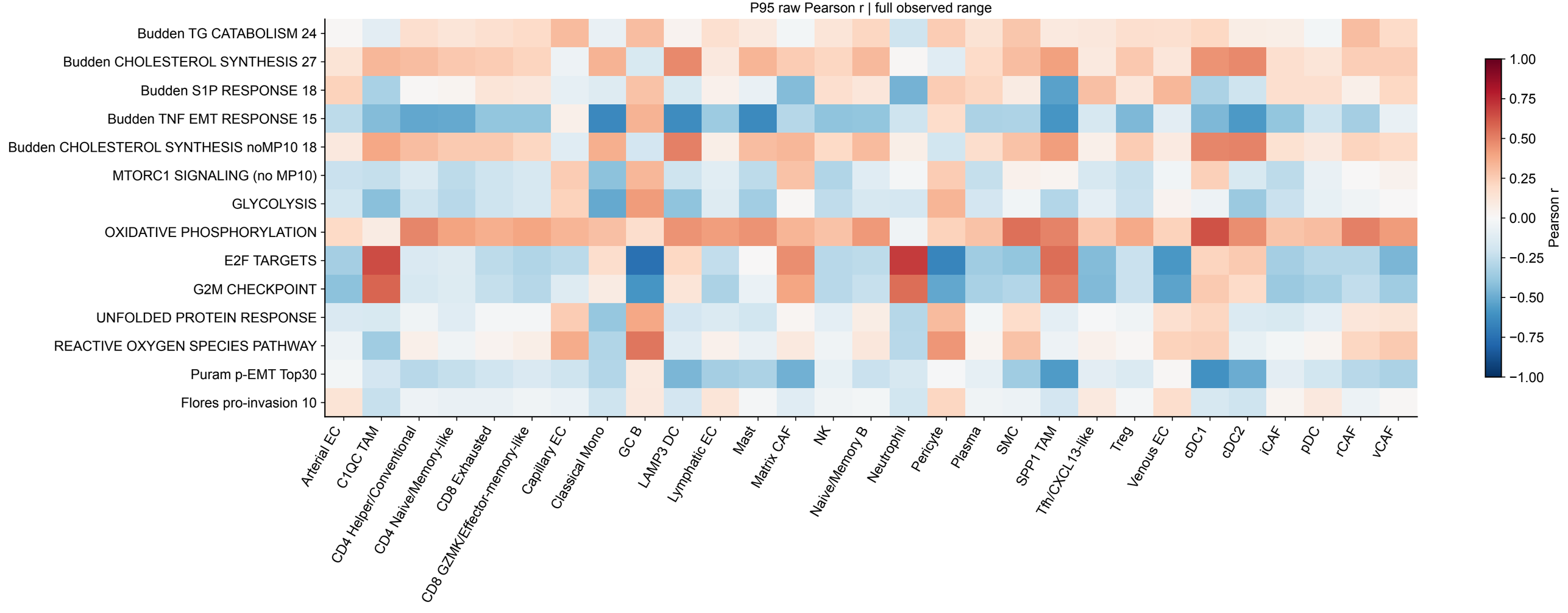


Tissue 5864 | Epi 1066 | internal 815 | P95 41
Exact front in Epi 31 | R12 labelled 822

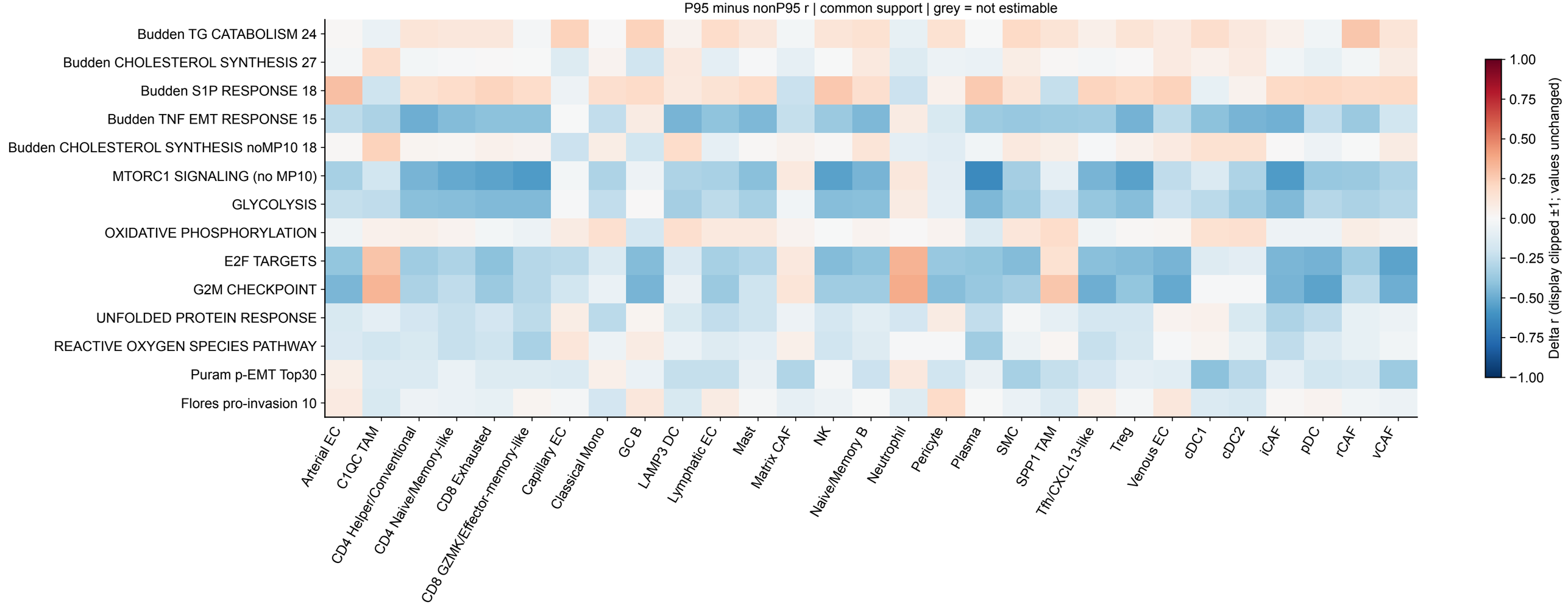


GSE319536_BC14 | all 29 candidates; fixed core programs

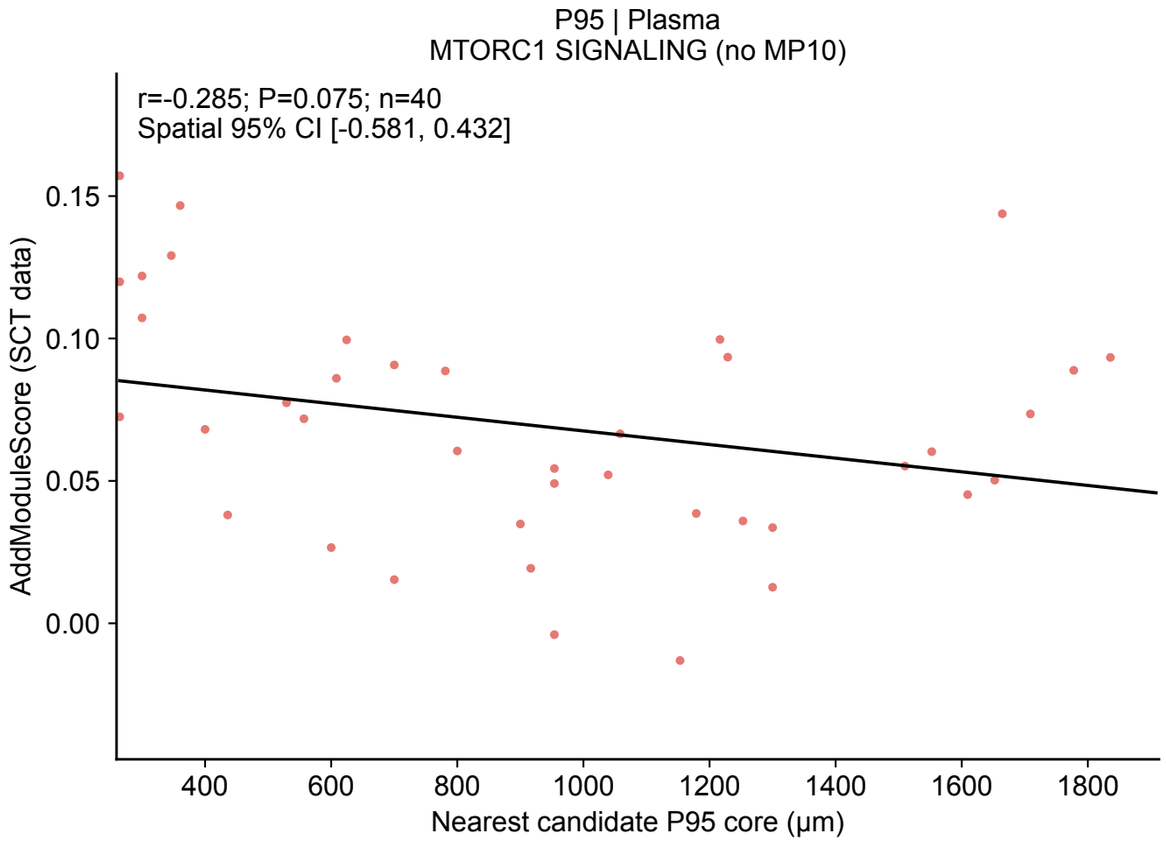
P95 raw Pearson r | full observed range

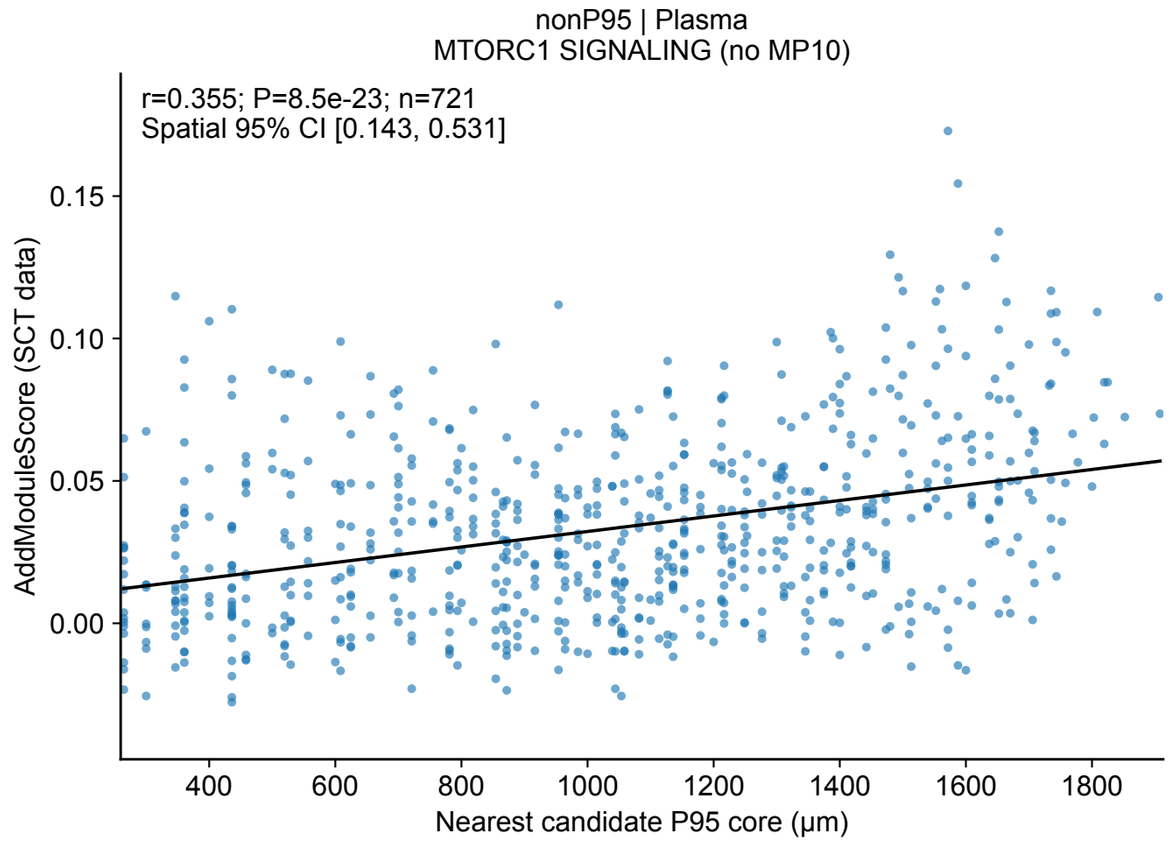


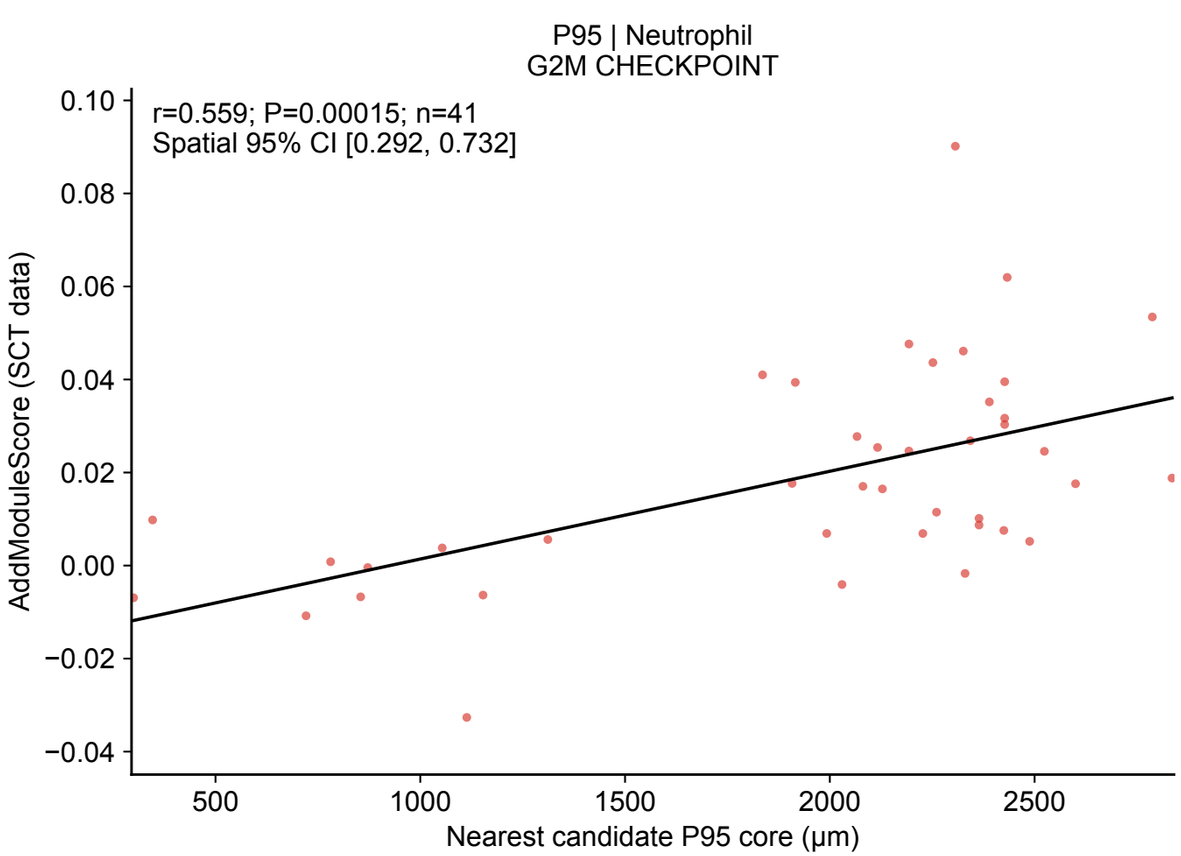
P95 minus nonP95 r | common support | grey = not estimable

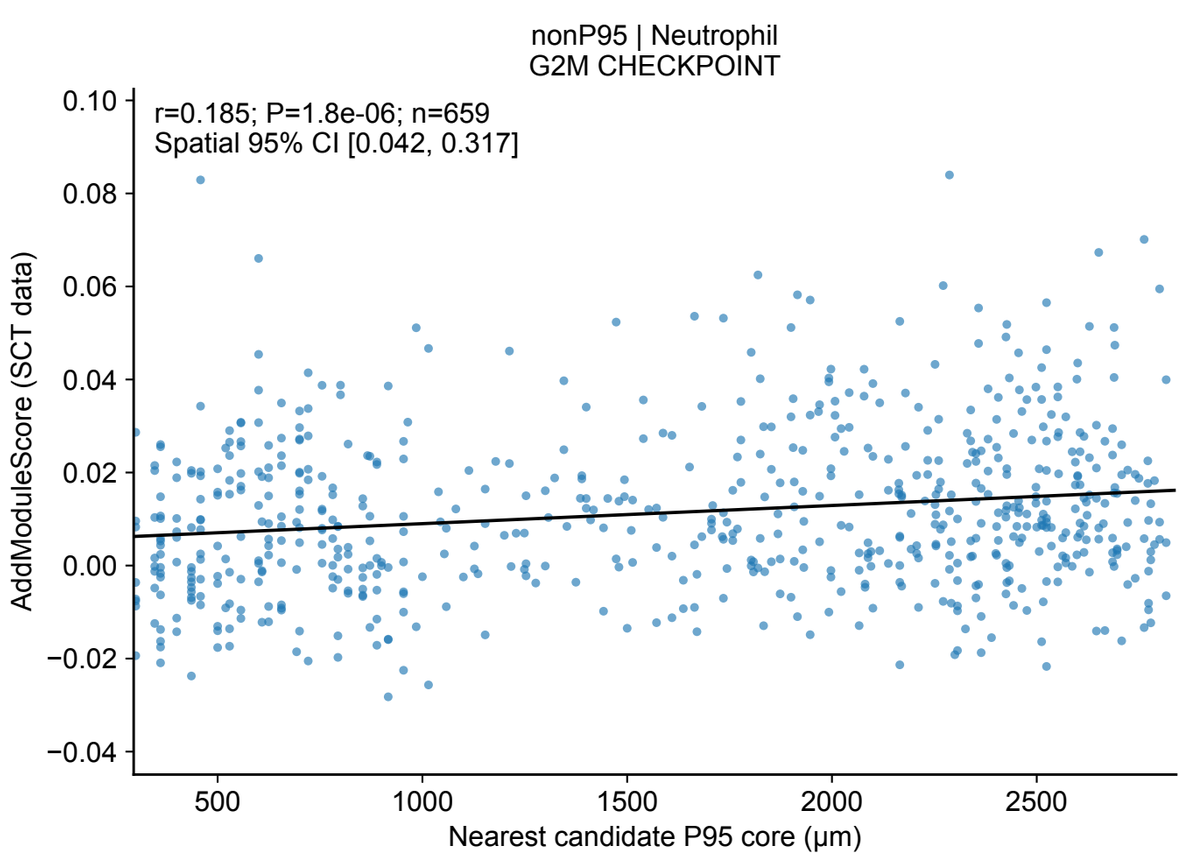


GSE319536_BC14 | exploratory scatter display selected by coverage and |delta r|, not P







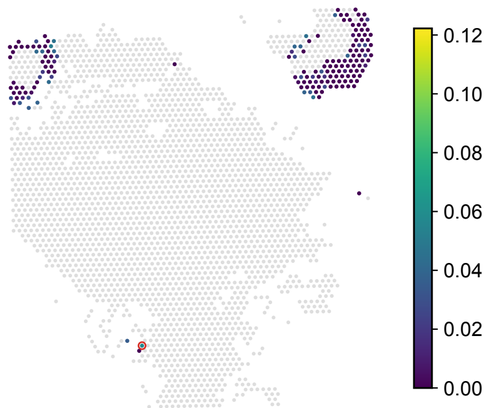


Plasma / MTORC1 SIGNALING (no MP10): delta r=-0.640 [-0.844, -0.033]; common 265–1908 µm; spatial P=2e-07
Neutrophil / G2M CHECKPOINT: delta r=0.374 [0.076, 0.614]; common 300–2835 µm; spatial P=0.011
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

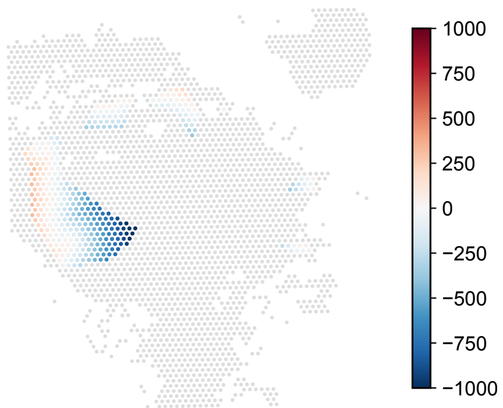
P95 nonP95 distance 0

GSE319536_BC15 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

MP10 AUCell and frozen P95



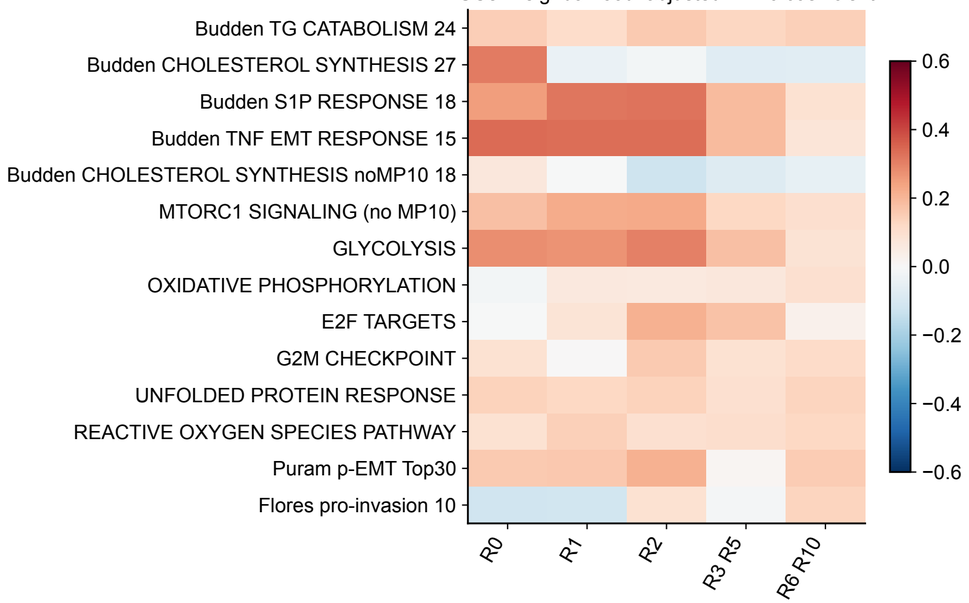
Strict v3.4 signed distance (µm)



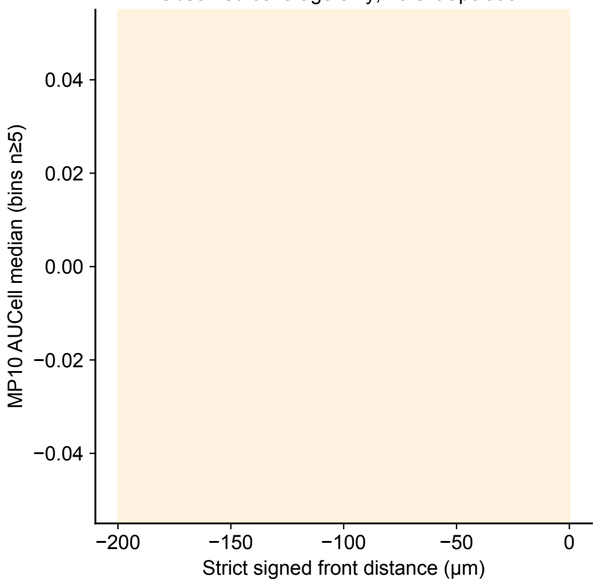
Formal R12 0–8; grey = unlabelled



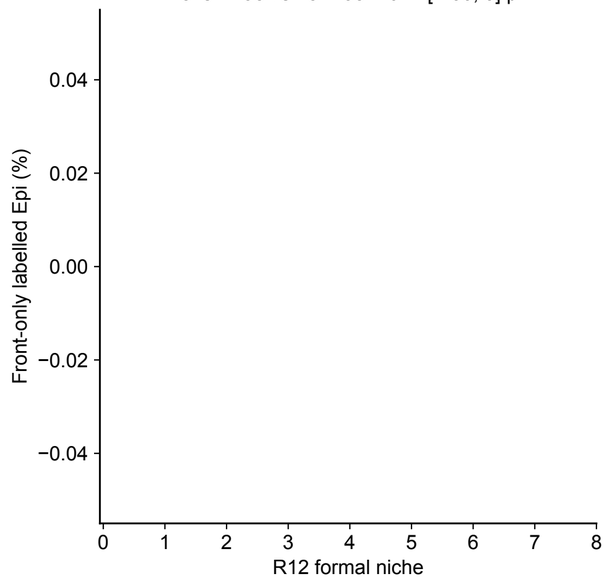
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

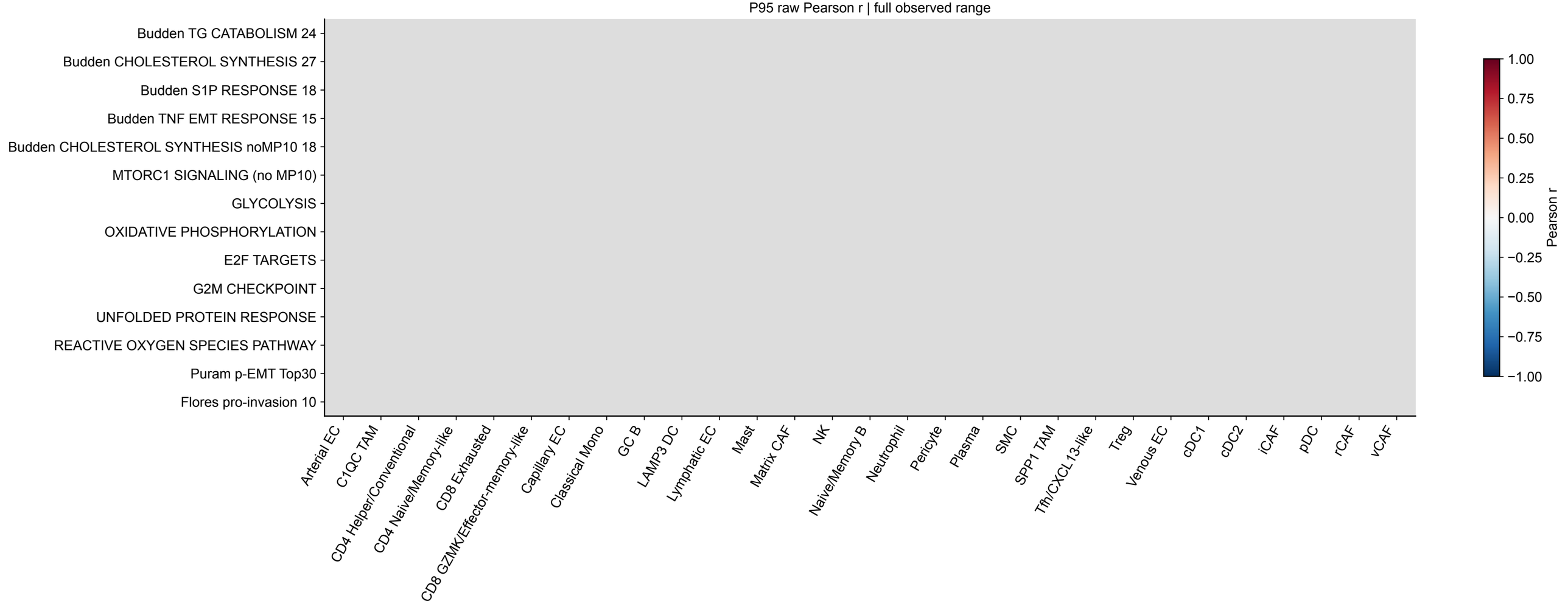


Tissue 2698 | Epi 167 | internal 4 | P95 1
Exact front in Epi 0 | R12 labelled 98

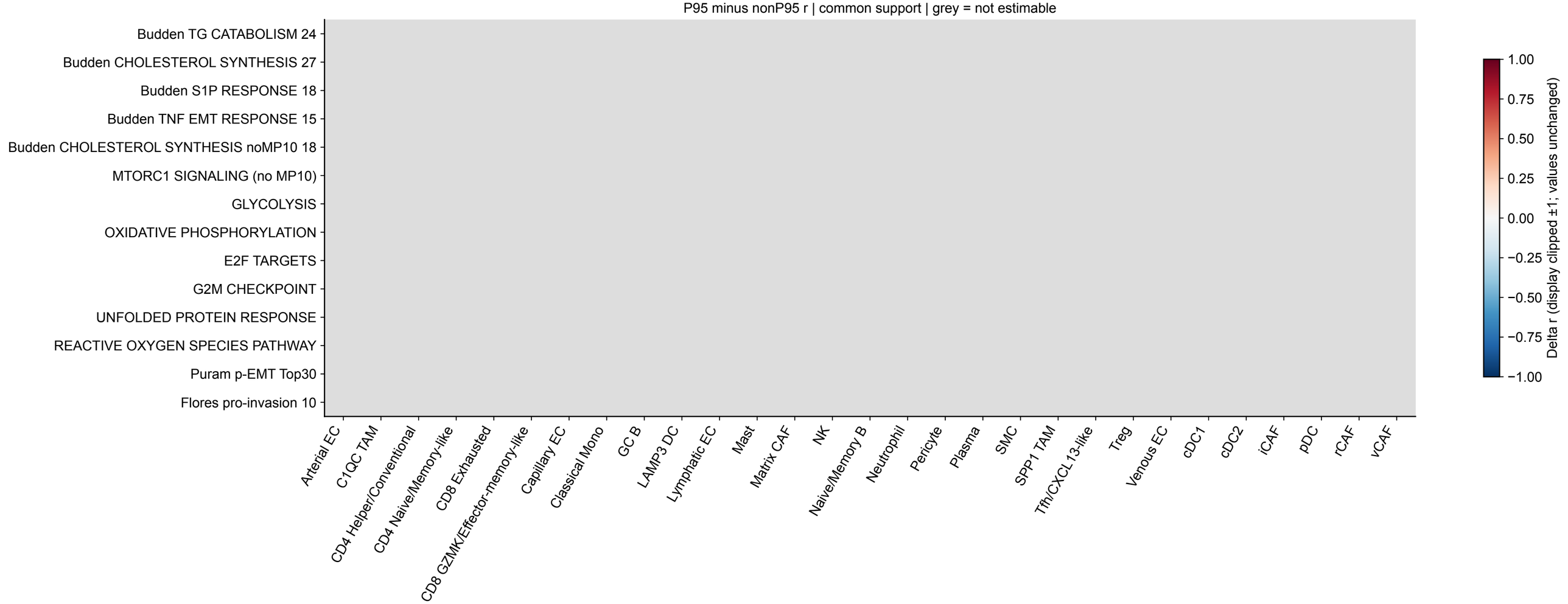
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC15 | all 29 candidates; fixed core programs

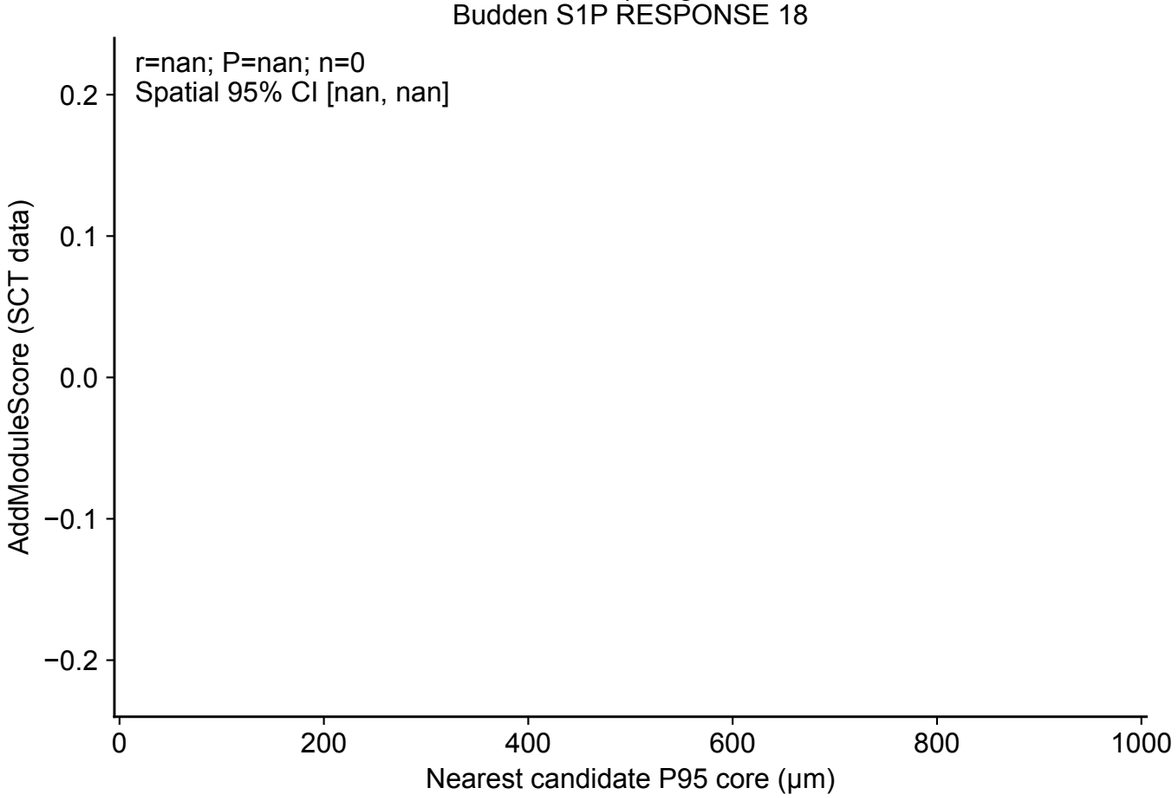
P95 raw Pearson r | full observed range

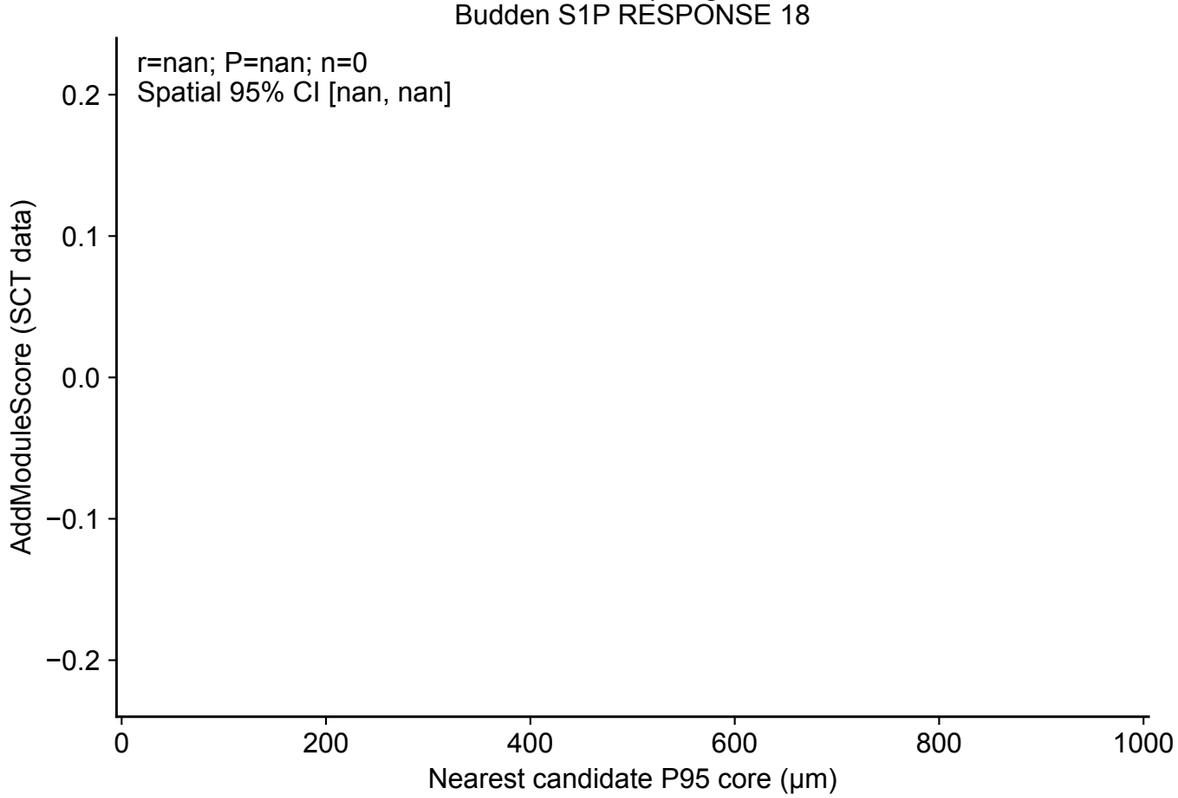


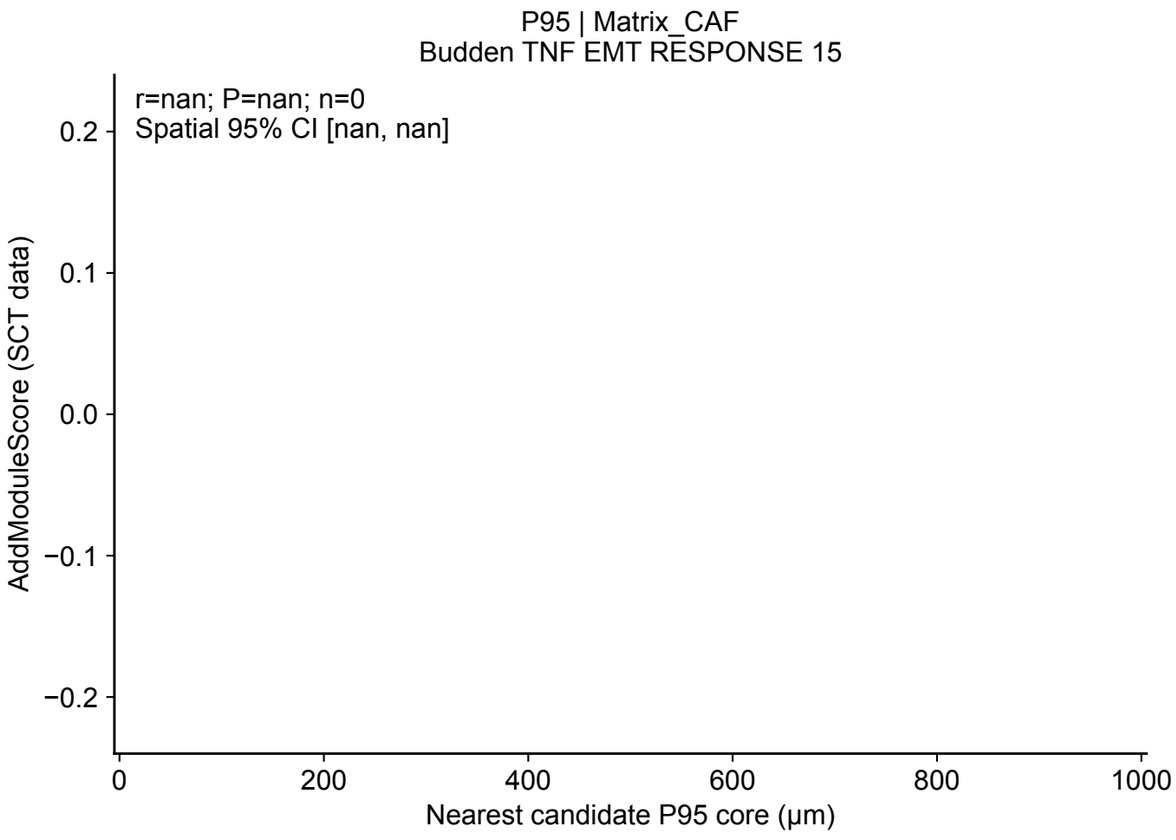
P95 minus nonP95 r | common support | grey = not estimable

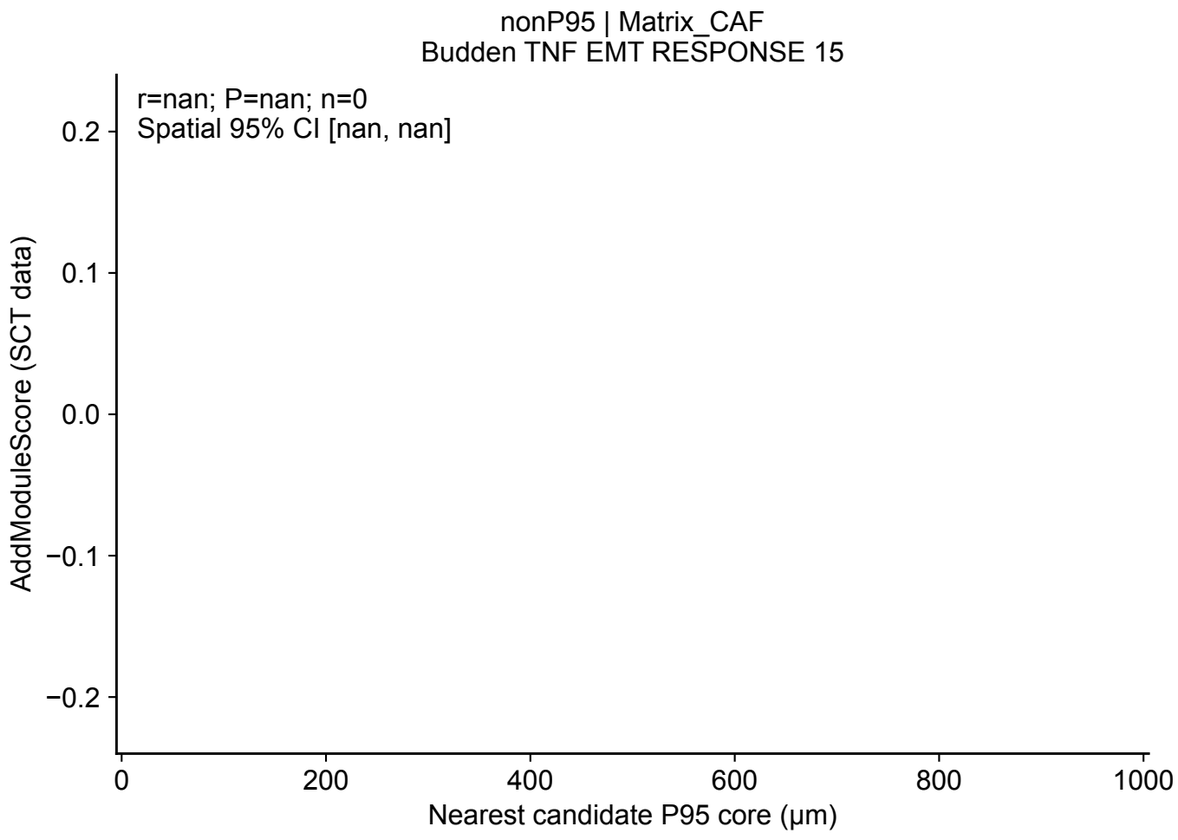


GSE319536_BC15 | exploratory scatter display selected by coverage and |delta r|, not P







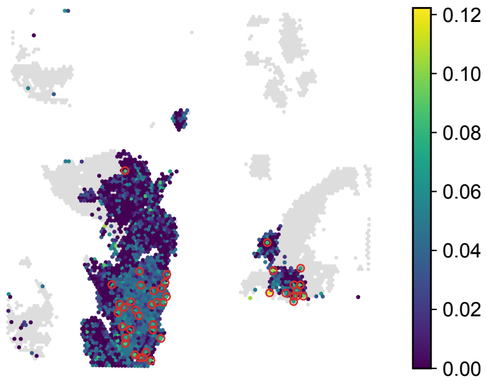


Treg / Budden S1P RESPONSE 18: delta r=nan [nan, nan]; common 346–265 μm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=nan [nan, nan]; common 361–300 μm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

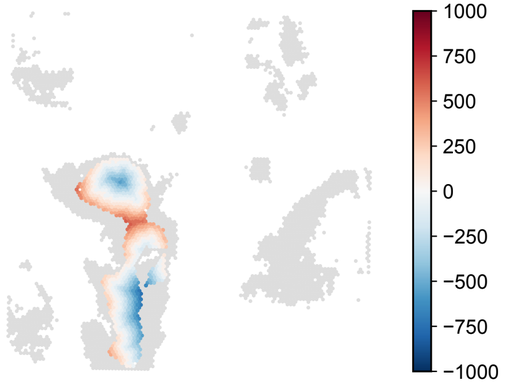
P95 nonP95 distance 0

GSE319536_BC16 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

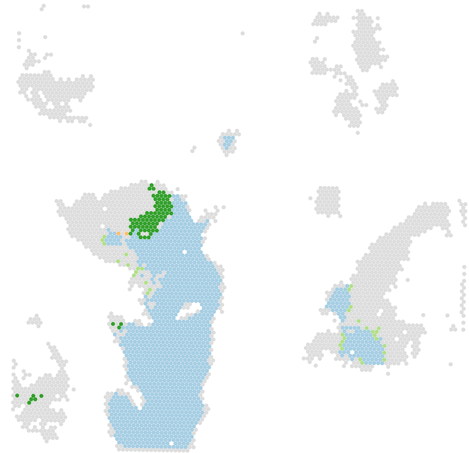
MP10 AUCell and frozen P95



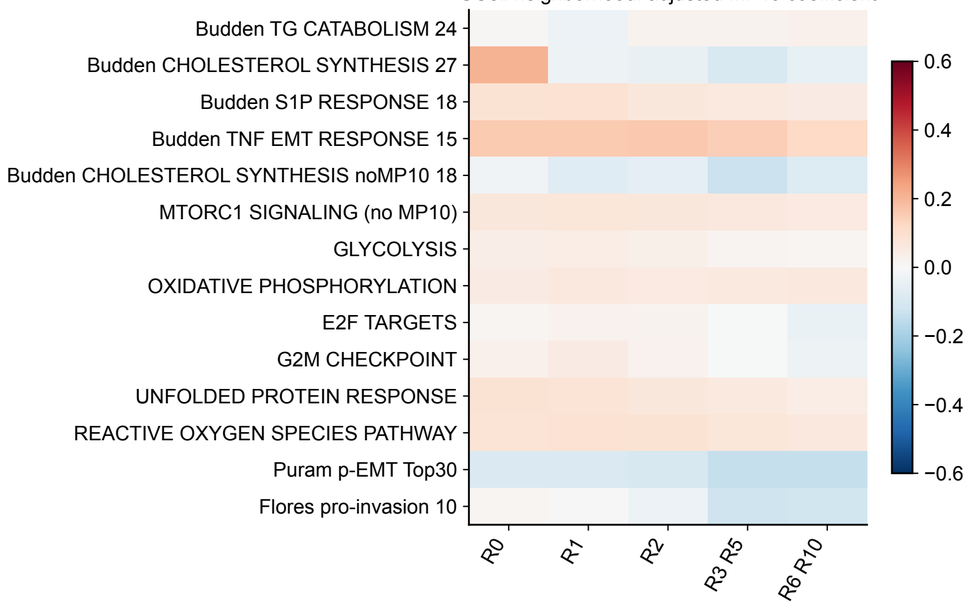
Strict v3.4 signed distance (µm)



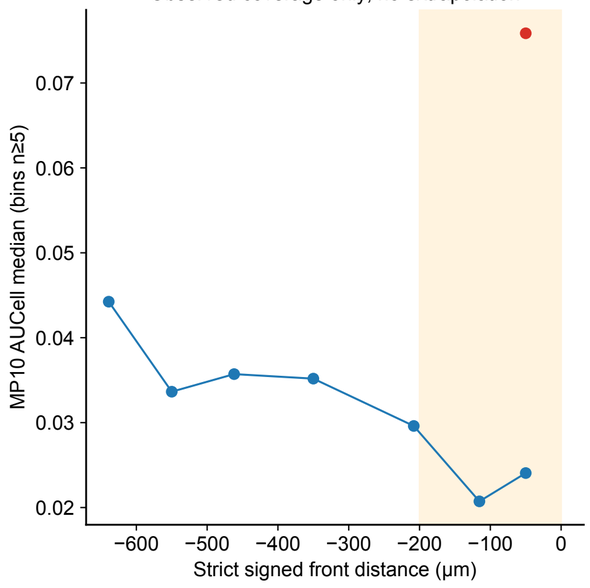
Formal R12 0–8; grey = unlabelled



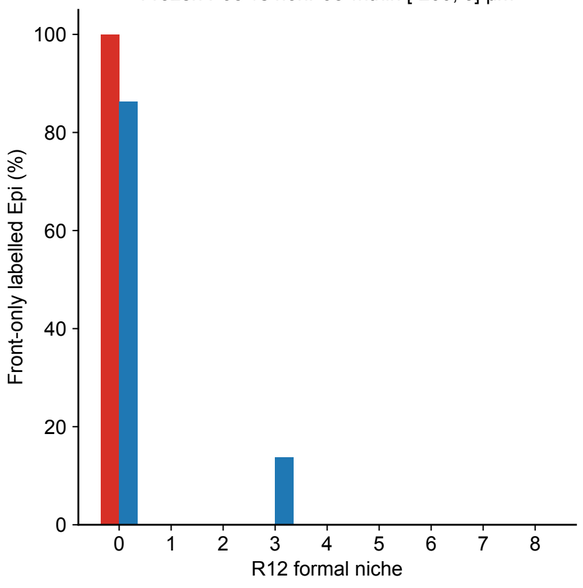
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

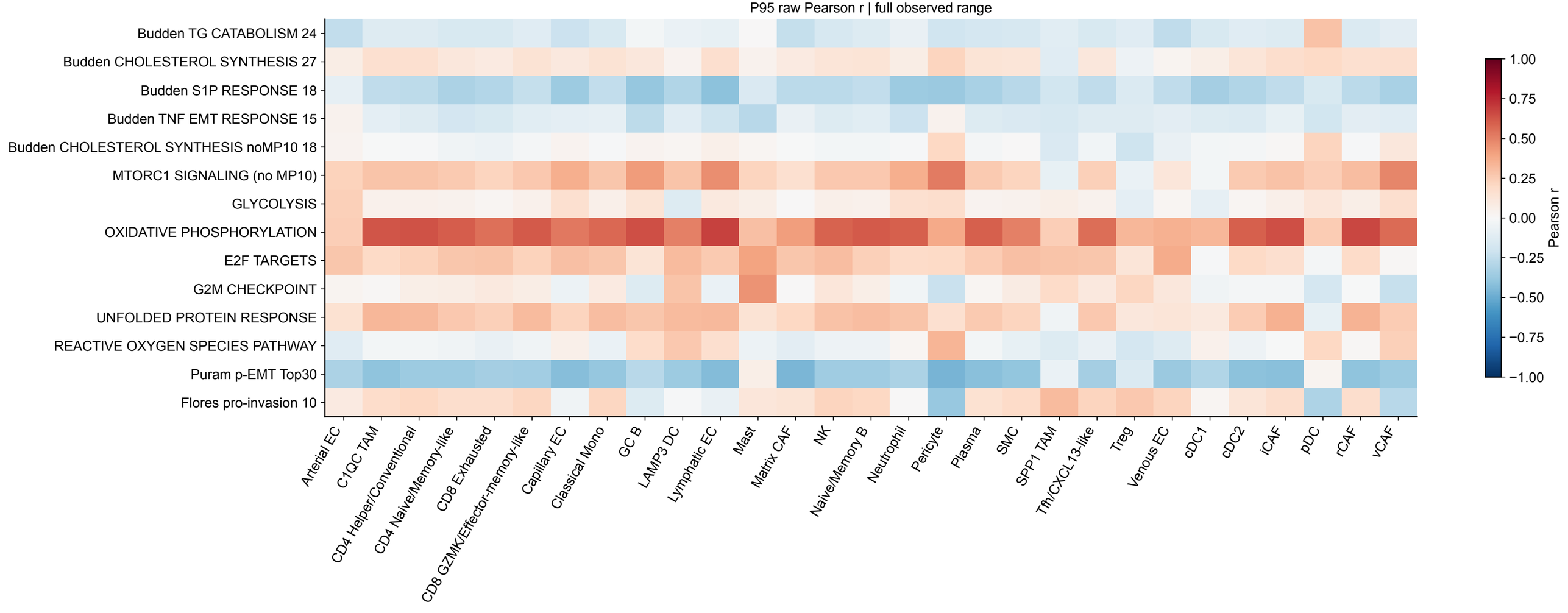


Tissue 3326 | Epi 1718 | internal 833 | P95 42
Exact front in Epi 737 | R12 labelled 1466

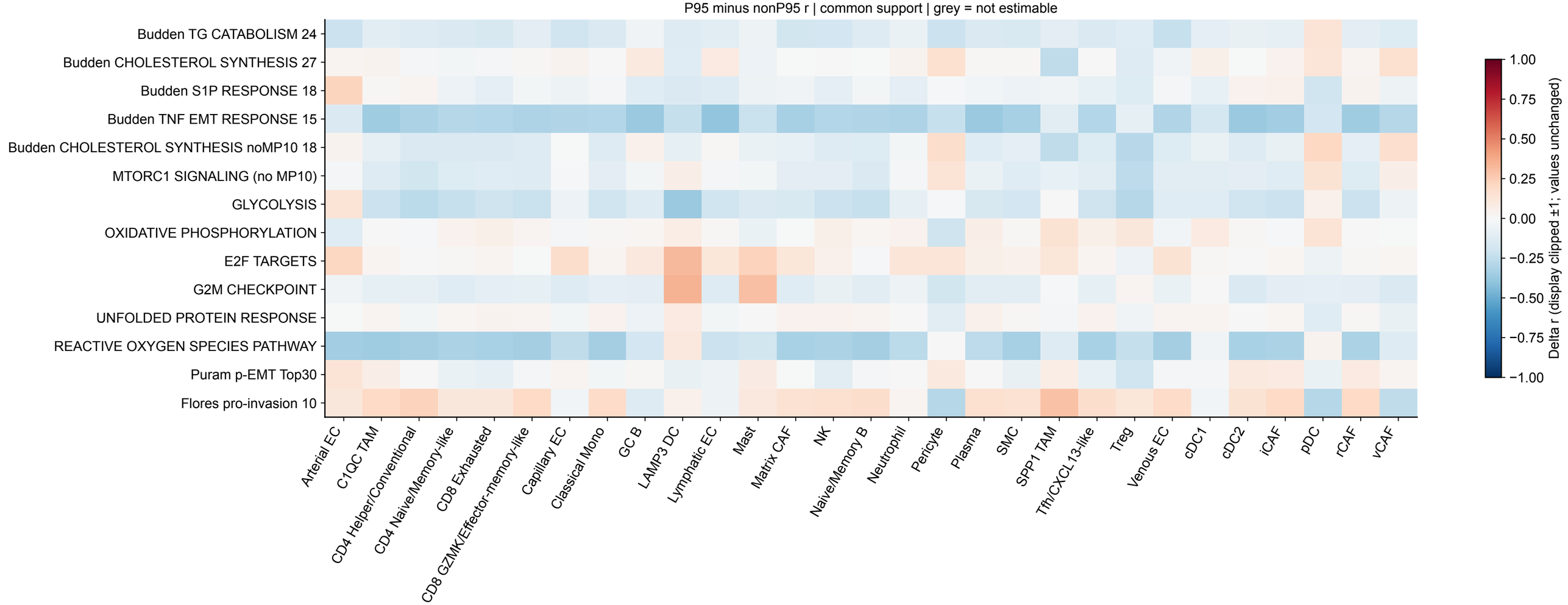
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC16 | all 29 candidates; fixed core programs

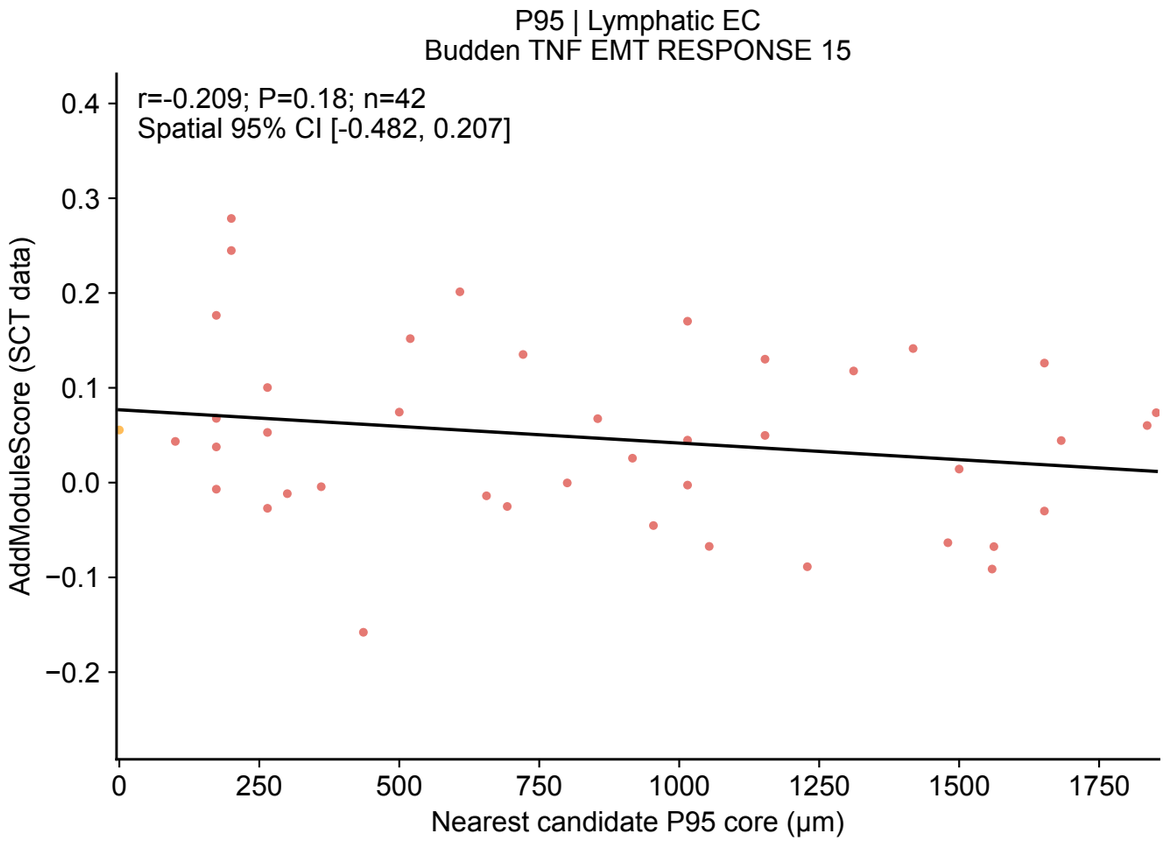
P95 raw Pearson r | full observed range

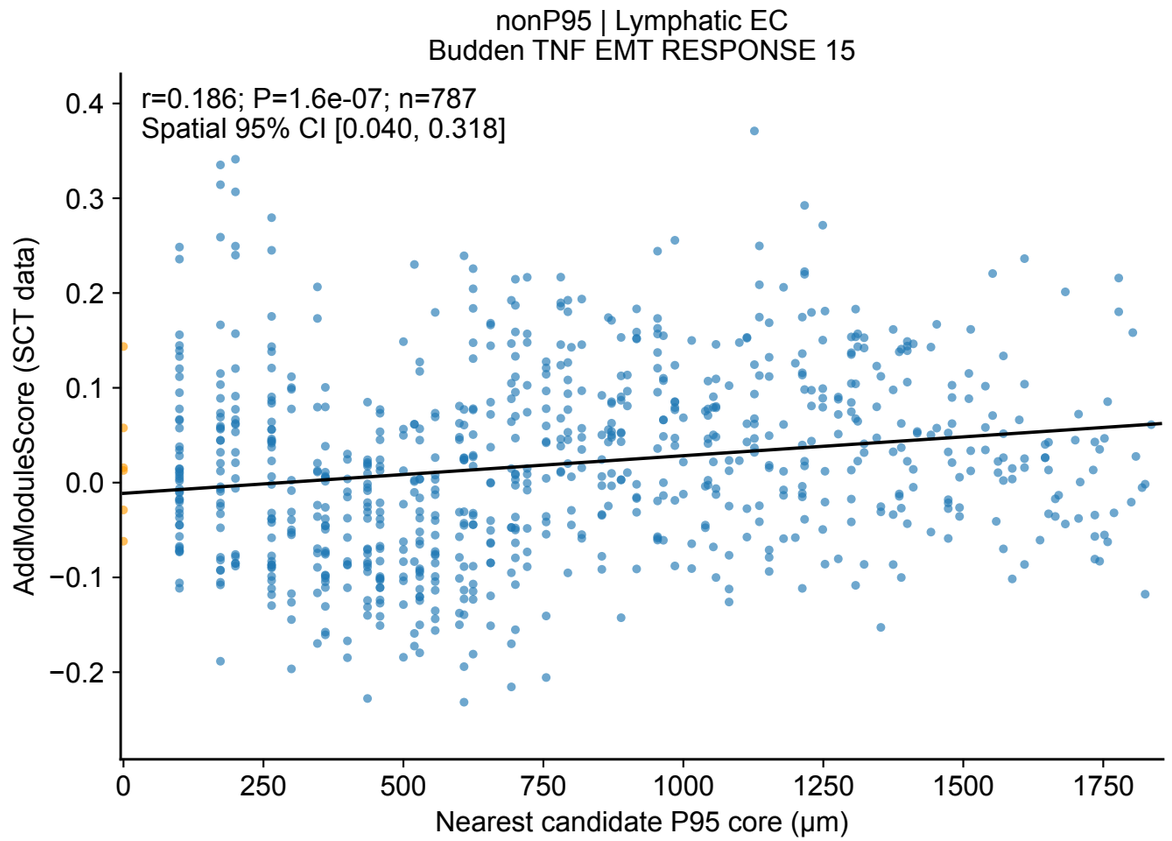


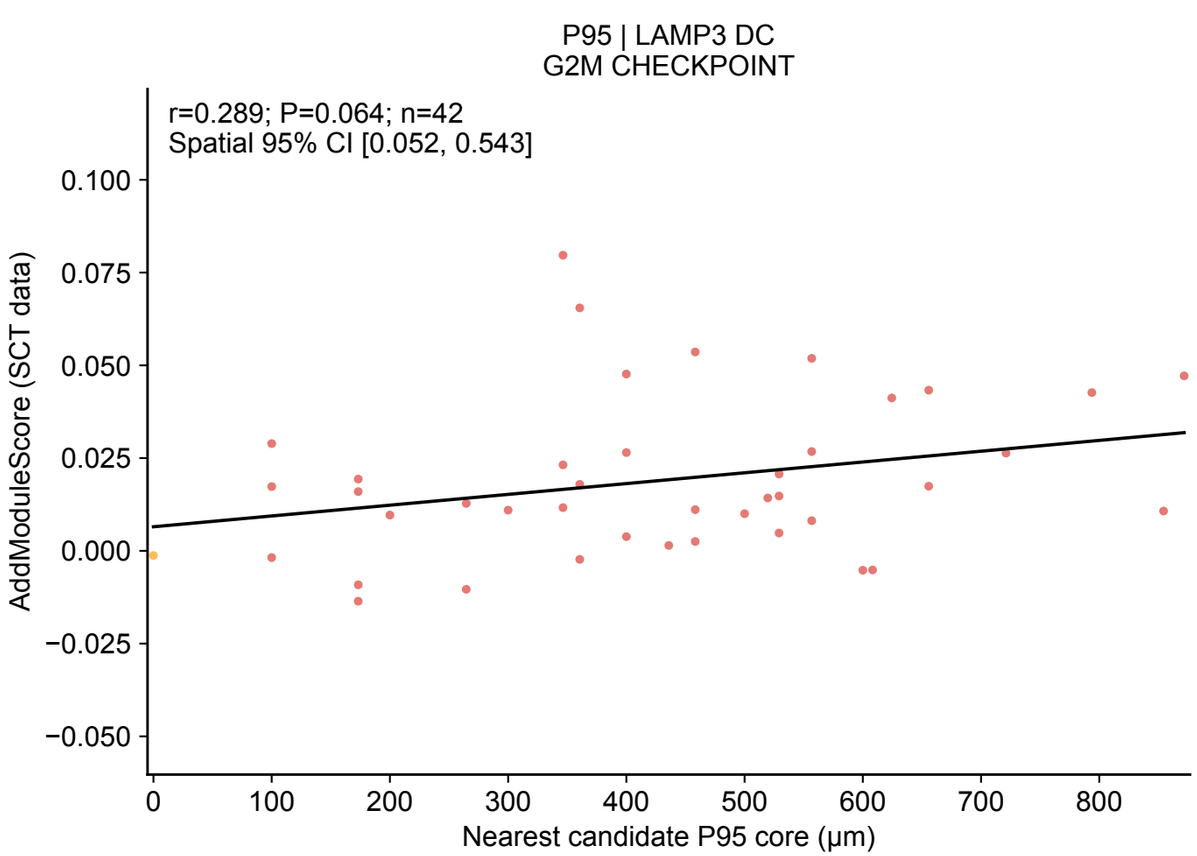
P95 minus nonP95 r | common support | grey = not estimable

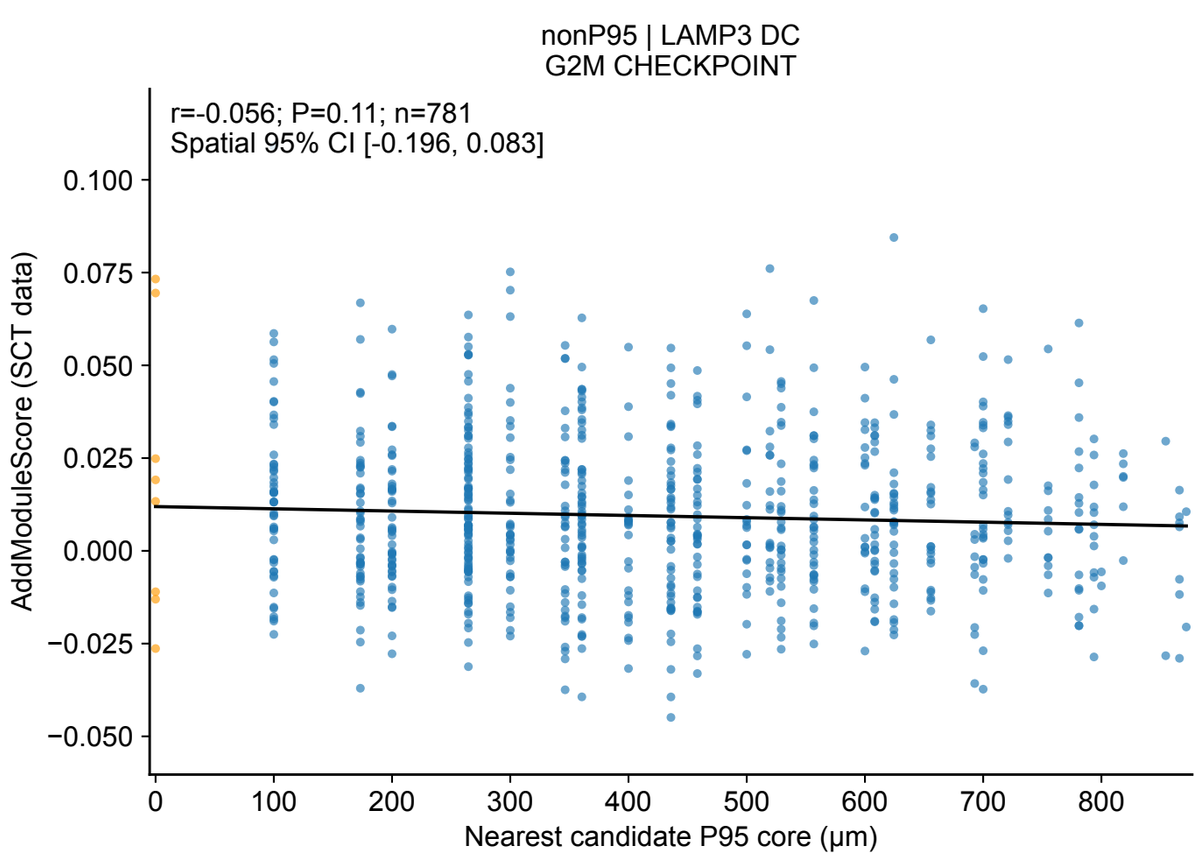


GSE319536_BC16 | exploratory scatter display selected by coverage and |delta r|, not P







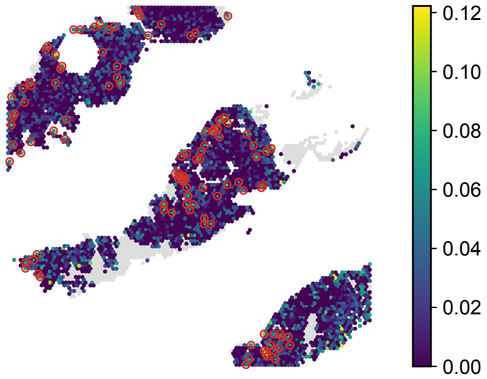


Lymphatic EC / Budden TNF EMT RESPONSE 15: delta r=-0.395 [-0.658, 0.014]; common 0–1852 µm; spatial P=0.019
LAMP3 DC / G2M CHECKPOINT: delta r=0.345 [0.073, 0.644]; common 0–872 µm; spatial P=0.024
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

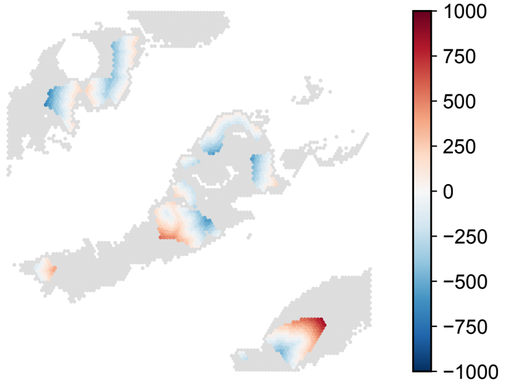
P95 nonP95 distance 0

GSE319536_BC17 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

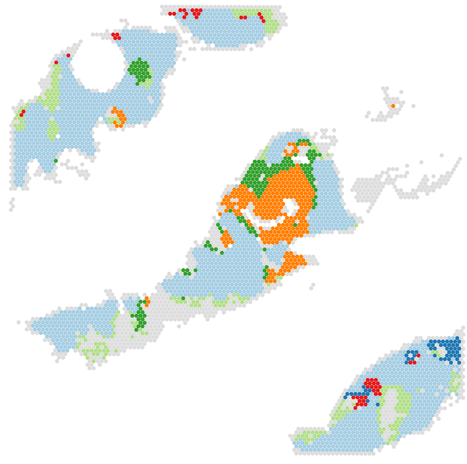
MP10 AUCell and frozen P95



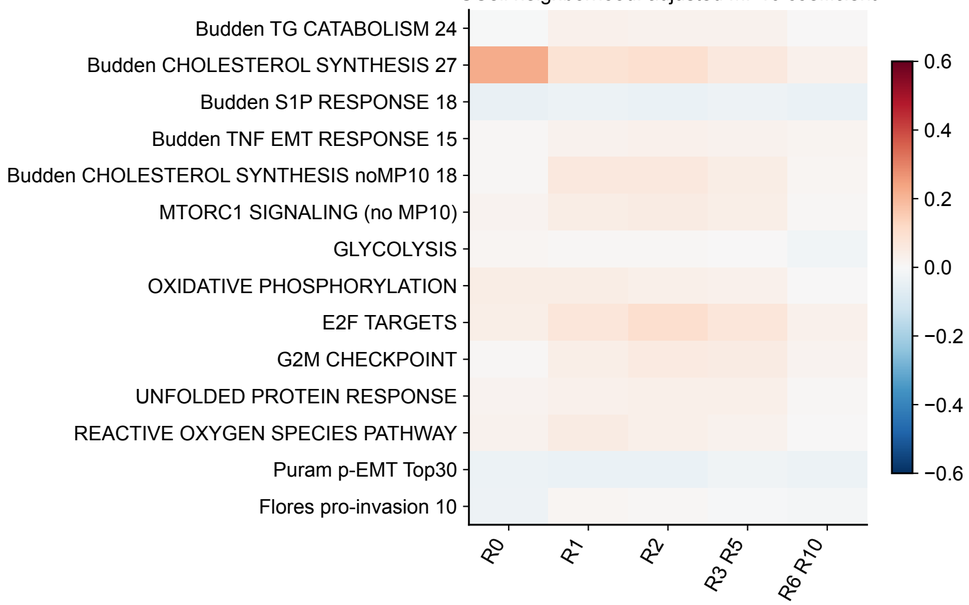
Strict v3.4 signed distance (µm)



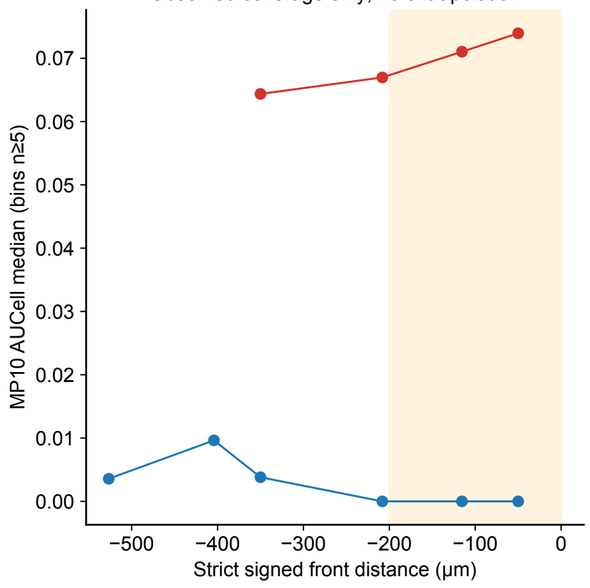
Formal R12 0–8; grey = unlabelled



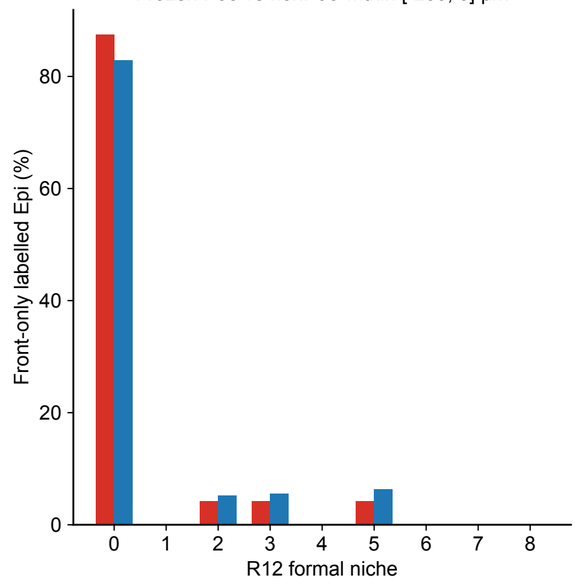
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

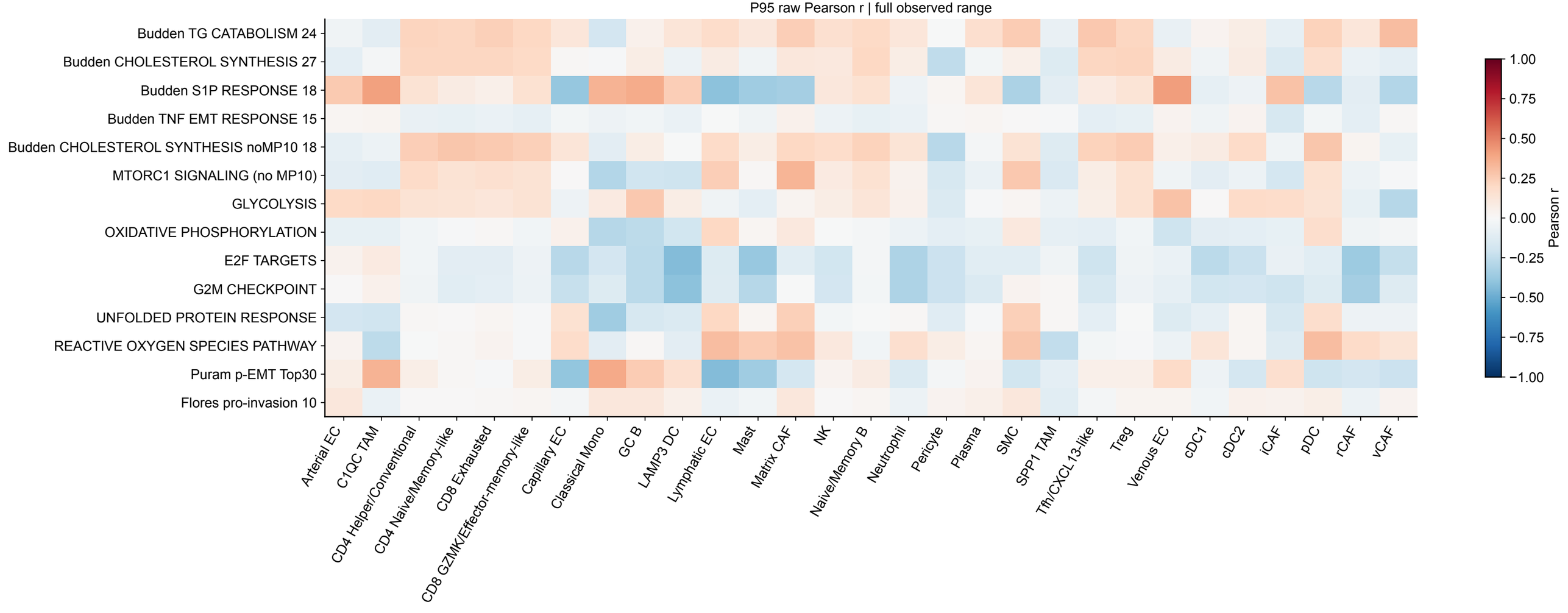


Tissue 4072 | Epi 3508 | internal 2130 | P95 107
Exact front in Epi 877 | R12 labelled 2881

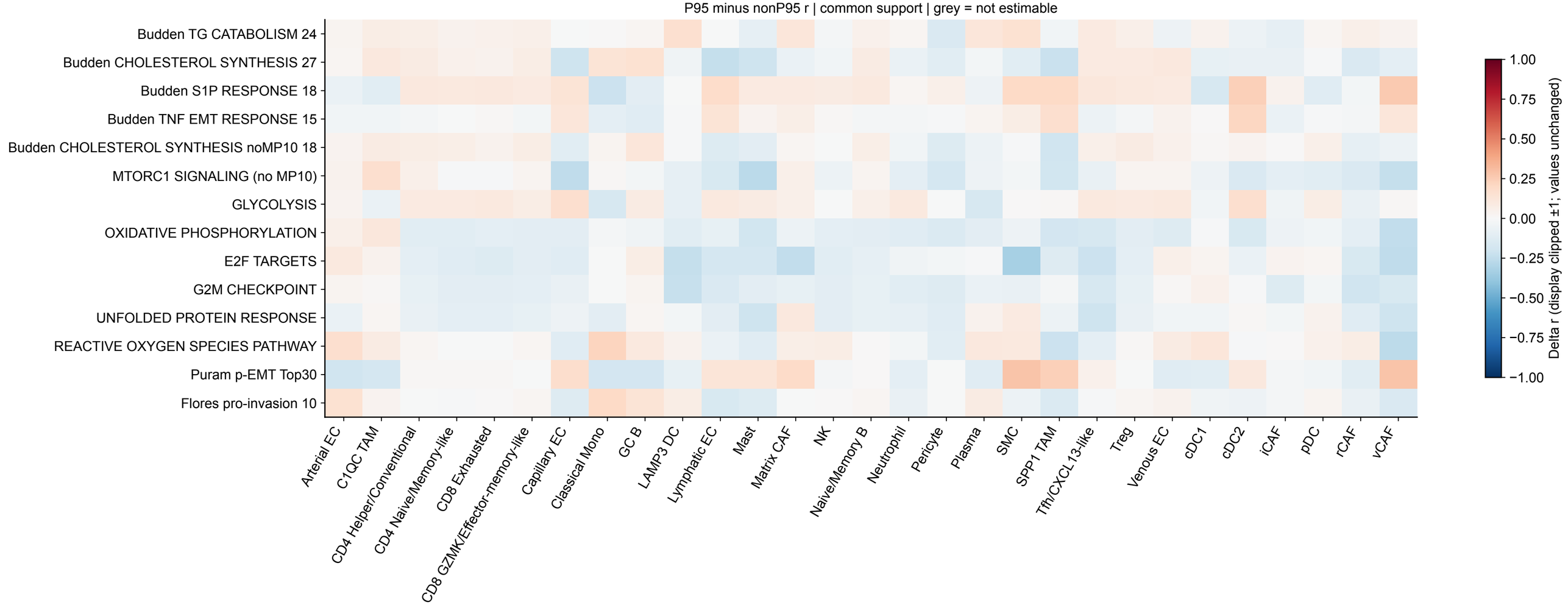
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC17 | all 29 candidates; fixed core programs

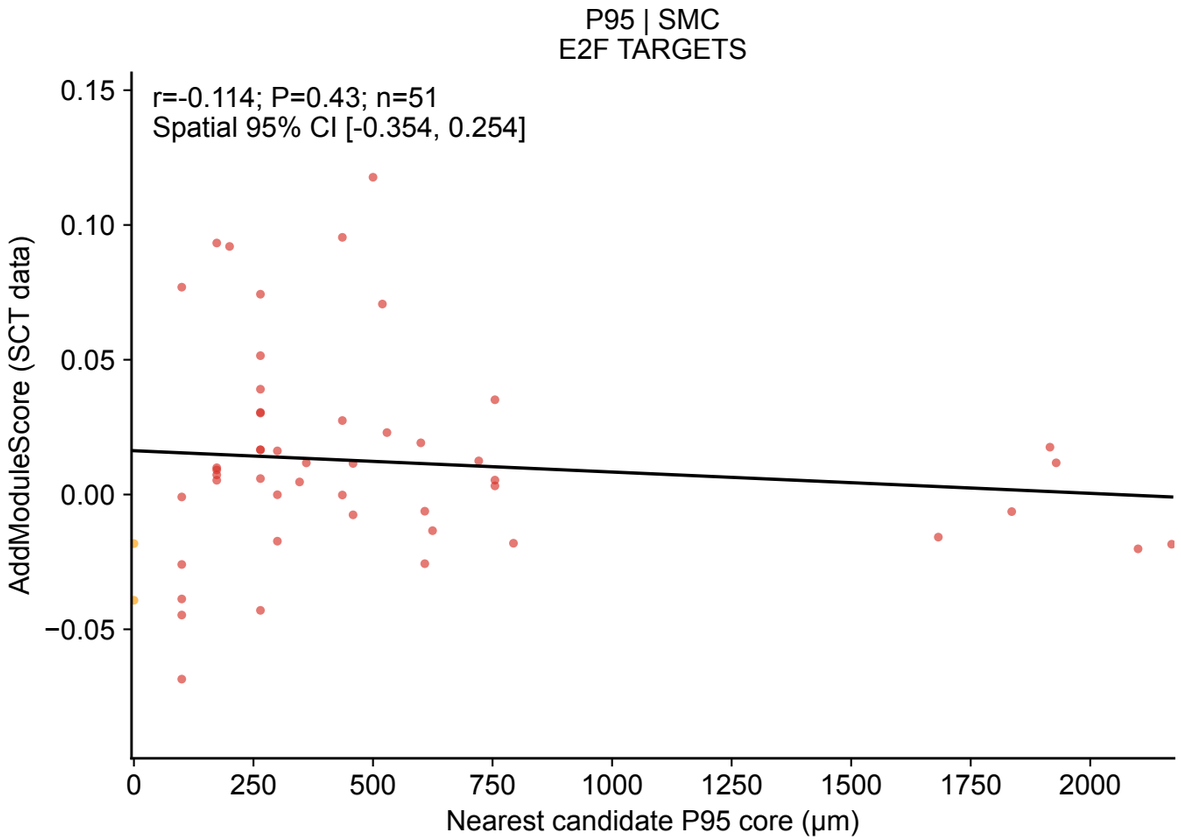
P95 raw Pearson r | full observed range

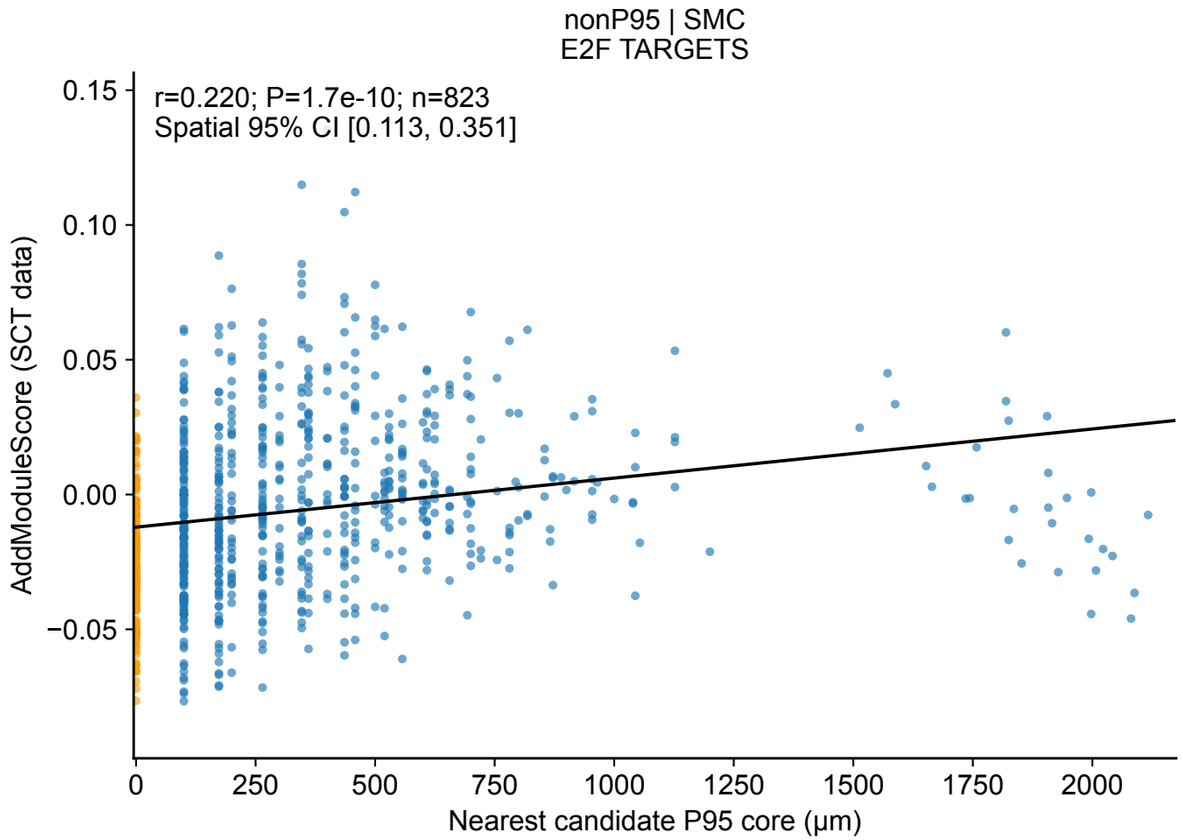


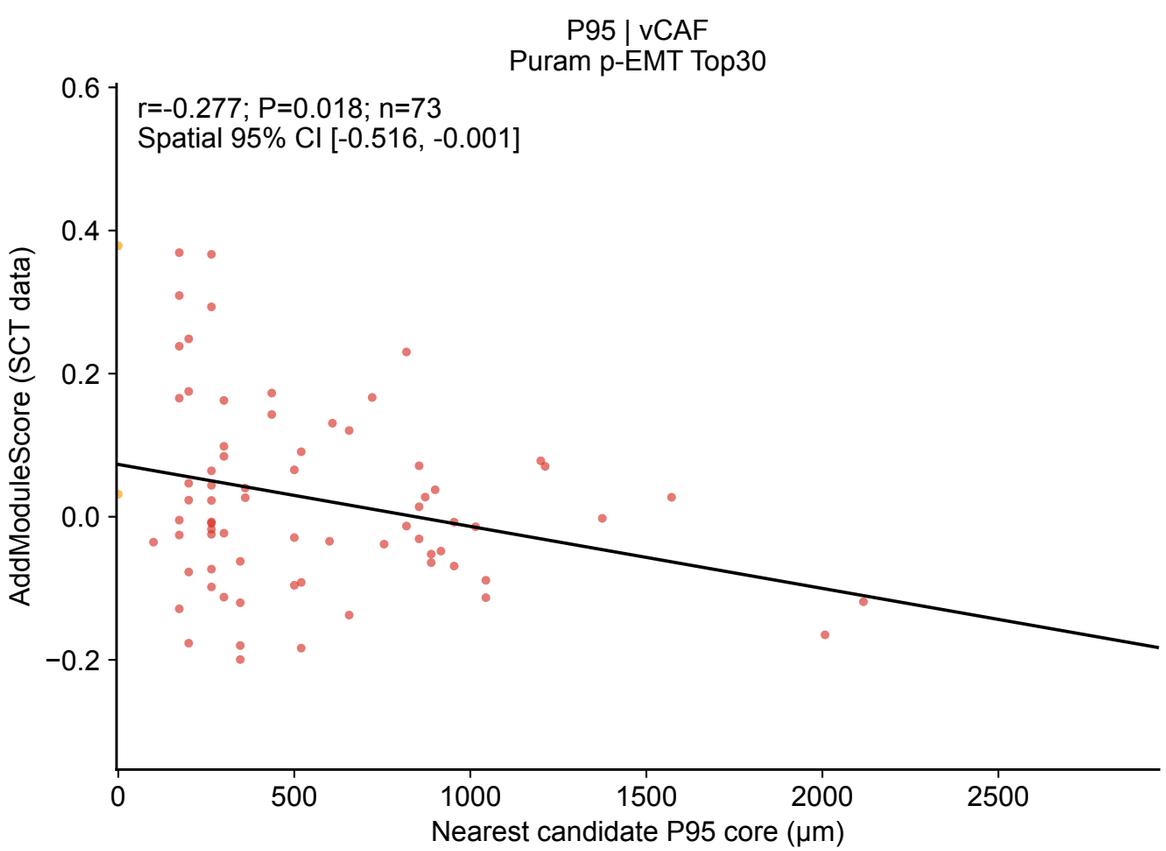
P95 minus nonP95 r | common support | grey = not estimable

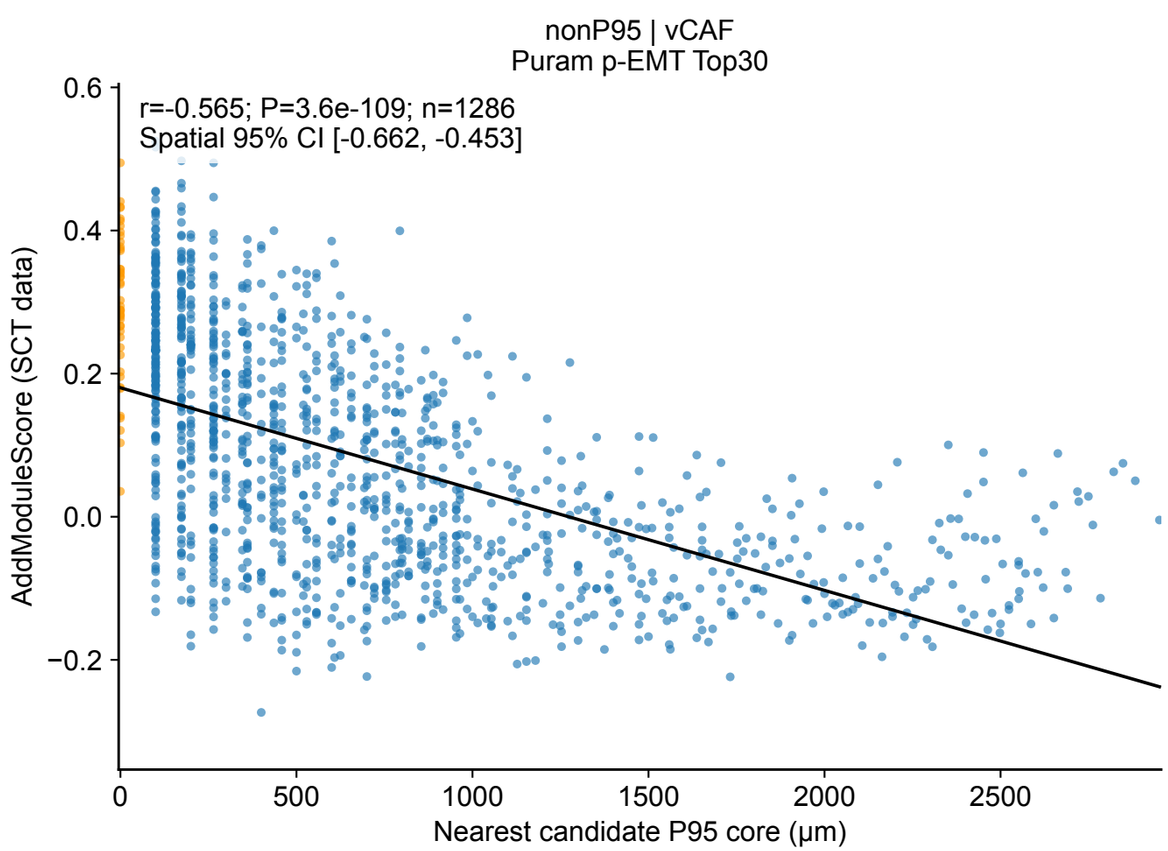


GSE319536_BC17 | exploratory scatter display selected by coverage and |delta r|, not P







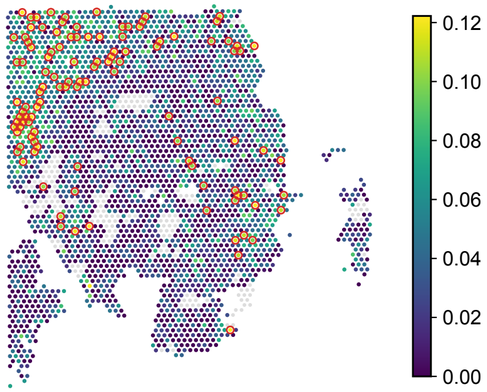


SMC / E2F TARGETS: delta r=-0.334 [-0.540, -0.037]; common 0–2170 μm; spatial P=0.0026
vCAF / Puram p-EMT Top30: delta r=0.287 [0.067, 0.551]; common 0–2951 μm; spatial P=0.0063
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

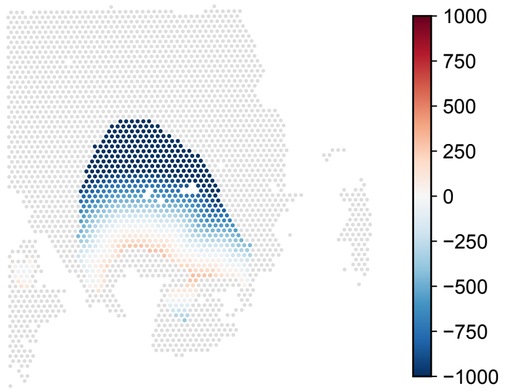
P95 nonP95 distance 0

GSE319536_BC18 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

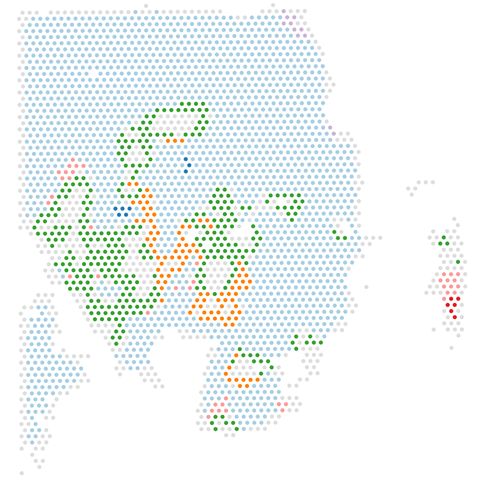
MP10 AUCell and frozen P95



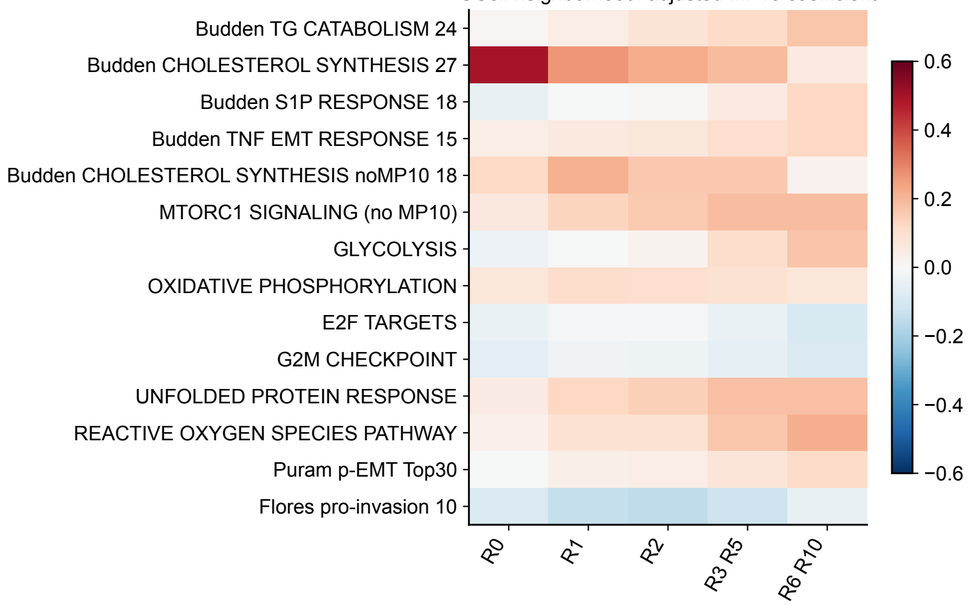
Strict v3.4 signed distance (µm)



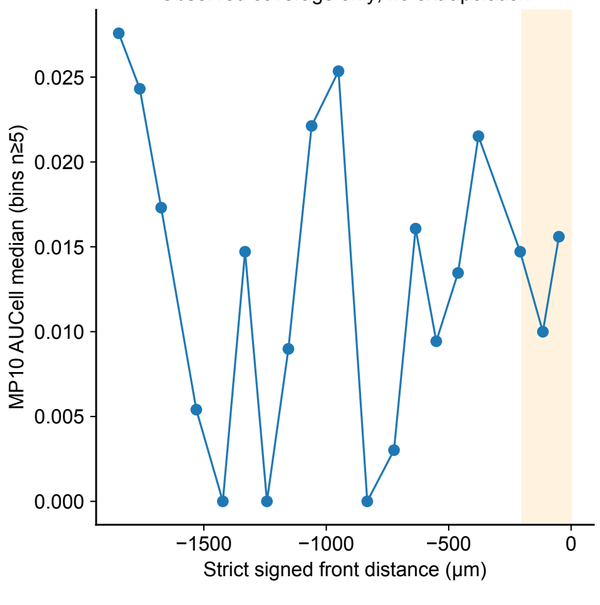
Formal R12 0–8; grey = unlabelled



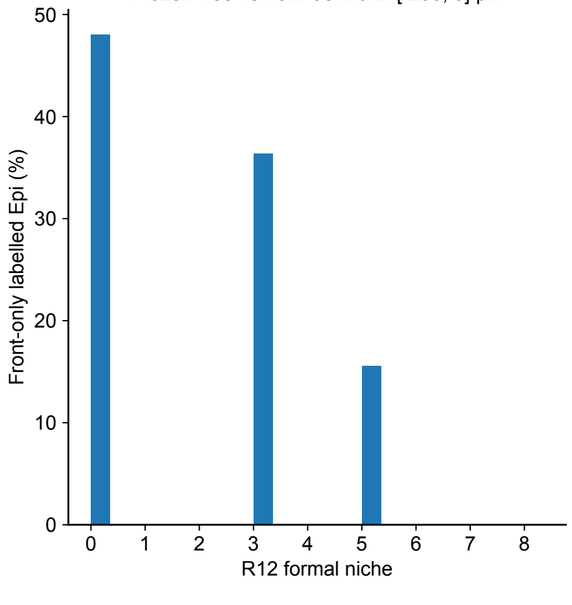
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

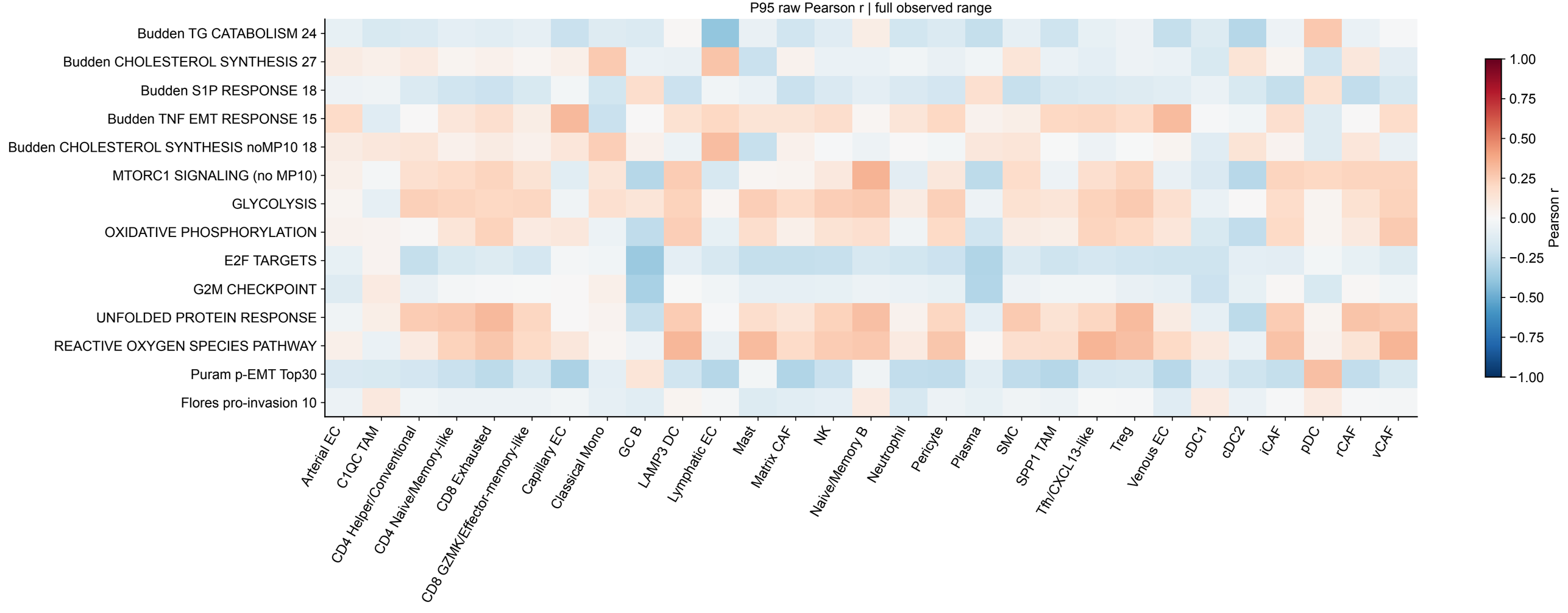


Tissue 2847 | Epi 2656 | internal 2204 | P95 111
Exact front in Epi 649 | R12 labelled 2297

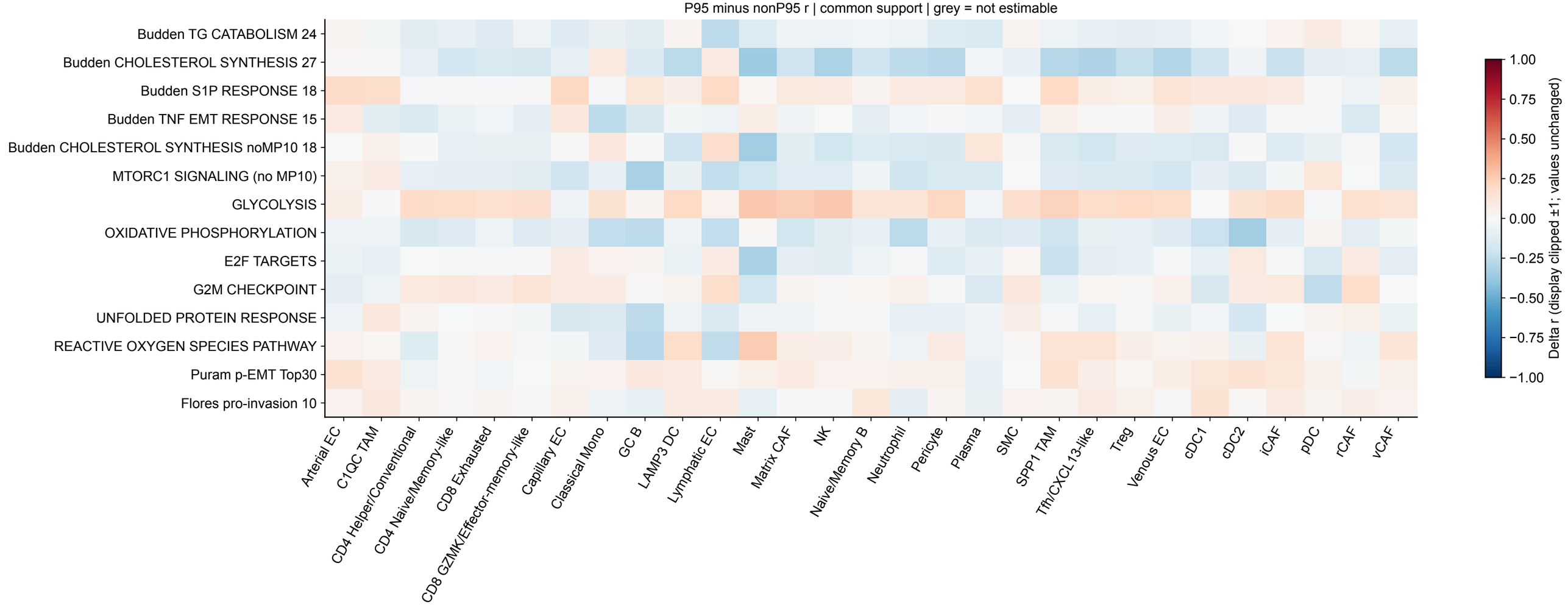
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE319536_BC18 | all 29 candidates; fixed core programs

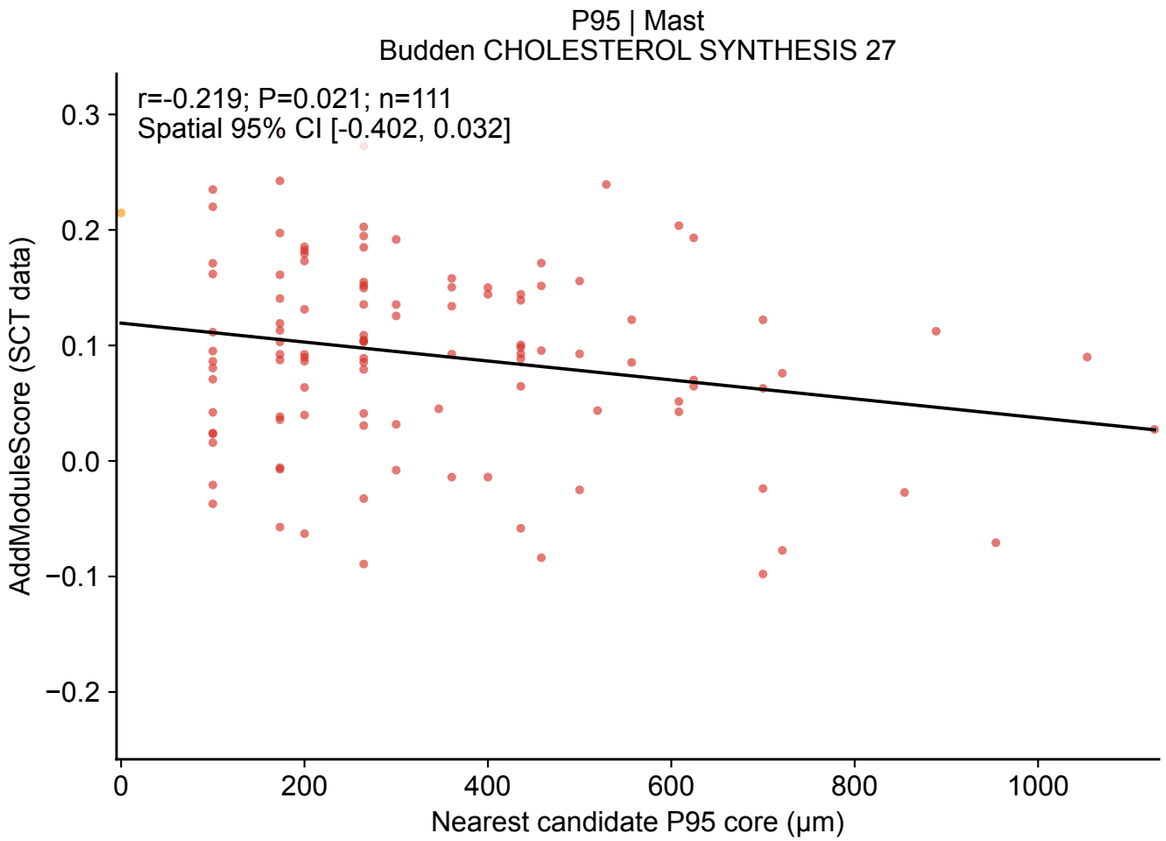
P95 raw Pearson r | full observed range

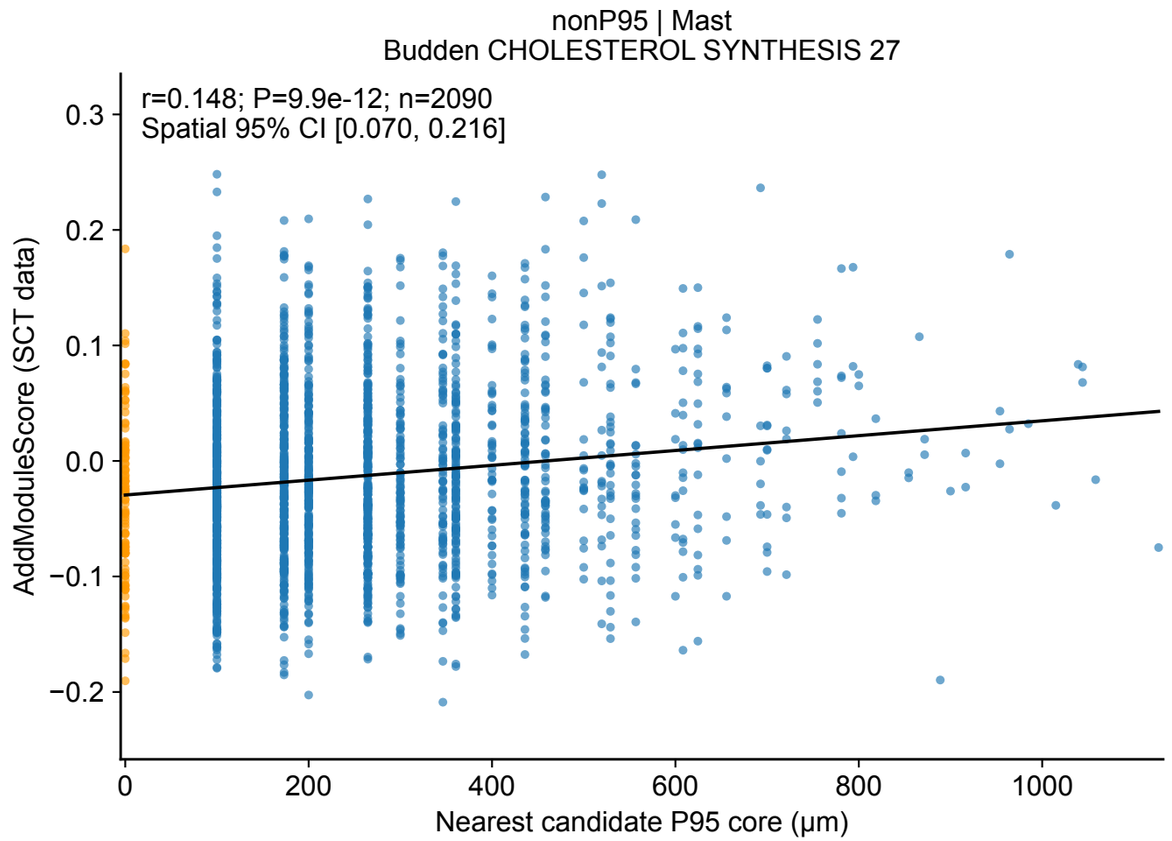


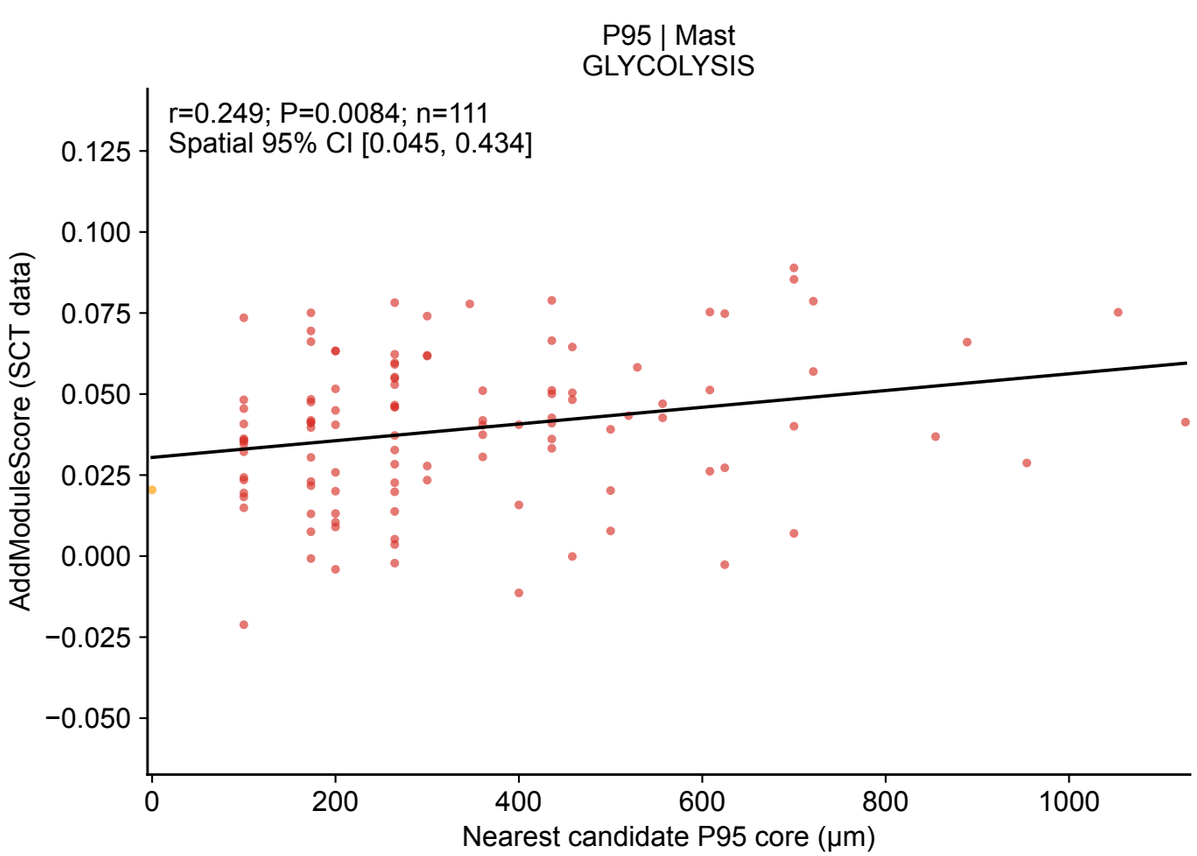
P95 minus nonP95 r | common support | grey = not estimable

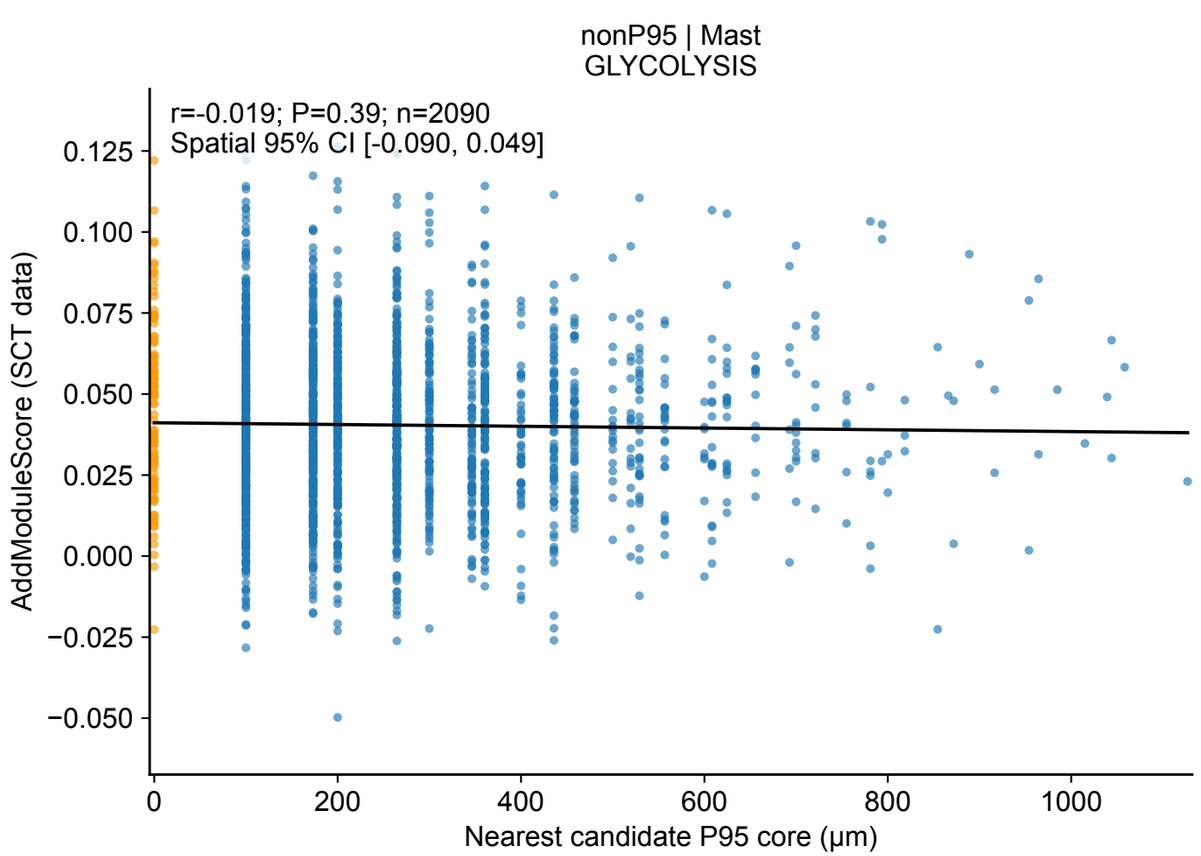


GSE319536_BC18 | exploratory scatter display selected by coverage and |delta r|, not P





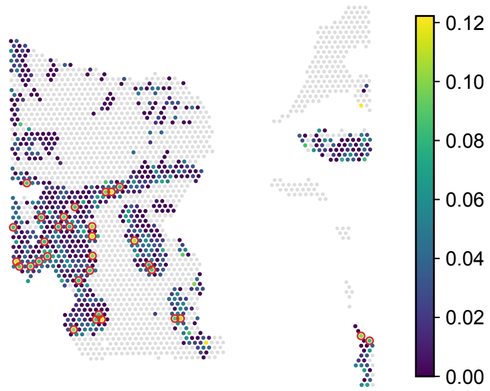


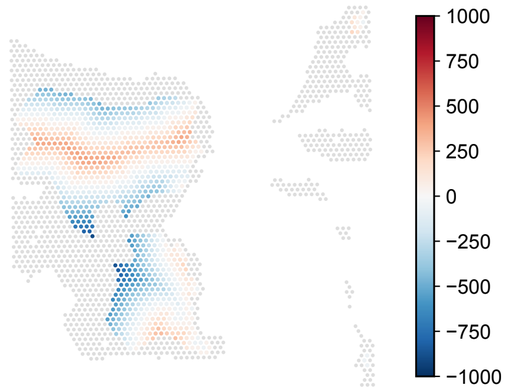


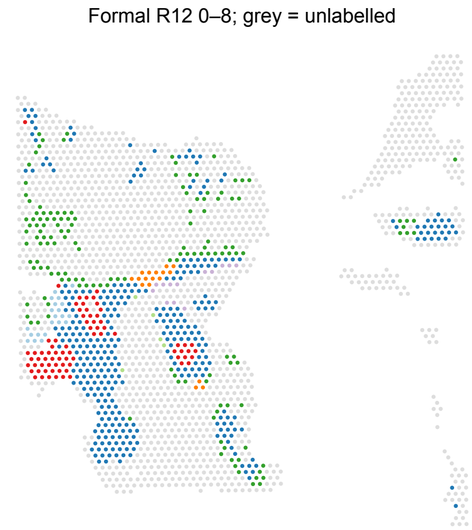
Mast / Budden CHOLESTEROL SYNTHESIS 27: delta r=-0.367 [-0.557, -0.099]; common 0–1127 µm; spatial P=0.0018
Mast / GLYCOLYSIS: delta r=0.268 [0.078, 0.439]; common 0–1127 µm; spatial P=0.0054
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

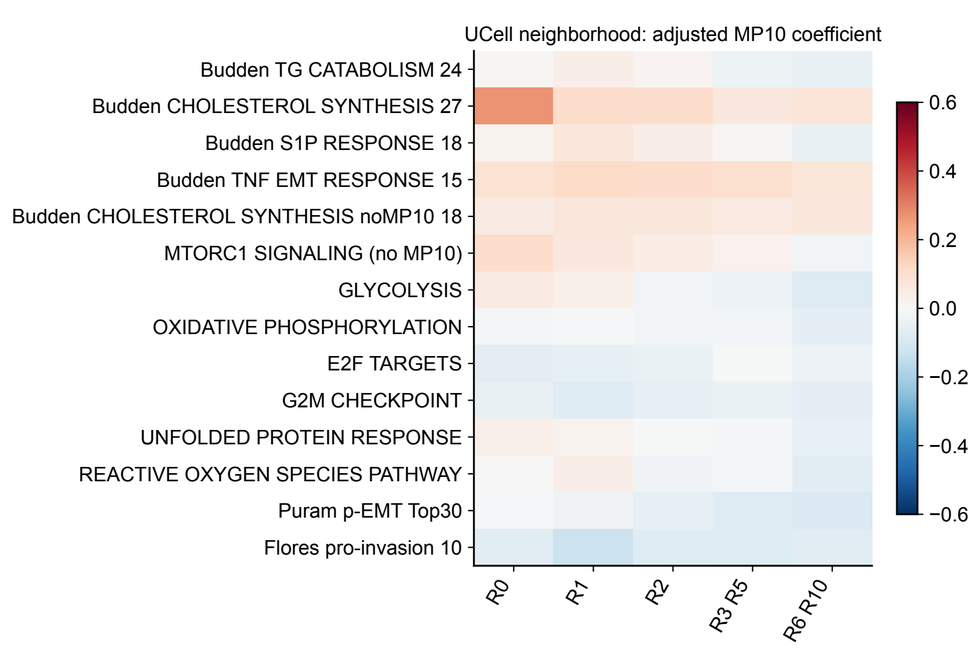
P95 nonP95 distance 0

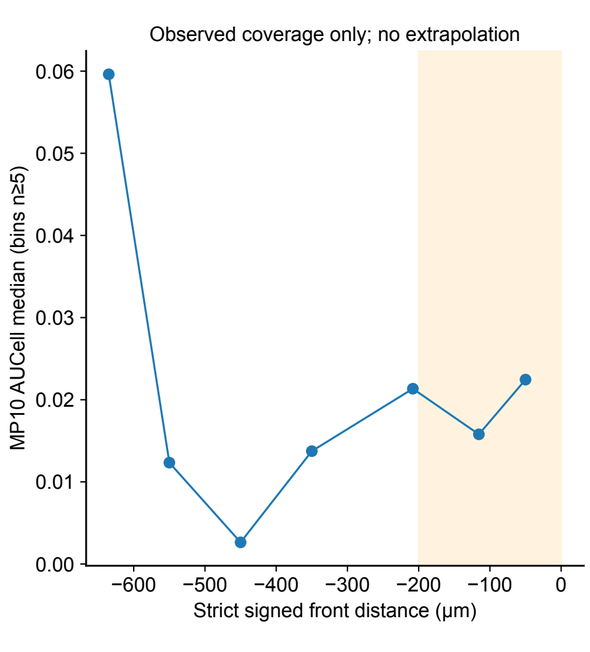
GSE319536_BC19 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

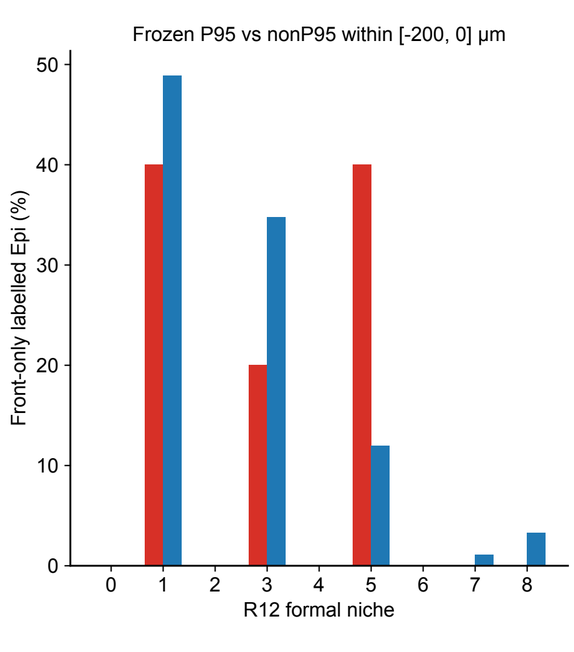








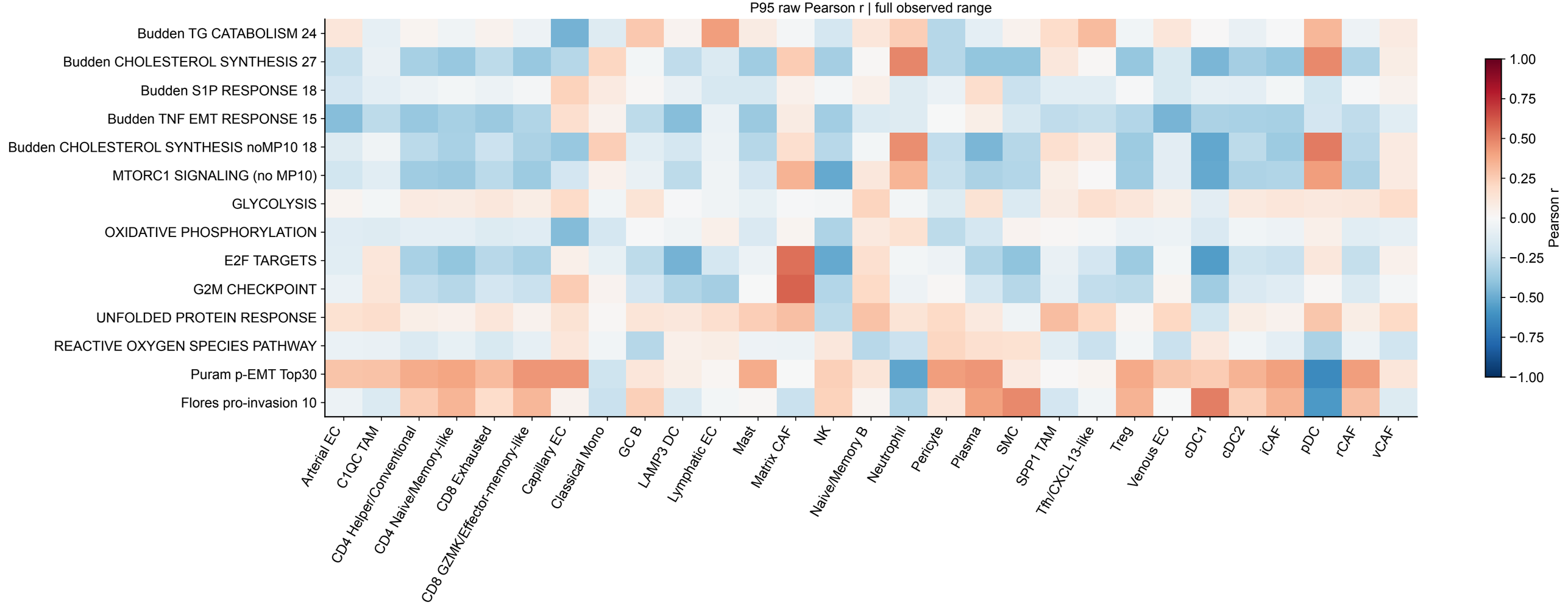




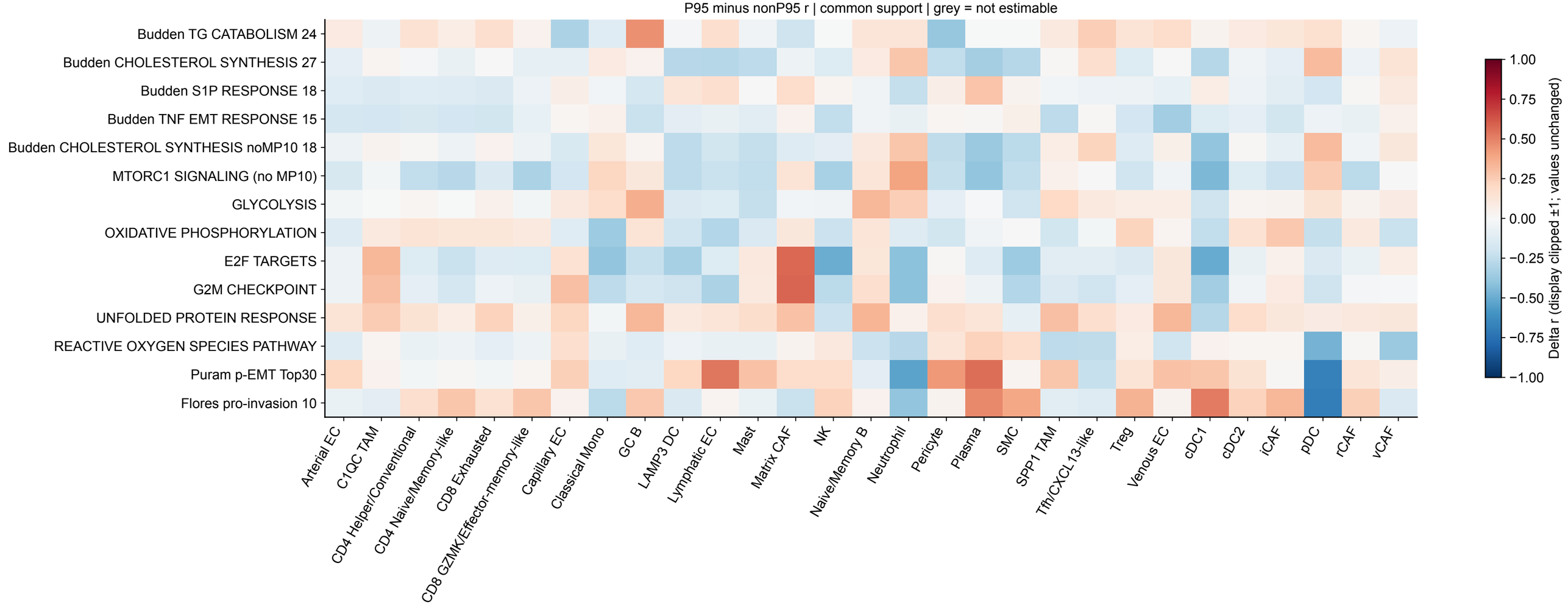
Tissue 2090 | Epi 784 | internal 624 | P95 32
Exact front in Epi 290 | R12 labelled 620

GSE319536_BC19 | all 29 candidates; fixed core programs

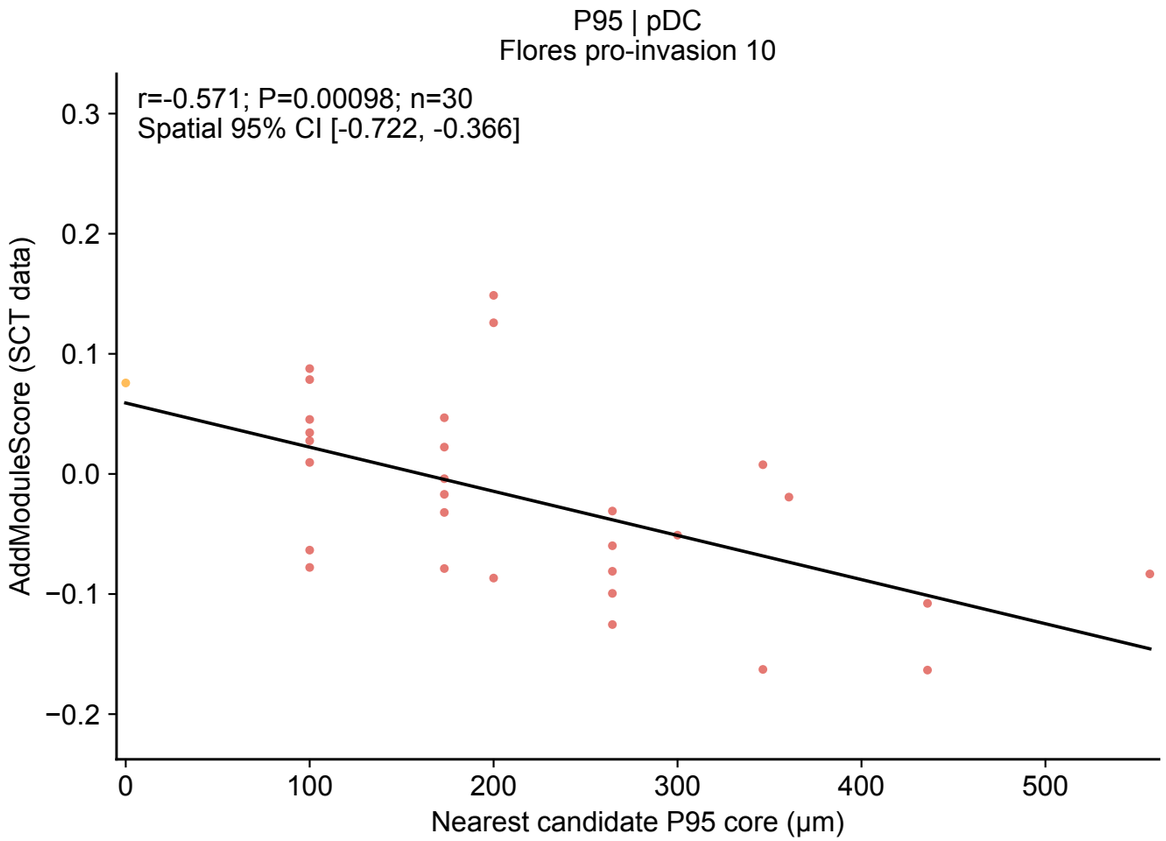
P95 raw Pearson r | full observed range

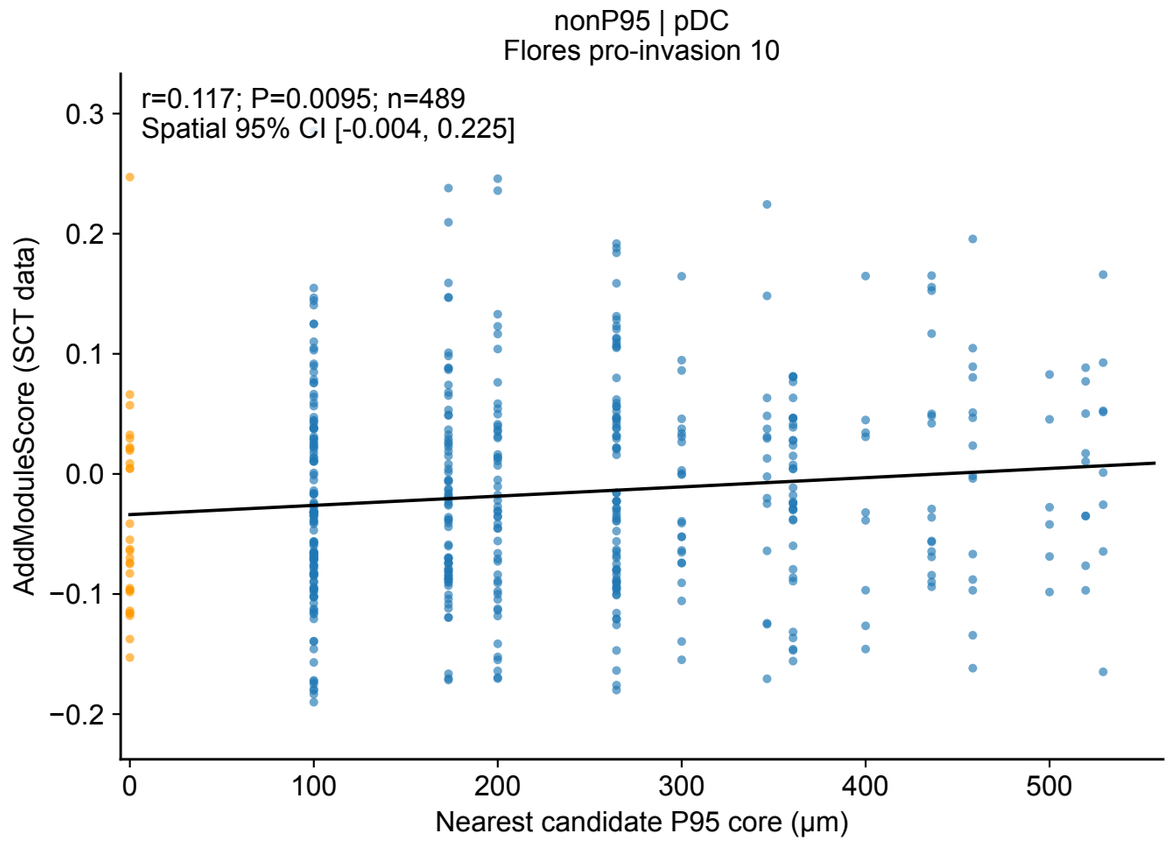


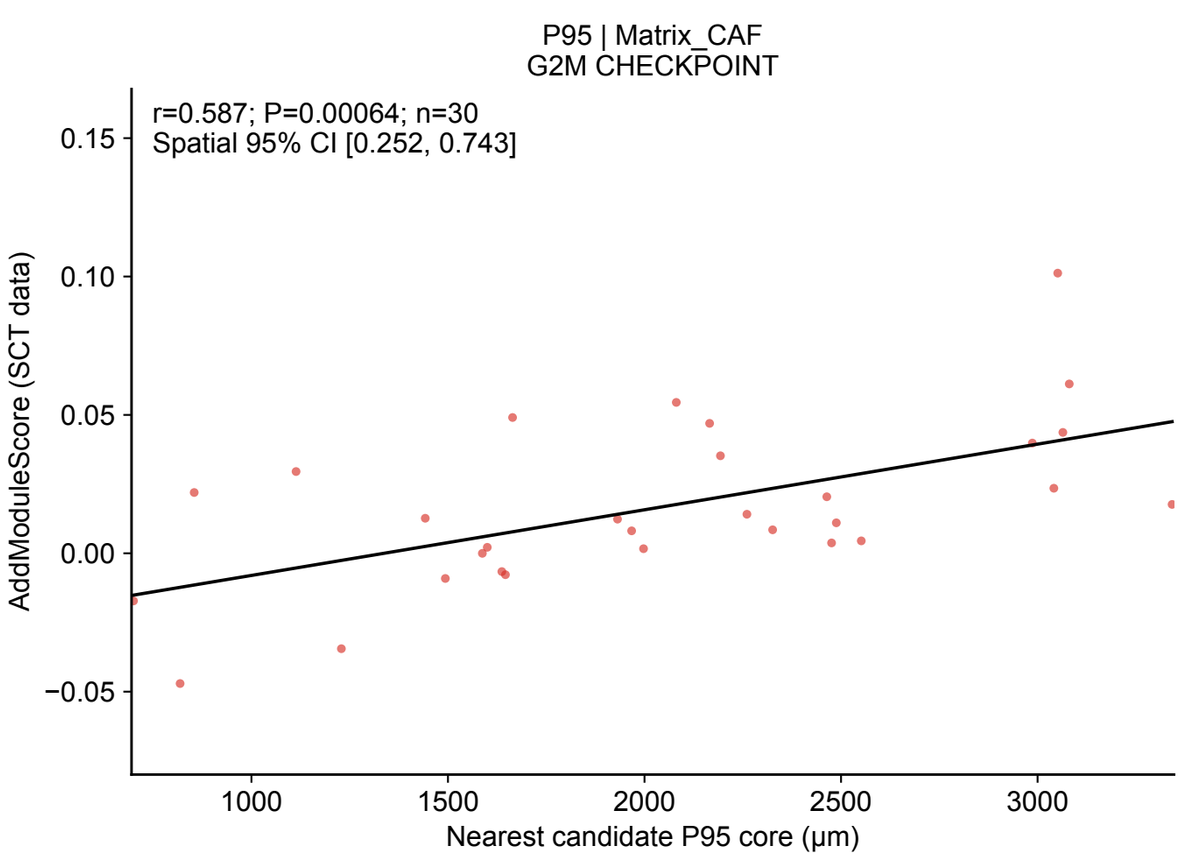
P95 minus nonP95 r | common support | grey = not estimable

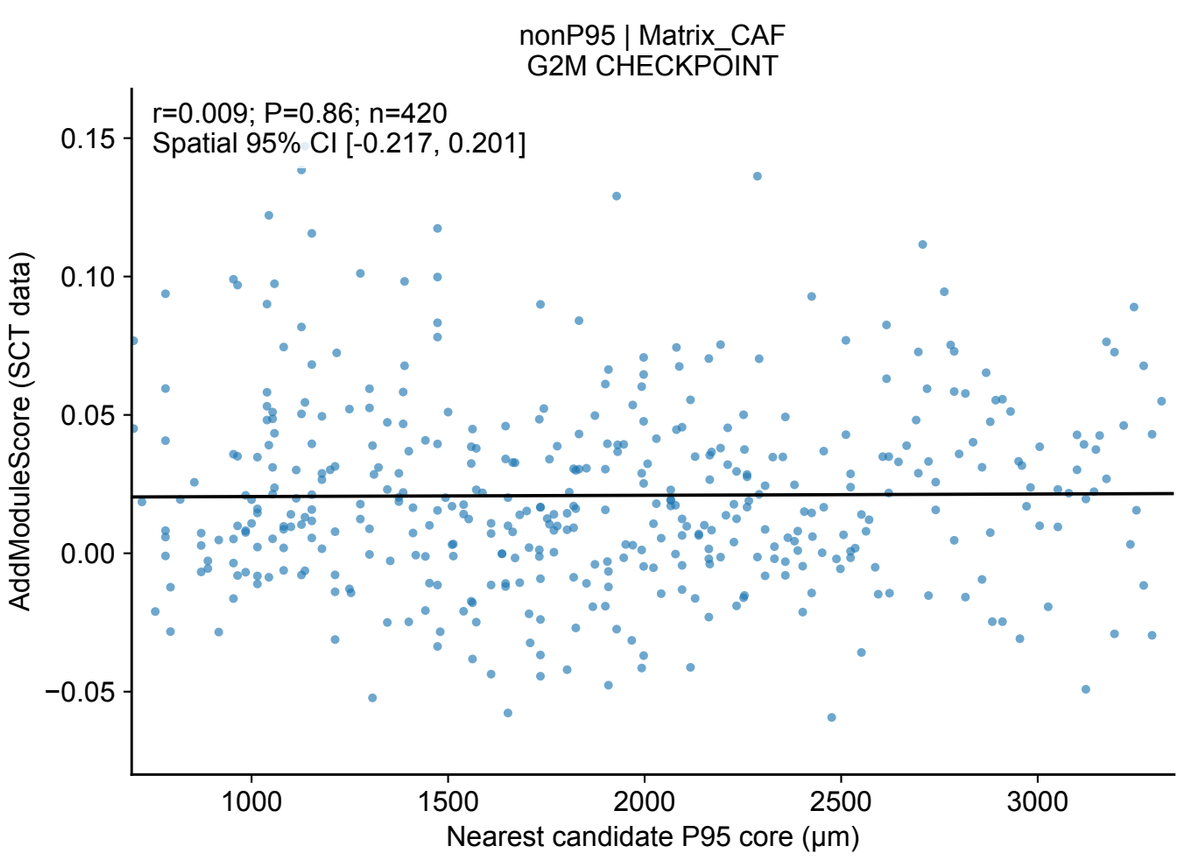


GSE319536_BC19 | exploratory scatter display selected by coverage and |delta r|, not P







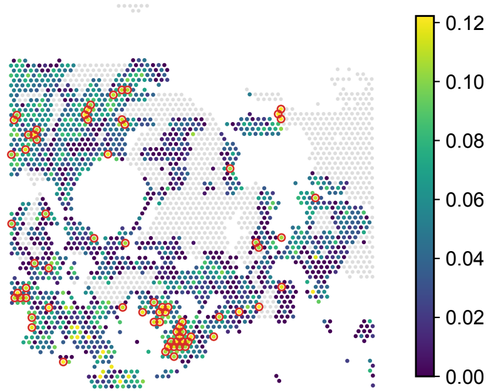


pDC / Flores pro-invasion 10: delta r=-0.688 [-0.849, -0.501]; common 0–557 μm; spatial P=4.6e-09
Matrix_CAF / G2M CHECKPOINT: delta r=0.579 [0.263, 0.811]; common 700–3342 μm; spatial P=0.00012
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

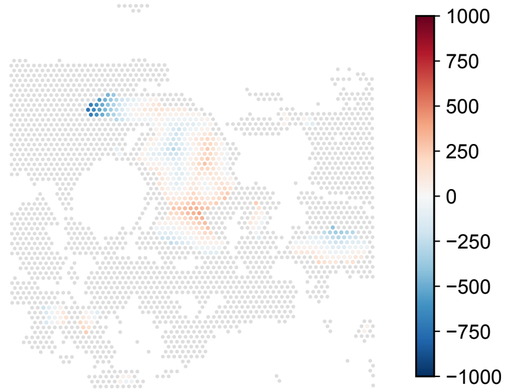
P95 nonP95 distance 0

GSE319536_BC21 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

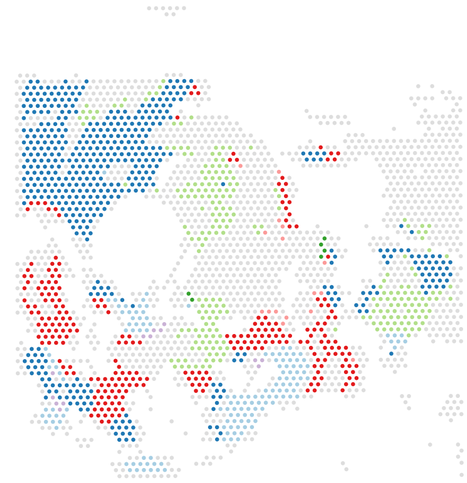
MP10 AUCell and frozen P95



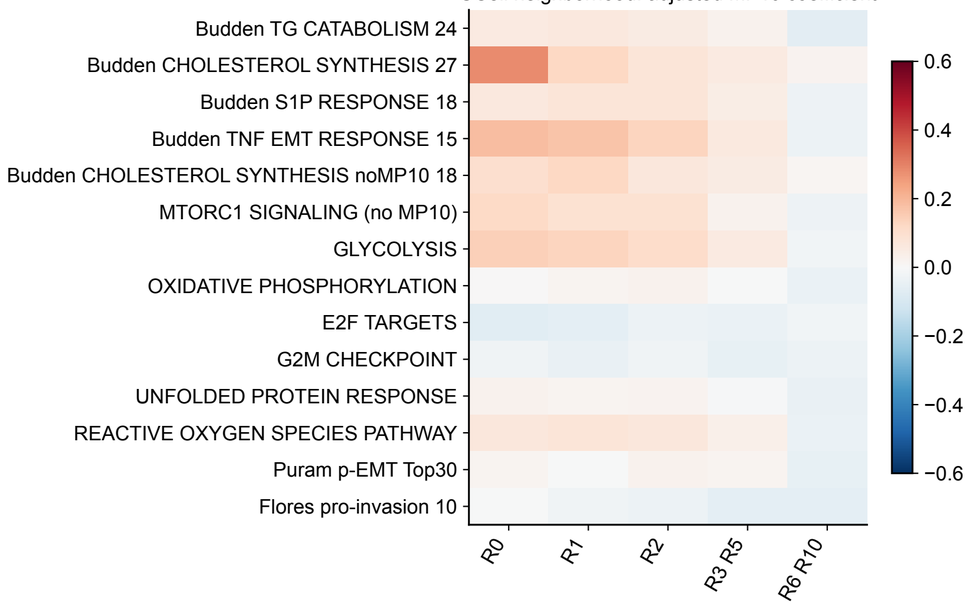
Strict v3.4 signed distance (µm)



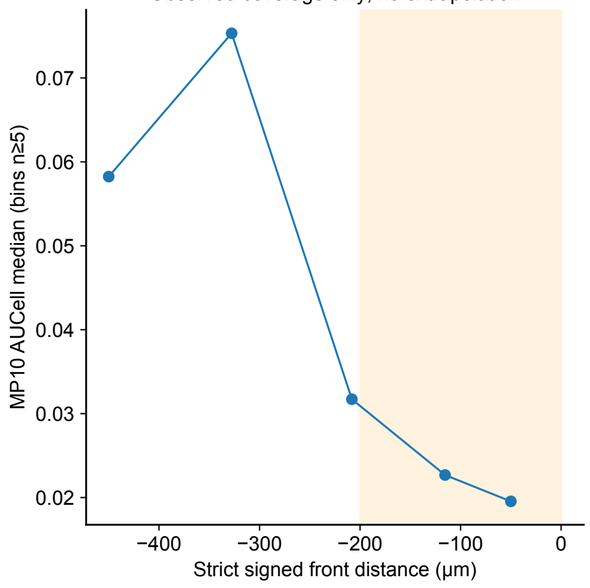
Formal R12 0–8; grey = unlabelled



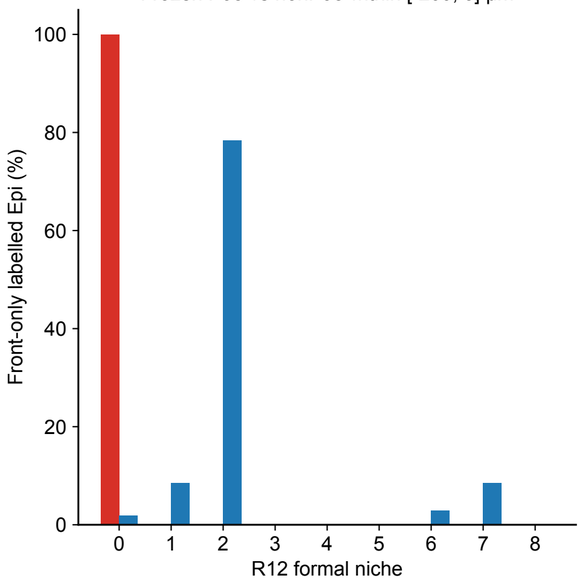
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

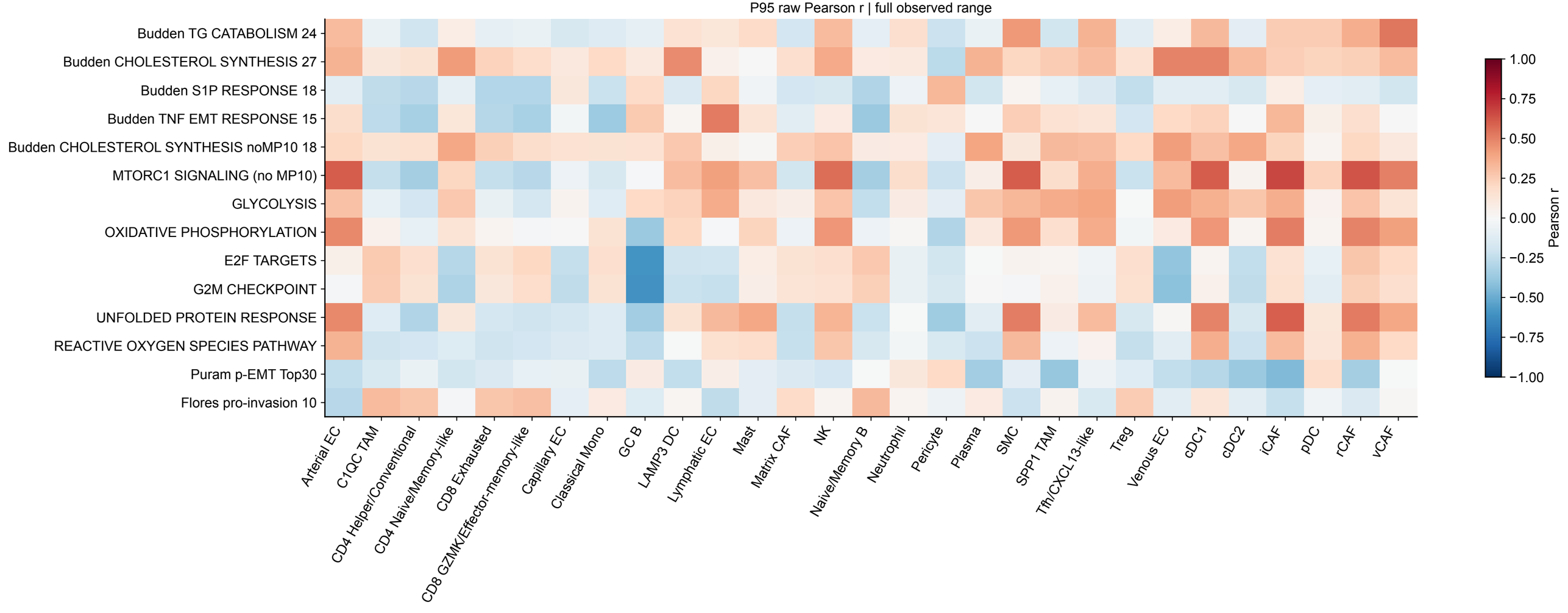


Tissue 2583 | Epi 1705 | internal 1410 | P95 71
Exact front in Epi 245 | R12 labelled 1169

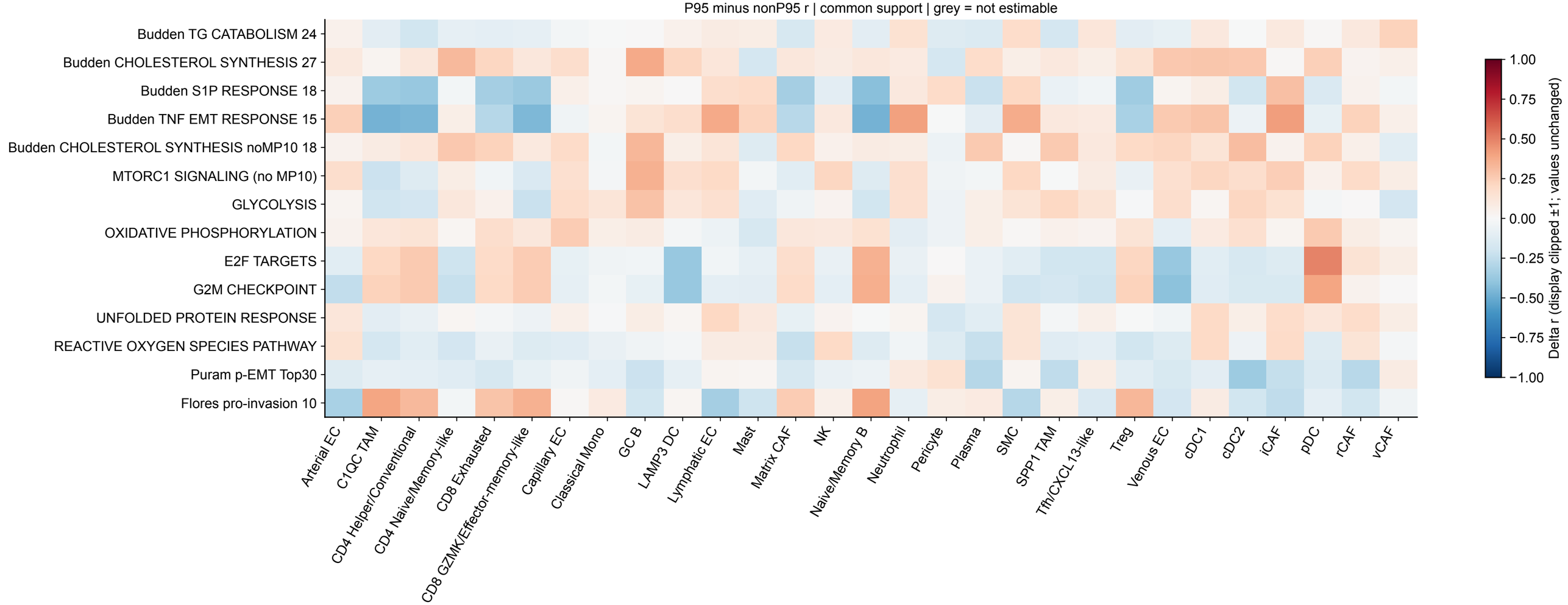


GSE319536_BC21 | all 29 candidates; fixed core programs

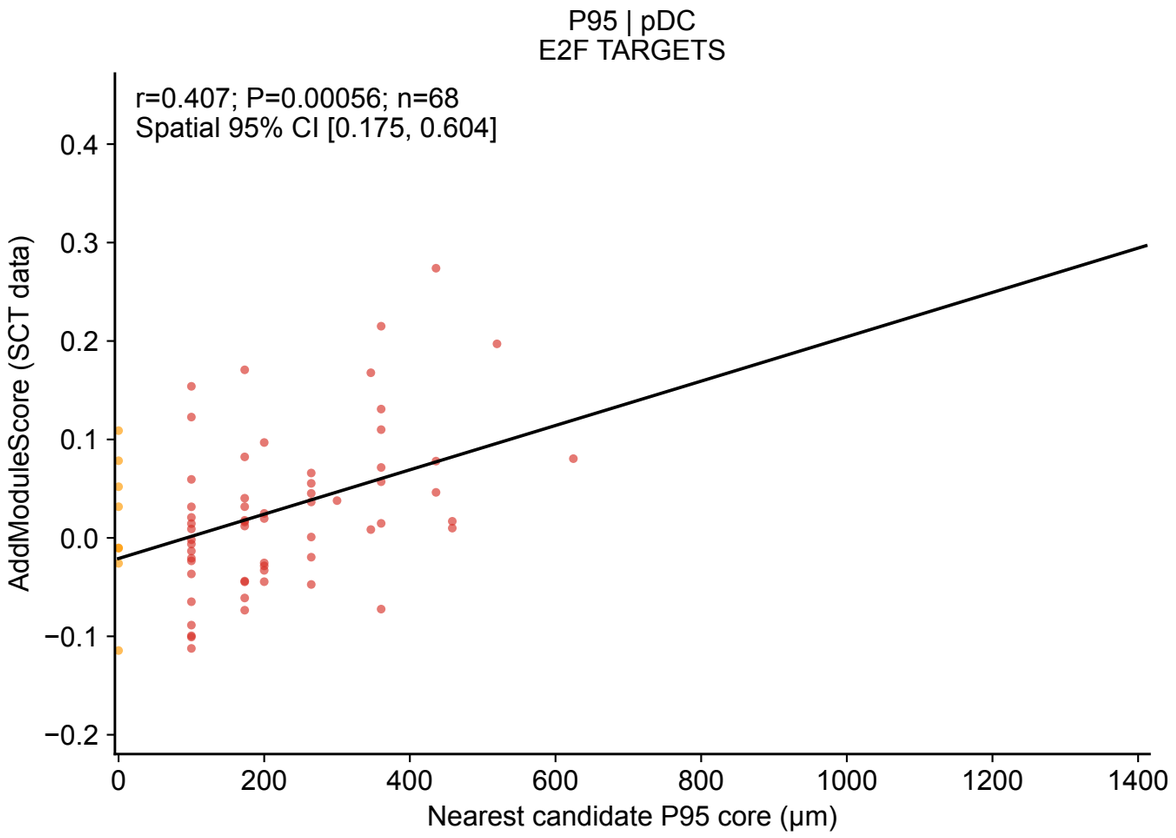
P95 raw Pearson r | full observed range

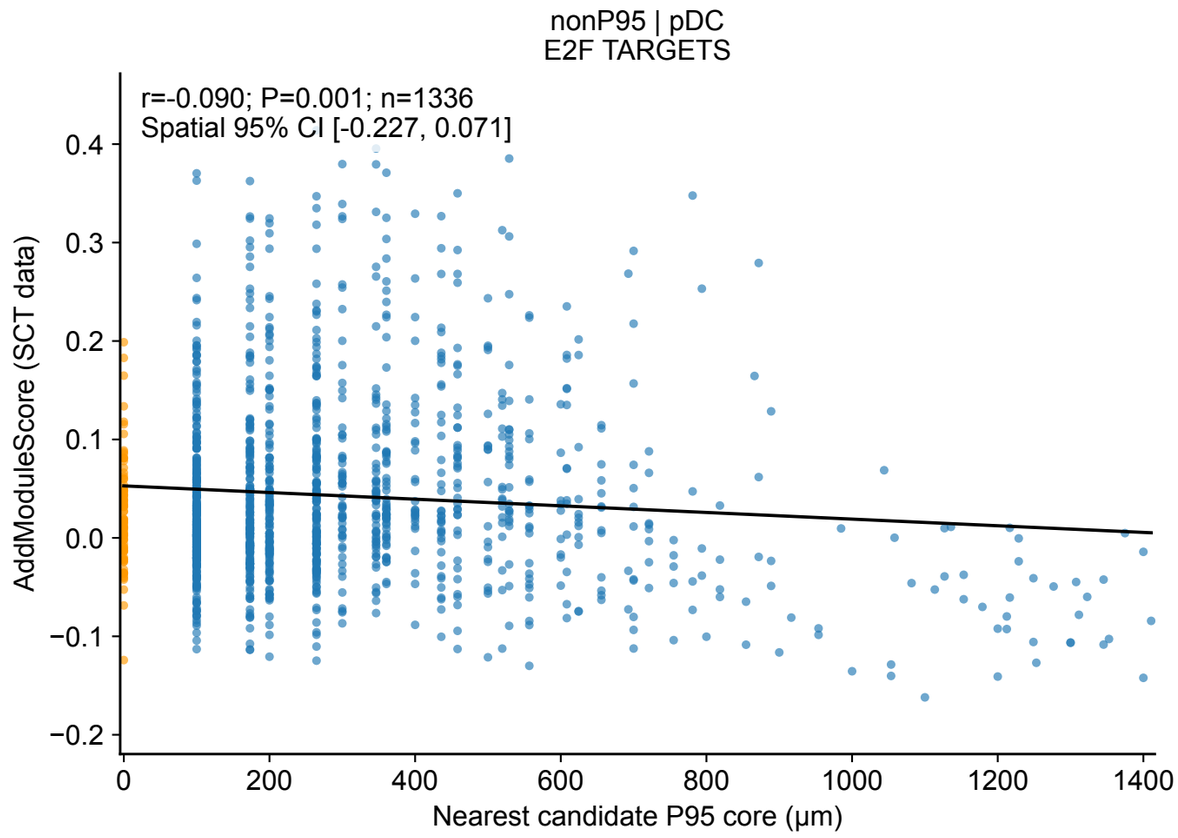


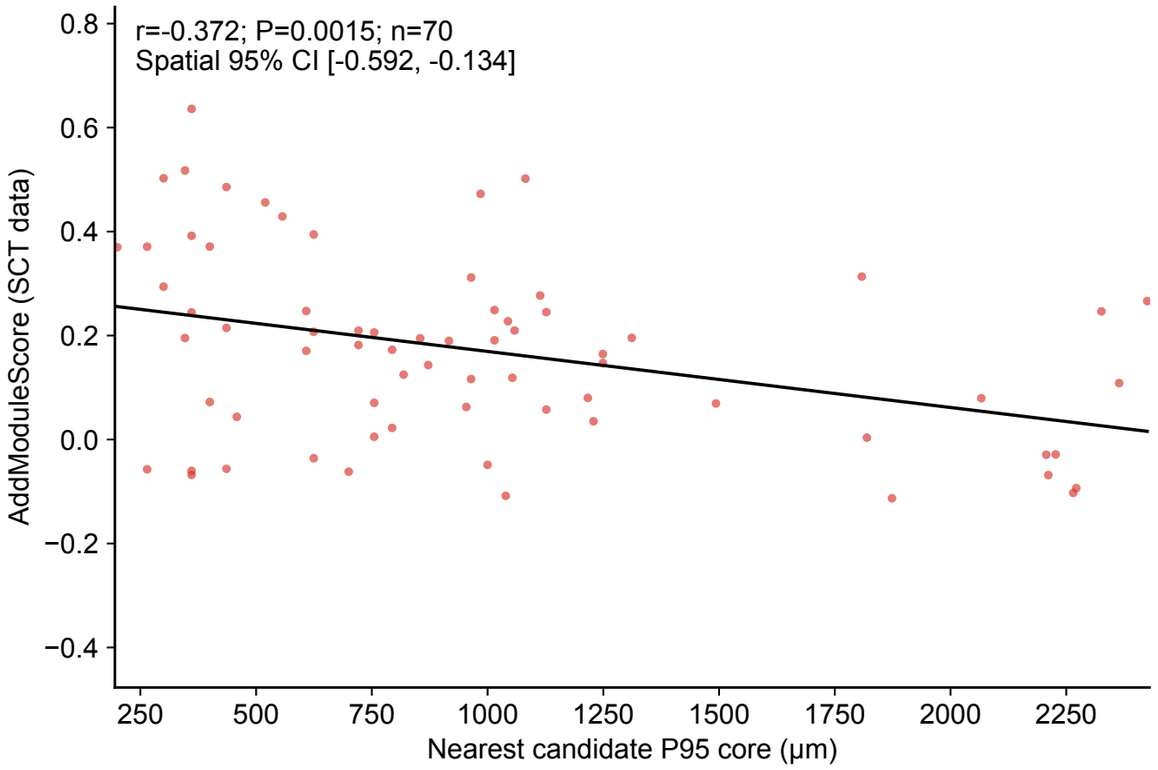
P95 minus nonP95 r | common support | grey = not estimable

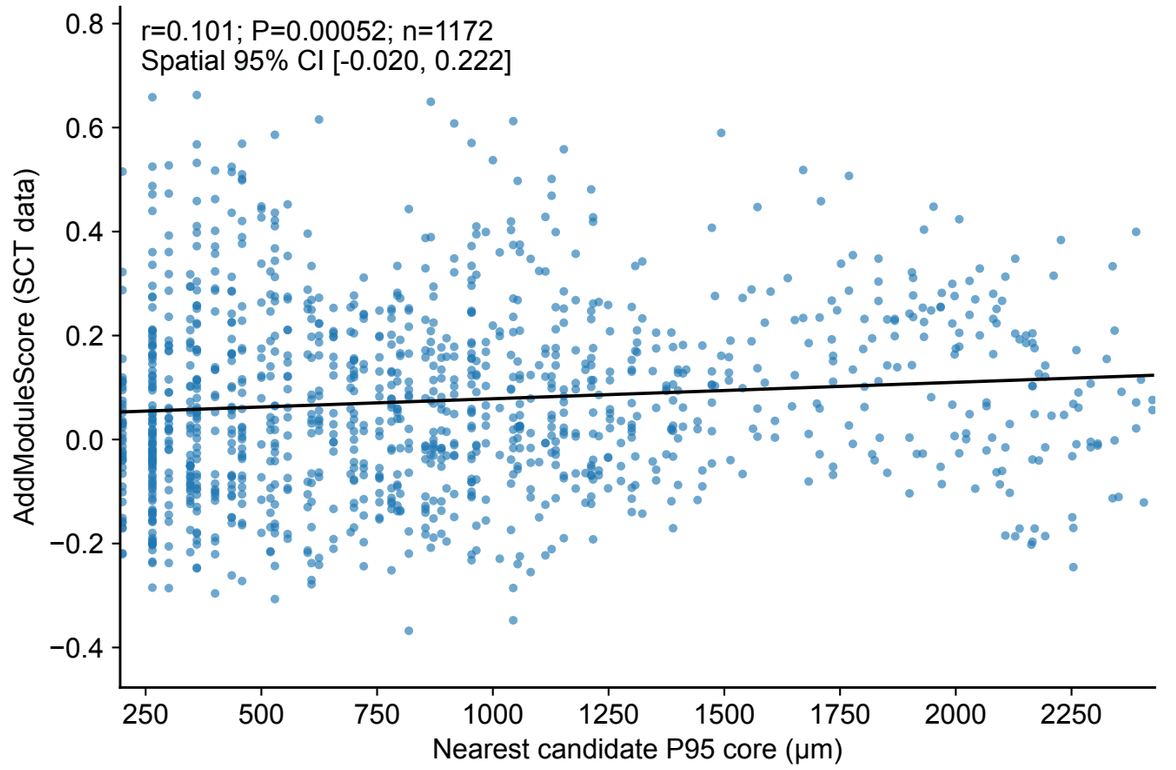


GSE319536_BC21 | exploratory scatter display selected by coverage and |delta r|, not P



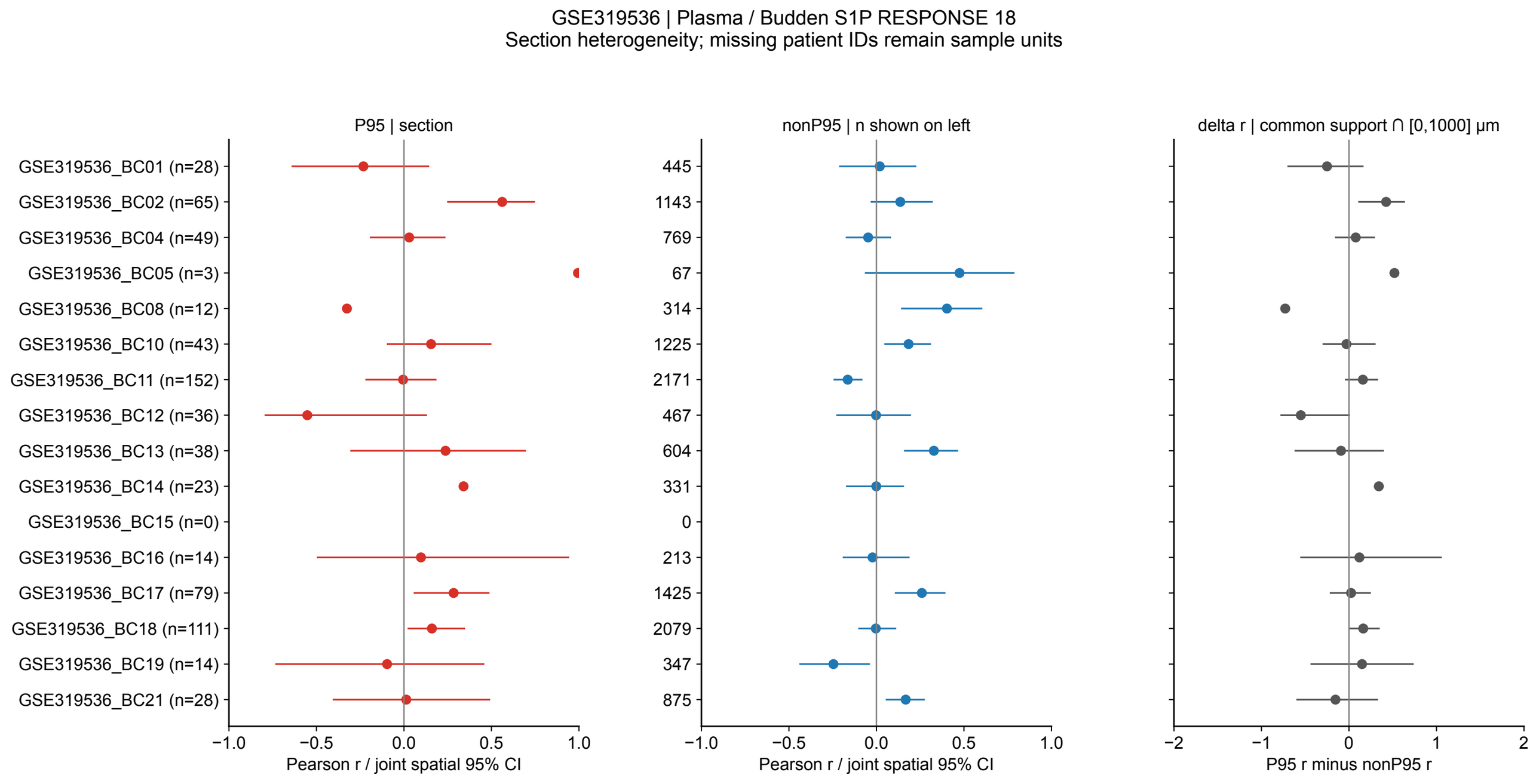


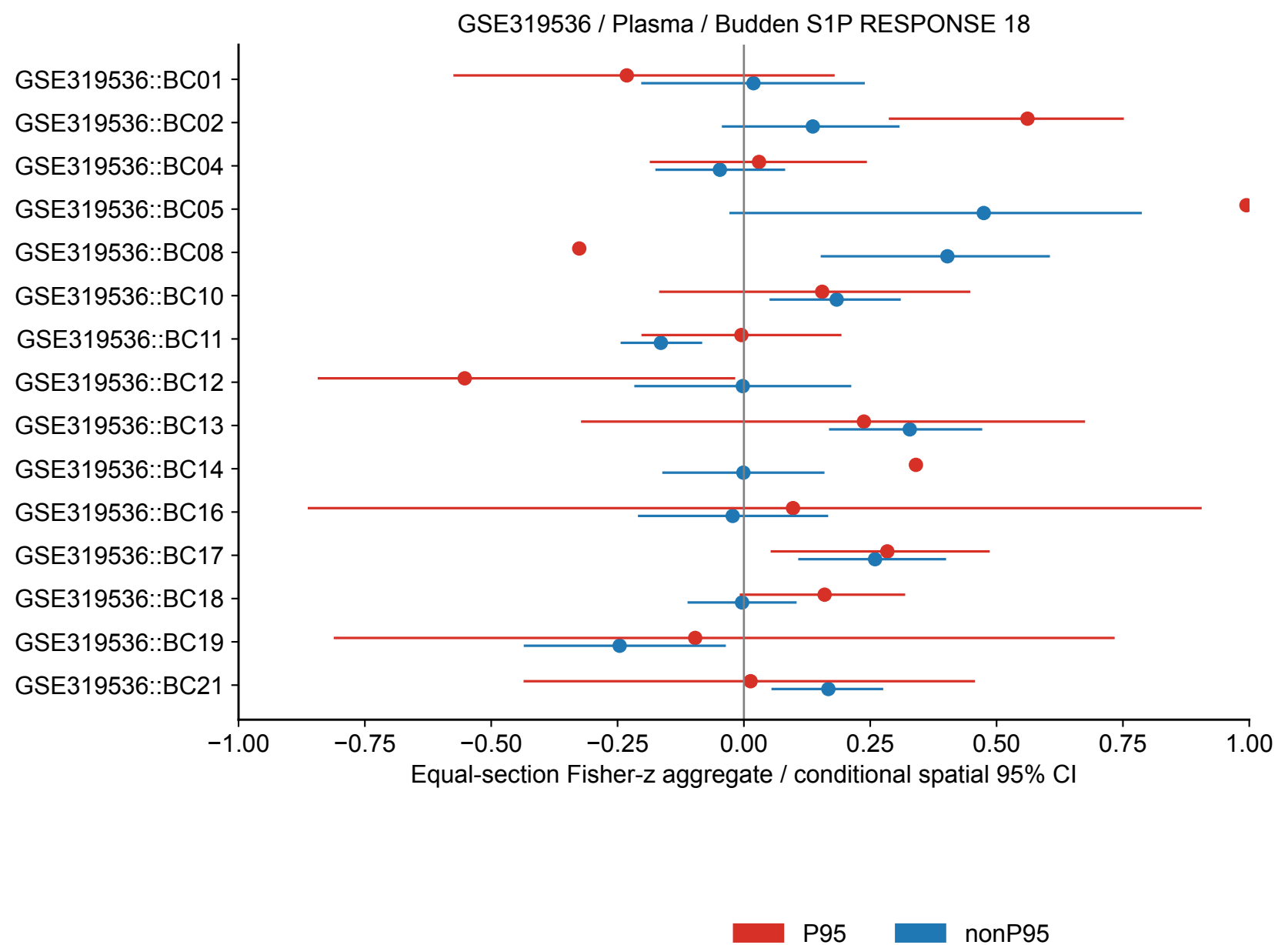


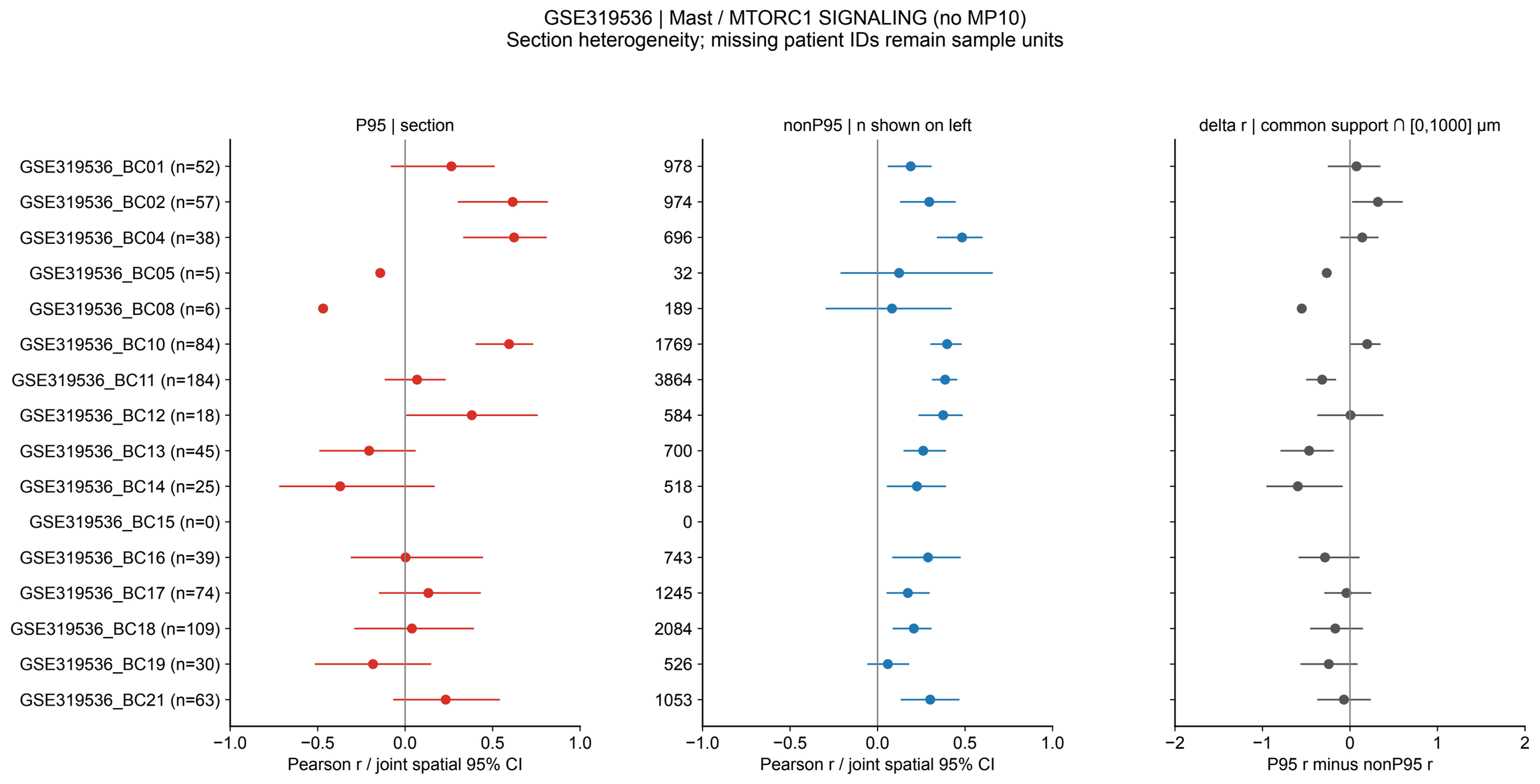


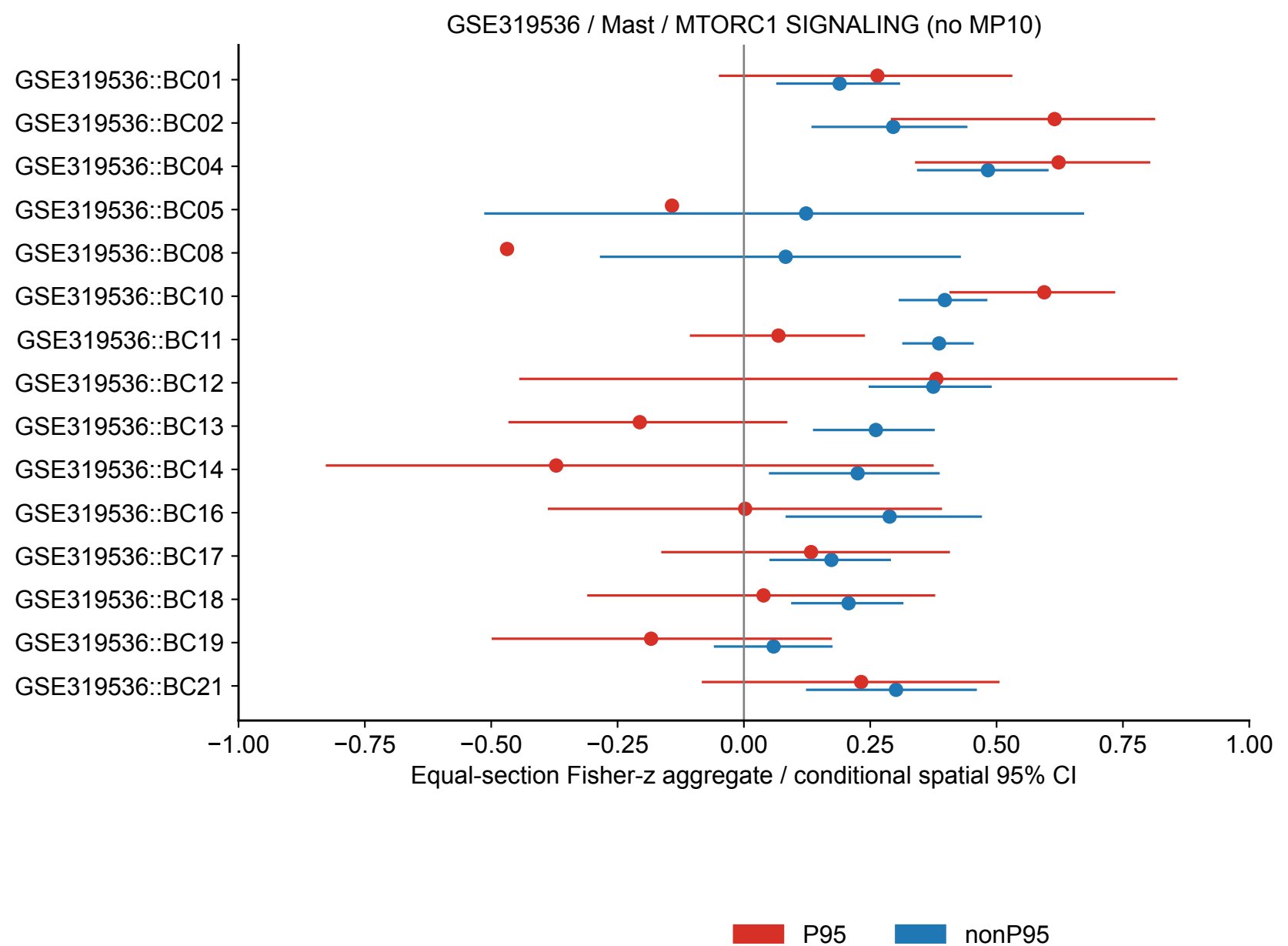
pDC / E2F TARGETS: delta r=0.497 [0.250, 0.737]; common 0–1411 µm; spatial P=0.00059
Naive/Memory B / Budden TNF EMT RESPONSE 15: delta r=-0.473 [-0.711, -0.219]; common 200–2425 µm; spatial P=0.00054
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

P95 nonP95 distance 0


GSE319536 | Plasma / Budden S1P RESPONSE 18
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
GSE319536_BC01 (n=28)
GSE319536_BC02 (n=65)
GSE319536_BC04 (n=49)
GSE319536_BC05 (n=3)
GSE319536_BC08 (n=12)
GSE319536_BC10 (n=43)
GSE319536_BC11 (n=152)
GSE319536_BC12 (n=36)
GSE319536_BC13 (n=38)
GSE319536_BC14 (n=23)
GSE319536_BC15 (n=0)
GSE319536_BC16 (n=14)
GSE319536_BC17 (n=79)
GSE319536_BC18 (n=111)
GSE319536_BC19 (n=14)
GSE319536_BC21 (n=28)
445
1143
769
67
314
1225
2171
467
604
331
0
213
1425
2079
347
875
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


GSE319536 / Plasma / Budden S1P RESPONSE 18
GSE319536::BC01
GSE319536::BC02
GSE319536::BC04
GSE319536::BC05
GSE319536::BC08
GSE319536::BC10
GSE319536::BC11
GSE319536::BC12
GSE319536::BC13
GSE319536::BC14
GSE319536::BC16
GSE319536::BC17
GSE319536::BC18
GSE319536::BC19
GSE319536::BC21
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95


GSE319536 | Mast / MTORC1 SIGNALING (no MP10)
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
GSE319536_BC01 (n=52)
GSE319536_BC02 (n=57)
GSE319536_BC04 (n=38)
GSE319536_BC05 (n=5)
GSE319536_BC08 (n=6)
GSE319536_BC10 (n=84)
GSE319536_BC11 (n=184)
GSE319536_BC12 (n=18)
GSE319536_BC13 (n=45)
GSE319536_BC14 (n=25)
GSE319536_BC15 (n=0)
GSE319536_BC16 (n=39)
GSE319536_BC17 (n=74)
GSE319536_BC18 (n=109)
GSE319536_BC19 (n=30)
GSE319536_BC21 (n=63)
978
974
696
32
189
1769
3864
584
700
518
0
743
1245
2084
526
1053
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


GSE319536 / Mast / MTORC1 SIGNALING (no MP10)
GSE319536::BC01
GSE319536::BC02
GSE319536::BC04
GSE319536::BC05
GSE319536::BC08
GSE319536::BC10
GSE319536::BC11
GSE319536::BC12
GSE319536::BC13
GSE319536::BC14
GSE319536::BC16
GSE319536::BC17
GSE319536::BC18
GSE319536::BC19
GSE319536::BC21
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95